

## Supplementary material

Throughout,  $m_i = \frac{1}{2\sqrt{y_i}}$ ,  $m_j = \frac{1}{2\sqrt{y_j}}$ ,  $\rho = \frac{x_{i1}}{x_{j1}}$ ,  $c = m_j - 1$  and  $d = m_i - m_j$ , as in the appendix of Chapter 1. Every polynomial below was generated and checked with computer algebra.

### Appendix A.3.4

Throughout,  $c = m_j - 1 > 0$  and  $d = m_i - m_j > 0$ , so that  $m_j = 1 + c$  and  $m_i = 1 + c + d$ . Polynomials in  $(c, d)$  have integer coefficients and are ordered by decreasing total degree.

$P_\Delta$

$\Delta S^2 = P_\Delta / (4m_i^2 m_j^2 \rho (m_i + \rho)^2 (1 + \rho)^2)$ . 62 terms.

$$\begin{aligned}
 P_\Delta(m_i, m_j, \rho) = & 64m_i^5 m_j^5 \rho^2 - 64m_i^5 m_j^4 \rho^2 - 8m_i^5 m_j^2 \rho^4 \\
 & - 32m_i^4 m_j^5 \rho^2 + 4m_i^4 m_j^3 \rho^4 - 16m_i^4 m_j^2 \rho^5 \\
 & + 32m_i^3 m_j^5 \rho^3 + 8m_i^3 m_j^3 \rho^5 - 8m_i^3 m_j^2 \rho^6 \\
 & + 32m_i^2 m_j^5 \rho^4 + 4m_i^2 m_j^3 \rho^6 + 40m_i^6 m_j^2 \rho^2 \\
 & + 12m_i^5 m_j^3 \rho^2 + 48m_i^5 m_j^2 \rho^3 + 24m_i^4 m_j^4 \rho^2 \\
 & - 40m_i^4 m_j^3 \rho^3 + 12m_i^4 m_j^2 \rho^4 + 52m_i^3 m_j^5 \rho^2 \\
 & - 20m_i^3 m_j^3 \rho^4 + 8m_i^3 m_j^2 \rho^5 + 24m_i^2 m_j^5 \rho^3 \\
 & - 8m_i^2 m_j^4 \rho^4 + 4m_i^2 m_j^2 \rho^6 - 8m_i m_j^5 \rho^4 \\
 & + 32m_i^6 m_j^2 \rho - 16m_i^6 m_j \rho^2 + 20m_i^5 m_j^2 \rho^2 \\
 & - 16m_i^5 m_j \rho^3 + 16m_i^4 m_j^2 \rho^3 - 24m_i^3 m_j^4 \rho^2 \\
 & + 12m_i^3 m_j^3 \rho^3 + 36m_i^3 m_j^2 \rho^4 - 12m_i^2 m_j^5 \rho^2 \\
 & - 16m_i^2 m_j^4 \rho^3 + 8m_i^2 m_j^3 \rho^4 + 16m_i^2 m_j^2 \rho^5 \\
 & - 4m_i m_j^5 \rho^3 + 4m_j^5 \rho^4 - 16m_i^6 m_j \rho \\
 & + 4m_i^6 \rho^2 - 8m_i^5 m_j^2 \rho - 15m_i^5 m_j \rho^2 \\
 & + 4m_i^5 \rho^3 + 20m_i^4 m_j^2 \rho^2 + 6m_i^3 m_j^3 \rho^2 \\
 & + 52m_i^3 m_j^2 \rho^3 + 4m_i^2 m_j^3 \rho^3 + 24m_i^2 m_j^2 \rho^4 \\
 & + 5m_i m_j^5 \rho^2 + 4m_j^5 \rho^3 + 4m_i^6 \rho \\
 & + 4m_i^5 m_j^2 + 4m_i^5 \rho^2 - 8m_i^4 m_j^3 \\
 & + 28m_i^4 m_j^2 \rho - 16m_i^3 m_j^3 \rho + 40m_i^3 m_j^2 \rho^2 \\
 & - 8m_i^2 m_j^3 \rho^2 + 16m_i^2 m_j^2 \rho^3 + 4m_i^4 m_j^2 \\
 & + 8m_i^3 m_j^2 \rho + 4m_i^2 m_j^2 \rho^2
 \end{aligned}$$

$U_\Delta$

The coefficient of  $\rho$  in  $R_1 \rho + R_0$  multiplied by  $(-A_2)^5$ . Its coefficients are of both signs. 405 terms.

$$\begin{aligned}
 U_\Delta(c, d) = & -2097152c^{30}d - 31457280c^{29}d^2 - 220200960c^{28}d^3 \\
 & - 954204160c^{27}d^4 - 2862612480c^{26}d^5 - 6297747456c^{25}d^6 \\
 & - 10496245760c^{24}d^7 - 13495173120c^{23}d^8 - 13495173120c^{22}d^9
 \end{aligned}$$

$$\begin{aligned}
& -10496245760c^{21}d^{10} - 6297747456c^{20}d^{11} - 2862612480c^{19}d^{12} \\
& - 954204160c^{18}d^{13} - 220200960c^{17}d^{14} - 31457280c^{16}d^{15} \\
& - 2097152c^{15}d^{16} - 52428800c^{29}d - 759693312c^{28}d^2 \\
& - 5132648448c^{27}d^3 - 21446262784c^{26}d^4 - 61971628032c^{25}d^5 \\
& - 131162701824c^{24}d^6 - 210021515264c^{23}d^7 - 259029467136c^{22}d^8 \\
& - 248051662848c^{21}d^9 - 184393662464c^{20}d^{10} - 105508896768c^{19}d^{11} \\
& - 45621706752c^{18}d^{12} - 14424997888c^{17}d^{13} - 3147300864c^{16}d^{14} \\
& - 423493632c^{15}d^{15} - 26476544c^{14}d^{16} - 636223488c^{28}d \\
& - 8893497344c^{27}d^2 - 57914556416c^{26}d^3 - 233013608448c^{25}d^4 \\
& - 647627767808c^{24}d^5 - 1316758716416c^{23}d^6 - 2022629081088c^{22}d^7 \\
& - 2389324726272c^{21}d^8 - 2187617304576c^{20}d^9 - 1551700525056c^{19}d^{10} \\
& - 845274349568c^{18}d^{11} - 347061714944c^{17}d^{12} - 103896416256c^{16}d^{13} \\
& - 21390000128c^{15}d^{14} - 2705358848c^{14}d^{15} - 158269440c^{13}d^{16} \\
& + 2097152c^{28} - 4964614144c^{27}d - 67053551616c^{26}d^2 \\
& - 420660666368c^{25}d^3 - 1628004925440c^{24}d^4 - 4346539032576c^{23}d^5 \\
& - 8477424320512c^{22}d^6 - 12472628682752c^{21}d^7 - 14088810479616c^{20}d^8 \\
& - 12311592140800c^{19}d^9 - 8317211312128c^{18}d^{10} - 4304838500352c^{17}d^{11} \\
& - 1674849959936c^{16}d^{12} - 473623707648c^{15}d^{13} - 91783004160c^{14}d^{14} \\
& - 10882048000c^{13}d^{15} - 593936384c^{12}d^{16} + 51118080c^{27} \\
& - 27816239104c^{26}d - 364811198464c^{25}d^2 - 2205927698432c^{24}d^3 \\
& - 8206677018624c^{23}d^4 - 21023616428032c^{22}d^5 - 39278376171520c^{21}d^6 \\
& - 55262609242112c^{20}d^7 - 59584449085440c^{19}d^8 - 49599391199232c^{18}d^9 \\
& - 31846100684800c^{17}d^{10} - 15625599066112c^{16}d^{11} - 5746302568448c^{15}d^{12} \\
& - 1530827409408c^{14}d^{13} - 278394001408c^{13}d^{14} - 30836389888c^{12}d^{15} \\
& - 1564061696c^{11}d^{16} + 600899584c^{26} - 118306406400c^{25}d \\
& - 1520646535168c^{24}d^2 - 8877520510976c^{23}d^3 - 31742601084928c^{22}d^4 \\
& - 77950634513920c^{21}d^5 - 139311749408768c^{20}d^6 - 187118253646848c^{19}d^7 \\
& - 192205060116480c^{18}d^8 - 152081643803648c^{17}d^9 - 92584716441600c^{16}d^{10} \\
& - 42951548612608c^{15}d^{11} - 14886623252480c^{14}d^{12} - 3723889698304c^{13}d^{13} \\
& - 633186027520c^{12}d^{14} - 65244855296c^{11}d^{15} - 3060080640c^{10}d^{16} \\
& + 4538925056c^{25} - 393329410048c^{24}d - 5031572600832c^{23}d^2 \\
& - 28449277747712c^{22}d^3 - 97804284679168c^{21}d^4 - 230062195707648c^{20}d^5 \\
& - 392773379845504c^{19}d^6 - 502745411696512c^{18}d^7 - 490939224367744c^{17}d^8 \\
& - 368350372137856c^{16}d^9 - 212044014915456c^{15}d^{10} - 92725378007168c^{14}d^{11} \\
& - 30184663425920c^{13}d^{12} - 7062359572352c^{12}d^{13} - 1117706929408c^{11}d^{14} \\
& - 106576571392c^{10}d^{15} - 4593011712c^9d^{16} + 24764563456c^{24} \\
& - 1033887625216c^{23}d - 13505696454656c^{22}d^2 - 74341465325056c^{21}d^3 \\
& - 245991927237504c^{20}d^4 - 553990048466560c^{19}d^5 - 902294388620544c^{18}d^6 \\
& - 1098532371496640c^{17}d^7 - 1017454574850304c^{16}d^8 - 721920288357312c^{15}d^9 \\
& - 391742918755712c^{14}d^{10} - 160901030175424c^{13}d^{11} - 48994102402944c^{12}d^{12} \\
& - 10671407619008c^{11}d^{13} - 1563308736128c^{10}d^{14} - 137036384896c^9d^{15} \\
& - 5382955776c^8d^{16} + 104004623360c^{23} - 2130176694272c^{22}d \\
& - 29770414891776c^{21}d^2 - 160858059677952c^{20}d^3 - 513374066245632c^{19}d^4 \\
& - 1106736729149888c^{18}d^5 - 1717457904959936c^{17}d^6 - 1984902316981568c^{16}d^7
\end{aligned}$$

$$\begin{aligned}
& -1739192030525088c^{15}d^8 - 1163384056804256c^{14}d^9 - 592946104154400c^{13}d^{10} \\
& - 227792202604128c^{12}d^{11} - 64565993328032c^{11}d^{12} - 13016952399776c^{10}d^{13} \\
& - 1753125745184c^9d^{14} - 140105427552c^8d^{15} - 4964512704c^7d^{16} \\
& + 349835929856c^{22} - 3276220856064c^{21}d - 54098952886528c^{20}d^2 \\
& - 290882787745792c^{19}d^3 - 898610748436288c^{18}d^4 - 1855138232841536c^{17}d^5 \\
& - 2739740615334944c^{16}d^6 - 2999366356472160c^{15}d^7 - 2479152431718304c^{14}d^8 \\
& - 1557940938900656c^{13}d^9 - 742675761782656c^{12}d^{10} - 265542561733520c^{11}d^{11} \\
& - 69651140090592c^{10}d^{12} - 12906572215312c^9d^{13} - 1584480880928c^8d^{14} \\
& - 114226024464c^7d^{15} - 3601589120c^6d^{16} + 968126717696c^{21} \\
& - 3055679530624c^{20}d - 80517002510464c^{19}d^2 - 441288039404480c^{18}d^3 \\
& - 1327715954304192c^{17}d^4 - 2628089522126112c^{16}d^5 - 3690281023827904c^{15}d^6 \\
& - 3818403488094736c^{14}d^7 - 2967826311515440c^{13}d^8 - 1745027026092304c^{12}d^9 \\
& - 774222993866320c^{11}d^{10} - 256115346901216c^{10}d^{11} - 61723507422520c^9d^{12} \\
& - 10421202448168c^8d^{13} - 1153583283368c^7d^{14} - 73983216792c^6d^{15} \\
& - 2037801920c^5d^{16} + 2246691922944c^{20} + 1036191027200c^{19}d \\
& - 95668956542976c^{18}d^2 - 560380773445376c^{17}d^3 - 1659837577410752c^{16}d^4 \\
& - 3158649253907328c^{15}d^5 - 4214948591512576c^{14}d^6 - 4112747304113792c^{13}d^7 \\
& - 2995243243206320c^{12}d^8 - 1640144212203232c^{11}d^9 - 673327610391008c^{10}d^{10} \\
& - 204603708238888c^9d^{11} - 44906371309624c^8d^{12} - 6832203797788c^7d^{13} \\
& - 672375883880c^6d^{14} - 37652495156c^5d^{15} - 882840168c^4d^{16} \\
& + 4433591905792c^{19} + 11589660284160c^{18}d - 83916900181632c^{17}d^2 \\
& - 588968620006464c^{16}d^3 - 1752076497252480c^{15}d^4 - 3222605971736000c^{14}d^5 \\
& - 4088072904298624c^{13}d^6 - 3753371357962416c^{12}d^7 - 2551270684219904c^{11}d^8 \\
& - 1293991166372048c^{10}d^9 - 488112521324688c^9d^{10} - 135051287234752c^8d^{11} \\
& - 26695532138832c^7d^{12} - 3607988625168c^6d^{13} - 309756300912c^5d^{14} \\
& - 14755577728c^4d^{15} - 283419136c^3d^{16} + 7517093122560c^{18} \\
& + 29262012011008c^{17}d - 37801038448640c^{16}d^2 - 497399125672704c^{15}d^3 \\
& - 1548777292150464c^{14}d^4 - 2782300261571904c^{13}d^5 - 3361889726523104c^{12}d^6 \\
& - 2898883652629056c^{11}d^7 - 1831140380164112c^{10}d^8 - 854776894784816c^9d^9 \\
& - 293764077093040c^8d^{10} - 73195900024680c^7d^{11} - 12845341661252c^6d^{12} \\
& - 1513074820020c^5d^{13} - 110373664598c^4d^{14} - 4303038124c^3d^{15} \\
& - 63640318c^2d^{16} + 11033835448832c^{17} + 51041283087616c^{16}d \\
& + 33434417910656c^{15}d^2 - 310966821304128c^{14}d^3 - 1124787005885632c^{13}d^4 \\
& - 2016380650562976c^{12}d^5 - 2332698112519104c^{11}d^6 - 1887043760003488c^{10}d^7 \\
& - 1102558002456304c^9d^8 - 470126071116792c^8d^9 - 145666880744424c^7d^{10} \\
& - 32228274758588c^6d^{11} - 4926588925724c^5d^{12} - 492538405394c^4d^{13} \\
& - 29363058812c^3d^{14} - 880388383c^2d^{15} - 8936865cd^{16} \\
& + 14097868334592c^{16} + 70570691594240c^{15}d + 105640189687808c^{14}d^2 \\
& - 99560029423360c^{13}d^3 - 642064849593408c^{12}d^4 - 1206781703672064c^{11}d^5 \\
& - 1352792016597504c^{10}d^6 - 1027476872590720c^9d^7 - 552617616023520c^8d^8 \\
& - 213307375417600c^7d^9 - 58801833432192c^6d^{10} - 11338939427640c^5d^{11} \\
& - 1470575463924c^4d^{12} - 120014238996c^3d^{13} - 5493199048c^2d^{14} \\
& - 112879634cd^{15} - 591675d^{16} + 15737116333568c^{15} \\
& + 81039878480128c^{14}d + 152664328682880c^{13}d^2 + 63397794413760c^{12}d^3
\end{aligned}$$

$$\begin{aligned}
& -253033056398976c^{11}d^4 - 576968345254208c^{10}d^5 - 644843727768896c^9d^6 \\
& - 462194375763072c^8d^7 - 227807863237888c^7d^8 - 78738318879664c^6d^9 \\
& - 18980064736072c^5d^{10} - 3109206797836c^4d^{11} - 329139267808c^3d^{12} \\
& - 20599383416c^2d^{13} - 644939456cd^{14} - 6831335d^{15} \\
& + 15380192359424c^{14} + 78922471921664c^{13}d + 161202325199360c^{12}d^2 \\
& + 140846862224128c^{11}d^3 - 26643328119488c^{10}d^4 - 203378175076288c^9d^5 \\
& - 245086195031392c^8d^6 - 168412038410592c^7d^7 - 75860765121440c^6d^8 \\
& - 23167556862256c^5d^9 - 4769867885760c^4d^{10} - 639358189064c^3d^{11} \\
& - 51929760052c^2d^{12} - 2220852164cd^{13} - 35762510d^{14} \\
& + 13169390565376c^{13} + 65840341232640c^{12}d + 136453088647296c^{11}d^2 \\
& + 142250050163776c^{10}d^3 + 56773542569408c^9d^4 - 38650834209632c^8d^5 \\
& - 69659335631424c^7d^6 - 48100118351056c^6d^7 - 19859073890288c^5d^8 \\
& - 5271339306320c^4d^9 - 896028155880c^3d^{10} - 92555455220c^2d^{11} \\
& - 5144507036cd^{12} - 113083030d^{13} + 9872860726784c^{12} \\
& + 47251732950016c^{11}d + 95626198645760c^{10}d^2 + 103562594216192c^9d^3 \\
& + 59227377736384c^8d^4 + 8682448927872c^7d^5 - 12210600287232c^6d^6 \\
& - 10121870455552c^5d^7 - 3914500332880c^4d^8 - 885114290208c^3d^9 \\
& - 117723451936c^2d^{10} - 8385081440cd^{11} - 240304188d^{12} \\
& + 6466841821696c^{11} + 29185161714432c^{10}d + 56157562231936c^9d^2 \\
& + 59318908266816c^8d^3 + 36217181742208c^7d^4 + 11379434623424c^6d^5 \\
& + 146766762240c^5d^6 - 1344752589232c^4d^7 - 538760425440c^3d^8 \\
& - 101481823904c^2d^9 - 9565271144cd^{10} - 355958340d^{11} \\
& + 3687712095744c^{10} + 15473394128384c^9d + 27701878394880c^8d^2 \\
& + 27437530192640c^7d^3 + 16180297021632c^6d^4 + 5564140678720c^5d^5 \\
& + 905369448160c^4d^6 - 41360255488c^3d^7 - 44000236336c^2d^8 \\
& - 6857135920cd^9 - 356415504d^{10} + 1821103146496c^9 \\
& + 7003177435904c^8d + 11432318744192c^7d^2 + 10290438269504c^6d^3 \\
& + 5532469476288c^5d^4 + 1783399152160c^4d^5 + 319128833344c^3d^6 \\
& + 22705226752c^2d^7 - 1117860848cd^8 - 189349224d^9 \\
& + 773077053952c^8 + 2682408337408c^7d + 3909337786368c^6d^2 \\
& + 3102417034496c^5d^3 + 1449726406080c^4d^4 + 400731218560c^3d^5 \\
& + 61660866816c^2d^6 + 4404846784cd^7 + 74232672d^8 \\
& + 279302903296c^7 + 858652048128c^6d + 1090207694720c^5d^2 \\
& + 736872304960c^4d^3 + 284019905152c^3d^4 + 61647078144c^2d^5 \\
& + 6828713088cd^6 + 285362208d^7 + 84724314880c^6 \\
& + 225639462656c^5d + 241972113152c^4d^2 + 133137939712c^3d^3 \\
& + 39395592192c^2d^4 + 5890582016cd^5 + 342845312d^6 \\
& + 21180852480c^5 + 47439704704c^4d + 41163666944c^3d^2 \\
& + 17231920768c^2d^3 + 3462845440cd^4 + 265398016d^5 \\
& + 4250705408c^4 + 7677488128c^3d + 5045466112c^2d^2 \\
& + 1425239552cd^3 + 145405952d^4 + 658556928c^3 \\
& + 898427904c^2d + 397012992cd^2 + 56665600d^3 \\
& + 73965568c^2 + 67698688cd + 15073280d^2 \\
& + 5361664c + 2467840d + 188416
\end{aligned}$$

$V_\Delta$

The constant term of  $R_1\rho + R_0$  multiplied by  $(-A_2)^5$ . Every coefficient is strictly positive. 376 terms.

$$\begin{aligned}
V_\Delta(c, d) = & 2097152c^{30} + 31457280c^{29}d + 220200960c^{28}d^2 \\
& + 954204160c^{27}d^3 + 2862612480c^{26}d^4 + 6297747456c^{25}d^5 \\
& + 10496245760c^{24}d^6 + 13495173120c^{23}d^7 + 13495173120c^{22}d^8 \\
& + 10496245760c^{21}d^9 + 6297747456c^{20}d^{10} + 2862612480c^{19}d^{11} \\
& + 954204160c^{18}d^{12} + 220200960c^{17}d^{13} + 31457280c^{16}d^{14} \\
& + 2097152c^{15}d^{15} + 51904512c^{29} + 751828992c^{28}d \\
& + 5077598208c^{27}d^2 + 21207711744c^{26}d^3 + 61255974912c^{25}d^4 \\
& + 129588264960c^{24}d^5 + 207397453824c^{23}d^6 + 255655673856c^{22}d^7 \\
& + 244677869568c^{21}d^8 + 181769601024c^{20}d^9 + 103934459904c^{19}d^{10} \\
& + 44906053632c^{18}d^{11} + 14186446848c^{17}d^{12} + 3092250624c^{16}d^{13} \\
& + 415629312c^{15}d^{14} + 25952256c^{14}d^{15} + 623378432c^{28} \\
& + 8707637248c^{27}d + 56660754432c^{26}d^2 + 227783245824c^{25}d^3 \\
& + 632540397568c^{24}d^4 + 1284886462464c^{23}d^5 + 1971698139136c^{22}d^6 \\
& + 2326647865344c^{21}d^7 + 2127741124608c^{20}d^8 + 1507307749376c^{19}d^9 \\
& + 819946749952c^{18}d^{10} + 336145317888c^{17}d^{11} + 100457152512c^{16}d^{12} \\
& + 20642627584c^{15}d^{13} + 2605252608c^{14}d^{14} + 152043520c^{13}d^{15} \\
& + 4841472000c^{27} + 65123188736c^{26}d + 407682064384c^{25}d^2 \\
& + 1575113048064c^{24}d^3 + 4198761709568c^{23}d^4 + 8176560898048c^{22}d^5 \\
& + 12010965983232c^{21}d^6 + 13544930131968c^{20}d^7 + 11815632273408c^{19}d^8 \\
& + 7967243698176c^{18}d^9 + 4115395076096c^{17}d^{10} + 1597644292096c^{16}d^{11} \\
& + 450712387584c^{15}d^{12} + 87113990144c^{14}d^{13} + 10298564608c^{13}d^{14} \\
& + 560283648c^{12}d^{15} + 27337695232c^{26} + 353561600000c^{25}d \\
& + 2126039848960c^{24}d^2 + 7881508292608c^{23}d^3 + 20134379042816c^{22}d^4 \\
& + 37524417693696c^{21}d^5 + 52671980089344c^{20}d^6 + 56660831440896c^{19}d^7 \\
& + 47055936737280c^{18}d^8 + 30140404707328c^{17}d^9 + 14751473776640c^{16}d^{10} \\
& + 5410422482944c^{15}d^{11} + 1437273511936c^{14}d^{12} + 260591593472c^{13}d^{13} \\
& + 28771418112c^{12}d^{14} + 1454309376c^{11}d^{15} + 119623137280c^{25} \\
& + 1485098504192c^{24}d + 8563442896896c^{23}d^2 + 30407067214848c^{22}d^3 \\
& + 74307951257088c^{21}d^4 + 132287441589248c^{20}d^5 + 177087649860608c^{19}d^6 \\
& + 181342439497728c^{18}d^7 + 143065597946880c^{17}d^8 + 86845113567232c^{16}d^9 \\
& + 40172803665920c^{15}d^{10} + 13882921276416c^{14}d^{11} + 3462495340032c^{13}d^{12} \\
& + 586960376832c^{12}d^{13} + 60297503744c^{11}d^{14} + 2819581952c^{10}d^{15} \\
& + 422179013632c^{24} + 5022475833344c^{23}d + 27720946864128c^{22}d^2 \\
& + 94101698148864c^{21}d^3 + 219546934132096c^{20}d^4 + 372574758624384c^{19}d^5 \\
& + 474606228502912c^{18}d^6 + 461573766913664c^{17}d^7 + 345064342660736c^{16}d^8 \\
& + 197979142184320c^{15}d^9 + 86304777530496c^{14}d^{10} + 28011276489088c^{13}d^{11} \\
& + 6535379006976c^{12}d^{12} + 1031581366272c^{11}d^{13} + 98133085184c^{10}d^{14} \\
& + 4221236224c^9d^{15} + 1234852761600c^{23} + 14051056760064c^{22}d \\
& + 74087123809408c^{21}d^2 + 239935742718208c^{20}d^3 + 533264579367040c^{19}d^4 \\
& + 860643153641984c^{18}d^5 + 1040683812150208c^{17}d^6 + 958670485354752c^{16}d^7 \\
& + 677176616577216c^{15}d^8 + 366068699873152c^{14}d^9 + 149861830940736c^{13}d^{10} \\
& + 45502600182784c^{12}d^{11} + 9886951959744c^{11}d^{12} + 1445625520768c^{10}d^{13}
\end{aligned}$$

$$\begin{aligned}
& + 126567569792c^9d^{14} + 4971084032c^8d^{15} + 3052368571136c^{22} \\
& + 33153992500736c^{21}d + 166642965853952c^{20}d^2 + 513705091728128c^{19}d^3 \\
& + 1084984634614720c^{18}d^4 + 1660984805086016c^{17}d^5 + 1901143490274528c^{16}d^6 \\
& + 1653785092567200c^{15}d^7 + 1100082230308640c^{14}d^8 + 558218775425248c^{13}d^9 \\
& + 213707028478880c^{12}d^{10} + 60412008015072c^{11}d^{11} + 12156610928928c^{10}d^{12} \\
& + 1635672482272c^9d^{13} + 130751479168c^8d^{14} + 4642705280c^7d^{15} \\
& + 6469043334400c^{21} + 66928323835520c^{20}d + 319940504757376c^{19}d^2 \\
& + 936441443824512c^{18}d^3 + 1874444920640608c^{17}d^4 + 2713906631210880c^{16}d^5 \\
& + 2930869737234496c^{15}d^6 + 2398994182140960c^{14}d^7 + 1496810629011056c^{13}d^8 \\
& + 709777222430432c^{12}d^9 + 252813052530192c^{11}d^{10} + 66142341198336c^{10}d^{11} \\
& + 12239824274864c^9d^{12} + 1502654688224c^8d^{13} + 108525008304c^7d^{14} \\
& + 3437462432c^6d^{15} + 11884262656000c^{20} + 116843214423040c^{19}d \\
& + 529865863465600c^{18}d^2 + 1468428419154560c^{17}d^3 + 2777164062788352c^{16}d^4 \\
& + 3790043605993600c^{15}d^5 + 3847513872470960c^{14}d^6 + 2951018012280240c^{13}d^7 \\
& + 1718917392199984c^{12}d^8 + 757604011122864c^{11}d^9 + 249492550506760c^{10}d^{10} \\
& + 59963996595752c^9d^{11} + 10113621929624c^8d^{12} + 1120461893064c^7d^{13} \\
& + 72095335648c^6d^{14} + 1999889808c^5d^{15} + 19083054022144c^{19} \\
& + 177845876839680c^{18}d + 762945730634112c^{17}d^2 + 1995757662705280c^{16}d^3 \\
& + 3553979972343360c^{15}d^4 + 4554129811960320c^{14}d^5 + 4327086017067904c^{13}d^6 \\
& + 3094661358404272c^{12}d^7 + 167337780486064c^{11}d^8 + 681029075686000c^{10}d^9 \\
& + 205755944896576c^9d^{10} + 45007332362184c^8d^{11} + 6839525212236c^7d^{12} \\
& + 673913252328c^6d^{13} + 37903639900c^5d^{14} + 897092232c^4d^{15} \\
& + 26952273062400c^{18} + 237442131315200c^{17}d + 960616090687872c^{16}d^2 \\
& + 2363631928087296c^{15}d^3 + 3947685273750976c^{14}d^4 + 4728821351964480c^{13}d^5 \\
& + 4184072133485904c^{12}d^6 + 2774038692376800c^{11}d^7 + 1383066341427136c^{10}d^8 \\
& + 51559903501184c^9d^9 + 141538378685968c^8d^{10} + 27844596844544c^7d^{11} \\
& + 3755696996880c^6d^{12} + 322731584824c^5d^{13} + 15447631856c^4d^{14} \\
& + 300054664c^3d^{15} + 33636716585472c^{17} + 279269880427776c^{16}d \\
& + 1061843431748224c^{15}d^2 + 2447943586339072c^{14}d^3 + 3817443983795744c^{13}d^4 \\
& + 4252731310920000c^{12}d^5 + 3483251022505824c^{11}d^6 + 2126042510134528c^{10}d^7 \\
& + 969333203732272c^9d^8 + 327741473421000c^8d^9 + 80755049827784c^7d^{10} \\
& + 14069844064584c^6d^{11} + 1650943204558c^5d^{12} + 120389034024c^4d^{13} \\
& + 4713830498c^3d^{14} + 70582064c^2d^{15} + 37210825858560c^{16} \\
& + 290178475039744c^{15}d + 1032916771124864c^{14}d^2 + 2221164584671616c^{13}d^3 \\
& + 3217509174892928c^{12}d^4 + 3313529853912832c^{11}d^5 + 2494676376116576c^{10}d^6 \\
& + 1390064431973216c^9d^7 + 573741193242432c^8d^8 + 173765267507232c^7d^9 \\
& + 37832849958632c^6d^{10} + 5719683573752c^5d^{11} + 567863131412c^4d^{12} \\
& + 33757884904c^3d^{13} + 1014867381c^2d^{14} + 10432541cd^{15} \\
& + 36555748548096c^{15} + 266746103347968c^{14}d + 885033642961536c^{13}d^2 \\
& + 1766168166335360c^{12}d^3 + 2362320455786112c^{11}d^4 + 2233130137611200c^{10}d^5 \\
& + 1532419620892736c^9d^6 + 771606727251456c^8d^7 + 284702619884672c^7d^8 \\
& + 76020371799984c^6d^9 + 14325859265752c^5d^{10} + 1827243060832c^4d^{11} \\
& + 147393499512c^3d^{12} + 6700539412c^2d^{13} + 137633636cd^{14} \\
& + 729953d^{15} + 31909696856064c^{14} + 216947667751424c^{13}d
\end{aligned}$$

$$\begin{aligned}
& + 667579270369152c^{12}d^2 + 1229042255143936c^{11}d^3 + 1507278128303296c^{10}d^4 \\
& + 1296928434163392c^9d^5 + 802927743917696c^8d^6 + 360744806687136c^7d^7 \\
& + 117102256191104c^6d^8 + 26998595095040c^5d^9 + 4280622887656c^4d^{10} \\
& + 442182681136c^3d^{11} + 27172514856c^2d^{12} + 840091408cd^{13} \\
& + 8852557d^{14} + 24733542050816c^{13} + 155911958197760c^{12}d \\
& + 442380160120704c^{11}d^2 + 746187362304768c^{10}d^3 + 832098799973024c^9d^4 \\
& + 645100641443840c^8d^5 + 355798190575232c^7d^6 + 140372274921056c^6d^7 \\
& + 39258561284528c^5d^8 + 7596019106528c^4d^9 + 972503521512c^3d^{10} \\
& + 76263742376c^2d^{11} + 3173760270cd^{12} + 50072404d^{13} \\
& + 16991000067584c^{12} + 98741291780096c^{11}d + 256583665560448c^{10}d^2 \\
& + 393279109515136c^9d^3 + 394783537146880c^8d^4 + 272333804600448c^7d^5 \\
& + 131695663581936c^6d^6 + 44682179541680c^5d^7 + 10464212335760c^4d^8 \\
& + 1630669821808c^3d^9 + 158000780432c^2d^{10} + 8359703504cd^{11} \\
& + 176708364d^{12} + 10311431512576c^{11} + 54874356488960c^{10}d \\
& + 129536489587840c^9d^2 + 178635389925248c^8d^3 + 159430703654976c^7d^4 \\
& + 96324043441024c^6d^5 + 40001298037888c^5d^6 + 11344485599472c^4d^7 \\
& + 2135112977936c^3d^8 + 251145094688c^2d^9 + 16365627368cd^{10} \\
& + 438082920d^{11} + 5502424151552c^{10} + 26600148719104c^9d \\
& + 56483952038016c^8d^2 + 69226224124672c^7d^3 + 54076610669120c^6d^4 \\
& + 28030299719616c^5d^5 + 9717510783184c^4d^6 + 2211080645984c^3d^7 \\
& + 313452126368c^2d^8 + 24731625952cd^9 + 812708968d^{10} \\
& + 2565468030464c^9 + 11156374696704c^8d + 21051511656320c^7d^2 \\
& + 22575289767168c^6d^3 + 15121598856416c^5d^4 + 6538101759296c^4d^5 \\
& + 1816436561120c^3d^6 + 310836658432c^2d^7 + 29498976688cd^8 \\
& + 1170707896d^9 + 1036386678272c^8 + 4005132614656c^7d \\
& + 6612340892544c^6d^2 + 6079095765120c^5d^3 + 3395150776320c^4d^4 \\
& + 1175604190080c^3d^5 + 245390776960c^2d^6 + 28071947584cd^7 \\
& + 1337493120d^8 + 358811204096c^7 + 1213277662208c^6d \\
& + 1717197615872c^5d^2 + 1316120774528c^4d^3 + 588470761088c^3d^4 \\
& + 153008129216c^2d^5 + 21332411648cd^6 + 1223644384d^7 \\
& + 104938771712c^6 + 304204453888c^5d + 358929796992c^4d^2 \\
& + 220214971136c^3d^3 + 73918701440c^2d^4 + 12831378176cd^5 \\
& + 896349824d^6 + 25427728128c^5 + 61449659264c^4d \\
& + 58037932544c^3d^2 + 26729203584c^2d^3 + 5988682752cd^4 \\
& + 520666624d^5 + 4968743424c^4 + 9611332608c^3d \\
& + 6813736960c^2d^2 + 2094356992cd^3 + 234973696d^4 \\
& + 752498688c^3 + 1092449280c^2d + 516827136cd^2 \\
& + 79537664d^3 + 82894848c^2 + 80293888cd \\
& + 19015680d^2 + 5910528c + 2865152d \\
& + 204800
\end{aligned}$$

$W_{\Delta}$

$W_{\Delta} := A_0U_{\Delta} - A_1V_{\Delta}$ . Every coefficient is strictly positive. 456 terms.

$$W_{\Delta}(c, d) = 16777216c^{32} + 268435456c^{31}d + 2047868928c^{30}d^2$$

$$\begin{aligned}
& + 9914286080c^{29}d^3 + 34177286144c^{28}d^4 + 89159368704c^{27}d^5 \\
& + 182443835392c^{26}d^6 + 299279319040c^{25}d^7 + 398557446144c^{24}d^8 \\
& + 433494949888c^{23}d^9 + 385362165760c^{22}d^{10} + 278859350016c^{21}d^{11} \\
& + 162787229696c^{20}d^{12} + 75521589248c^{19}d^{13} + 27204255744c^{18}d^{14} \\
& + 7336886272c^{17}d^{15} + 1394606080c^{16}d^{16} + 166723584c^{15}d^{17} \\
& + 9437184c^{14}d^{18} + 463470592c^{31} + 7185891328c^{30}d \\
& + 53059780608c^{29}d^2 + 248310136832c^{28}d^3 + 826325073920c^{27}d^4 \\
& + 2077949165568c^{26}d^5 + 4092496445440c^{25}d^6 + 6450931302400c^{24}d^7 \\
& + 8240827465728c^{23}d^8 + 8581992939520c^{22}d^9 + 7289830113280c^{21}d^{10} \\
& + 5029364760576c^{20}d^{11} + 2792301264896c^{19}d^{12} + 1228691210240c^{18}d^{13} \\
& + 418519449600c^{17}d^{14} + 106369122304c^{16}d^{15} + 18980798464c^{15}d^{16} \\
& + 2121007104c^{14}d^{17} + 111673344c^{13}d^{18} + 6203375616c^{30} \\
& + 93105946624c^{29}d + 664682627072c^{28}d^2 + 3003428306944c^{27}d^3 \\
& + 9636739809280c^{26}d^4 + 23329928511488c^{25}d^5 + 44163972923392c^{24}d^6 \\
& + 66797648019456c^{23}d^7 + 81728788365312c^{22}d^8 + 81358635663360c^{21}d^9 \\
& + 65919914672128c^{20}d^{10} + 43279320154112c^{19}d^{11} + 22807480631296c^{18}d^{12} \\
& + 9498664763392c^{17}d^{13} + 3052496814080c^{16}d^{14} + 729335136256c^{15}d^{15} \\
& + 121861177344c^{14}d^{16} + 12693798912c^{13}d^{17} + 619905024c^{12}d^{18} \\
& + 53608316928c^{29} + 778025828352c^{28}d + 5363846709248c^{27}d^2 \\
& + 23373034323968c^{26}d^3 + 72212314193920c^{25}d^4 + 168066899771392c^{24}d^5 \\
& + 305340456468480c^{23}d^6 + 442421658288128c^{22}d^7 + 517566145626112c^{21}d^8 \\
& + 491588397301760c^{20}d^9 + 379171051438080c^{19}d^{10} + 236395424743424c^{18}d^{11} \\
& + 117973360443392c^{17}d^{12} + 46386750423040c^{16}d^{13} + 14026064035840c^{15}d^{14} \\
& + 3141277679616c^{14}d^{15} + 489883959296c^{13}d^{16} + 47401664512c^{12}d^{17} \\
& + 2138832896c^{11}d^{18} + 336285859840c^{28} + 4713793880064c^{27}d \\
& + 31343814676480c^{26}d^2 + 131535131947008c^{25}d^3 + 390745811812352c^{24}d^4 \\
& + 872930019115008c^{23}d^5 + 1519513574014976c^{22}d^6 + 2105405066469376c^{21}d^7 \\
& + 2350392413552640c^{20}d^8 + 2125577129324544c^{19}d^9 + 1557225874784256c^{18}d^{10} \\
& + 919675636510720c^{17}d^{11} + 433485049823232c^{16}d^{12} + 160453386100736c^{15}d^{13} \\
& + 45504615432192c^{14}d^{14} + 9519025061888c^{13}d^{15} + 1380126097408c^{12}d^{16} \\
& + 123500507136c^{11}d^{17} + 5122863104c^{10}d^{18} + 1632181903360c^{27} \\
& + 22068899008512c^{26}d + 141341721291776c^{25}d^2 + 570395001178112c^{24}d^3 \\
& + 1626666788452352c^{23}d^4 + 3482220649893888c^{22}d^5 + 5796983064379392c^{21}d^6 \\
& + 7665549608245248c^{20}d^7 + 8148573728450560c^{19}d^8 + 6999980663004160c^{18}d^9 \\
& + 4858486096143360c^{17}d^{10} + 2710499304925184c^{16}d^{11} + 1202960245716992c^{15}d^{12} \\
& + 417753081190400c^{14}d^{13} + 110701385975808c^{13}d^{14} + 21538504415232c^{12}d^{15} \\
& + 2889272469504c^{11}d^{16} + 237781765120c^{10}d^{17} + 9008842752c^9d^{18} \\
& + 6379377926144c^{26} + 83089985715200c^{25}d + 511808738892288c^{24}d^2 \\
& + 1983061740809984c^{23}d^3 + 5419735045726720c^{22}d^4 + 11096638894576640c^{21}d^5 \\
& + 17630670336636416c^{20}d^6 + 22199953838283264c^{19}d^7 + 22416523088493568c^{18}d^8 \\
& + 18243531507314176c^{17}d^9 + 11961302770115328c^{16}d^{10} + 6283484087918336c^{15}d^{11} \\
& + 2616515188839936c^{14}d^{12} + 849108640613376c^{13}d^{13} + 209305695614976c^{12}d^{14} \\
& + 37683558694912c^{11}d^{15} + 4649496048640c^{10}d^{16} + 349477885440c^9d^{17} \\
& + 11993517312c^8d^{18} + 20632813707776c^{25} + 258492781087232c^{24}d
\end{aligned}$$

$$\begin{aligned}
& + 1528915479340544c^{23}d^2 + 5677803680530944c^{22}d^3 + 14842850030570368c^{21}d^4 \\
& + 29005865985696256c^{20}d^5 + 43884303228711552c^{19}d^6 + 52487333303930112c^{18}d^7 \\
& + 50206904711161472c^{17}d^8 + 38594517488338432c^{16}d^9 + 23824308954219136c^{15}d^{10} \\
& + 11741293502806912c^{14}d^{11} + 4568443778957568c^{13}d^{12} + 1378972173257600c^{12}d^{13} \\
& + 314515994929664c^{11}d^{14} + 52076005182336c^{10}d^{15} + 5866901964032c^9d^{16} \\
& + 399251213440c^8d^{17} + 12278481152c^7d^{18} + 56310721273344c^{24} \\
& + 677500458480896c^{23}d + 3841200417182720c^{22}d^2 + 13646067398210688c^{21}d^3 \\
& + 34051394859579776c^{20}d^4 + 63367177408495616c^{19}d^5 + 91062456343357248c^{18}d^6 \\
& + 103166841160034496c^{17}d^7 + 93198420032281088c^{16}d^8 + 67439441555160064c^{15}d^9 \\
& + 39046906618034112c^{14}d^{10} + 17976613858488768c^{13}d^{11} + 6504259677095808c^{12}d^{12} \\
& + 1816090194395968c^{11}d^{13} + 380815629095488c^{10}d^{14} + 57551336121600c^9d^{15} \\
& + 5866807730176c^8d^{16} + 357449383872c^7d^{17} + 9713142208c^6d^{18} \\
& + 131575845440000c^{23} + 1517655003848192c^{22}d + 8232486995327488c^{21}d^2 \\
& + 27919787624354688c^{20}d^3 + 66349016191216128c^{19}d^4 + 117280278496123584c^{18}d^5 \\
& + 159637085373356864c^{17}d^6 + 170780556824052224c^{16}d^7 + 145198317357552768c^{15}d^8 \\
& + 98520751411477184c^{14}d^9 + 53270348155731392c^{13}d^{10} + 22797412779004736c^{12}d^{11} \\
& + 7626978330127040c^{11}d^{12} + 1956982654545792c^{10}d^{13} + 374354751858176c^9d^{14} \\
& + 51158308054848c^8d^{15} + 4664995761344c^7d^{16} + 250803422464c^6d^{17} \\
& + 5908512512c^5d^{18} + 266129900703232c^{22} + 2937324010204928c^{21}d \\
& + 15212895974612224c^{20}d^2 + 49140727637955840c^{19}d^3 + 110932916548649984c^{18}d^4 \\
& + 185734132275181056c^{17}d^5 + 238712321552393792c^{16}d^6 + 240305508903916672c^{15}d^7 \\
& + 191529631015955136c^{14}d^8 + 121321480804017248c^{13}d^9 + 60952523488771648c^{12}d^{10} \\
& + 24107782494463056c^{11}d^{11} + 7407642832744960c^{10}d^{12} + 1732854535486848c^9d^{13} \\
& + 299524957087728c^8d^{14} + 36581505174480c^7d^{15} + 2939968021536c^6d^{16} \\
& + 136793254528c^5d^{17} + 2720714416c^4d^{18} + 469903007246336c^{21} \\
& + 4952612544943616c^{20}d + 24434882501073408c^{19}d^2 + 74988624581477120c^{18}d^3 \\
& + 160359302423844096c^{17}d^4 + 253517093438941120c^{16}d^5 + 306577719564739584c^{15}d^6 \\
& + 289265668883078080c^{14}d^7 + 215166037919182176c^{13}d^8 + 126588235775900512c^{12}d^9 \\
& + 58747770518630880c^{11}d^{10} + 21328073150073424c^{10}d^{11} + 5970648458120344c^9d^{12} \\
& + 1261050199881376c^8d^{13} + 194621982955848c^7d^{14} + 20924958438088c^6d^{15} \\
& + 1453197693040c^5d^{16} + 56960469832c^4d^{17} + 919717456c^3d^{18} \\
& + 729001070264320c^{20}d + 7320512316242944c^{19}d^2 + 34319932193078784c^{18}d^3 \\
& + 99787706591551488c^{17}d^4 + 201512690413197568c^{16}d^5 + 299761853228522240c^{15}d^6 \\
& + 339736278192348096c^{14}d^7 + 299098162145623680c^{13}d^8 + 206567552640316896c^{12}d^9 \\
& + 112207735330393552c^{11}d^{10} + 47770398301288832c^{10}d^{11} + 15789144350605688c^9d^{12} \\
& + 3987476090277592c^8d^{13} + 751241471315224c^7d^{14} + 101950142856036c^6d^{15} \\
& + 9458646917372c^5d^{16} + 552351879480c^4d^{17} + 17530127692c^3d^{18} \\
& + 215792276c^2d^{18} + 998566285559808c^{19} + 9530062345172992c^{18}d \\
& + 42337606870383104c^{17}d^2 + 116266984857157888c^{16}d^3 + 220950277451518208c^{15}d^4 \\
& + 308046878296482112c^{14}d^5 + 325735129361153600c^{13}d^6 + 266204404393091648c^{12}d^7 \\
& + 169686444192280800c^{11}d^8 + 84512804980297408c^{10}d^9 + 32734860886476160c^9d^{10} \\
& + 9752805494941640c^8d^{11} + 2194951298755168c^7d^{12} + 363217720754468c^6d^{13} \\
& + 42477717753276c^5d^{14} + 3308331845124c^4d^{15} + 156089382484c^3d^{16} \\
& + 3764982032c^2d^{17} + 31462704cd^{18} + 1212015760824320c^{18}
\end{aligned}$$

$$\begin{aligned}
& + 10963108535870464c^{17}d + 46009772077291008c^{16}d^2 + 118925055544980992c^{15}d^3 \\
& + 211845726725917440c^{14}d^4 + 275579851533476864c^{13}d^5 + 270486360388418496c^{12}d^6 \\
& + 203979609944304640c^{11}d^7 + 119170533053954080c^{10}d^8 + 53970571120877904c^9d^9 \\
& + 18830001527011056c^8d^{10} + 4995011470125504c^7d^{11} + 986337788972928c^6d^{12} \\
& + 140474658105768c^5d^{13} + 13769874481884c^4d^{14} + 864817789632c^3d^{15} \\
& + 30932726496c^2d^{16} + 504680596cd^{17} + 2151296d^{18} \\
& + 1306724820360192c^{17} + 11168113266422784c^{16}d + 44124625161889280c^{15}d^2 \\
& + 106930108499262720c^{14}d^3 + 177754259110680576c^{13}d^4 + 214649306581240768c^{12}d^5 \\
& + 194399974934723840c^{11}d^6 + 134337754465916544c^{10}d^7 + 71338768154074880c^9d^8 \\
& + 29084978363476544c^8d^9 + 9028042477895456c^7d^{10} + 2099186027839536c^6d^{11} \\
& + 356332390420016c^5d^{12} + 42476126449772c^4d^{13} + 3351733022136c^3d^{14} \\
& + 159239162420c^2d^{15} + 3841546296cd^{16} + 31837260d^{17} \\
& + 1253212763358208c^{16} + 10085368998483456c^{15}d + 37366675335980544c^{14}d^2 \\
& + 84522188695420160c^{13}d^3 + 130451332965015552c^{12}d^4 + 145367460209191168c^{11}d^5 \\
& + 120637059867540160c^{10}d^6 + 75760709757880576c^9d^7 + 36204285194767104c^8d^8 \\
& + 13124342446000160c^7d^9 + 3567971121445024c^6d^{10} + 712429877788592c^5d^{11} \\
& + 101089230392688c^4d^{12} + 9685110508072c^3d^{13} + 577000498180c^2d^{14} \\
& + 18444219724cd^{15} + 225287300d^{16} + 1069641539365888c^{15} \\
& + 8073903379461120c^{14}d + 27928118815157760c^{13}d^2 + 58665673073320960c^{12}d^3 \\
& + 83570693675349248c^{11}d^4 + 85343115053536448c^{10}d^5 + 64363381193324224c^9d^6 \\
& + 36367570960408320c^8d^7 + 15446924928291904c^7d^8 + 4901386010096320c^6d^9 \\
& + 1143304624692000c^5d^{10} + 190618603462848c^4d^{11} + 21708998796896c^3d^{12} \\
& + 1567636995356c^2d^{13} + 62704435972cd^{14} + 1012102168d^{15} \\
& + 812140927001600c^{14} + 5724368442851840c^{13}d + 18392487331466240c^{12}d^2 \\
& + 35667784350346752c^{11}d^3 + 46572568186198784c^{10}d^4 + 43227301867861504c^9d^5 \\
& + 29332235483064256c^8d^6 + 14728722329531008c^7d^7 + 5473887961419520c^6d^8 \\
& + 1489407375736320c^5d^9 + 289846118298368c^4d^{10} + 38742482399856c^3d^{11} \\
& + 3320739320256c^2d^{12} + 160703480088cd^{13} + 3238280012d^{14} \\
& + 547777333299200c^{13} + 3587046922910720c^{12}d + 10642416070379008c^{11}d^2 \\
& + 18921769522521856c^{10}d^3 + 22460576082559744c^9d^4 + 18759802928194368c^8d^5 \\
& + 11312867163871744c^7d^6 + 4969786016666304c^6d^7 + 1583292494859104c^5d^8 \\
& + 359199046169184c^4d^9 + 55992621699840c^3d^{10} + 5626182800448c^2d^{11} \\
& + 322687725496cd^{12} + 7851650676d^{13} + 327434681001984c^{12} \\
& + 1980271455171584c^{11}d + 5387742797136384c^{10}d^2 + 8710305726404608c^9d^3 \\
& + 9306032147364608c^8d^4 + 6908696634563840c^7d^5 + 3645066941974592c^6d^6 \\
& + 1372505323998592c^5d^7 + 364464092646880c^4d^8 + 66194524115728c^3d^9 \\
& + 7750071288608c^2d^{10} + 520462850968cd^{11} + 14984999416d^{12} \\
& + 172855912327168c^{11} + 958810513242112c^{10}d + 2372503728779776c^9d^2 \\
& + 3453035734679296c^8d^3 + 3279799406236416c^7d^4 + 2130650305049536c^6d^5 \\
& + 963584661412032c^5d^6 + 302394710685760c^4d^7 + 64265486590048c^3d^8 \\
& + 8761775312064c^2d^9 + 685021218592cd^{10} + 23065296888d^{11} \\
& + 80208840751104c^{10} + 404690763913216c^9d + 901661772611584c^8d^2 \\
& + 1166921515639552c^7d^3 + 970080036904704c^6d^4 + 540266257730560c^5d^5 \\
& + 203651892737088c^4d^6 + 51144608356096c^3d^7 + 8156950009888c^2d^8
\end{aligned}$$

$$\begin{aligned}
& + 742227175760cd^9 + 29070383056d^{10} + 32509771469312c^9 \\
& + 147710378787328c^8d + 292686983953408c^7d^2 + 331589328966912c^6d^3 \\
& + 236363797714048c^5d^4 + 109735442436416c^4d^5 + 33104047530624c^3d^6 \\
& + 6238555760128c^2d^7 + 663880458176cd^8 + 30246415616d^9 \\
& + 11415646359040c^8 + 46132517821696c^7d + 80030325599744c^6d^2 \\
& + 77756014645888c^5d^3 + 46211404887680c^4d^4 + 17172127604480c^3d^5 \\
& + 3887518081536c^2d^6 + 488787803712cd^7 + 26036785472d^8 \\
& + 3435823356416c^7 + 12156718472704c^6d + 18087888166912c^5d^2 \\
& + 14653993709696c^4d^3 + 6971666302208c^3d^4 + 1944332905344c^2d^5 \\
& + 293683250944cd^6 + 18482115712d^7 + 873889381888c^6 \\
& + 2652008965888c^5d + 3290496136960c^4d^2 + 2134190159104c^3d^3 \\
& + 762103829760c^2d^4 + 141820809216cd^5 + 10722042880d^6 \\
& + 184275927040c^5 + 466331602432c^4d + 463227099136c^3d^2 \\
& + 225520878080c^2d^3 + 53738254336cd^4 + 5005571584d^5 \\
& + 31364655104c^4 + 63541159936c^3d + 47376844800c^2d^2 \\
& + 15391331328cd^3 + 1835785216d^4 + 4142080000c^3 \\
& + 6297997312c^2d + 3133300736cd^2 + 509497344d^3 \\
& + 398401536c^2 + 404140032cd + 100626432d^2 \\
& + 24838144c + 12607488d + 753664
\end{aligned}$$

$P_K$

$\Delta - N_i = 8m_i^2m_j^4(m_i + \rho)P_K/(\rho(1 + \rho)^2A_0^2)$ . 28 terms.

$$\begin{aligned}
P_K(m_i, m_j, \rho) = & 8m_i^3m_j^3\rho^2 + 2m_i^3m_j\rho^4 + 16m_i^2m_j^3\rho^3 \\
& + 2m_i^2m_j\rho^5 - 2m_i^4m_j\rho^2 - 2m_i^3m_j\rho^3 \\
& + 6m_i^2m_j^3\rho^2 - 4m_i^2m_j^2\rho^3 - 4m_im_j^3\rho^3 \\
& + 4m_i^4\rho^2 - 1m_i^3m_j\rho^2 + 12m_i^3\rho^3 \\
& - 4m_i^2m_j^2\rho^2 + 8m_i^2\rho^4 - 1m_im_j^3\rho^2 \\
& + 2m_j^3\rho^3 + 4m_i^4\rho + 14m_i^3\rho^2 \\
& + 1m_i^2m_j\rho^2 + 10m_i^2\rho^3 + 1m_j^3\rho^2 \\
& + 2m_i^4 - 4m_i^3m_j + 6m_i^3\rho \\
& - 4m_i^2m_j\rho + 4m_i^2\rho^2 + 2m_i^3 \\
& + 2m_i^2\rho
\end{aligned}$$

$W_U$

$W_U := U_KA_0 - A_1V_K$ . Every coefficient is strictly positive. 296 terms.

$$\begin{aligned}
W_U(c, d) = & 524288c^{25} + 6160384c^{24}d + 33030144c^{23}d^2 \\
& + 106758144c^{22}d^3 + 231473152c^{21}d^4 + 354680832c^{20}d^5 \\
& + 395051008c^{19}d^6 + 326565888c^{18}d^7 + 207618048c^{17}d^8 \\
& + 111017984c^{16}d^9 + 57671680c^{15}d^{10} + 30081024c^{14}d^{11} \\
& + 13369344c^{13}d^{12} + 4194304c^{12}d^{13} + 786432c^{11}d^{14} \\
& + 65536c^{10}d^{15} + 10944512c^{24} + 123797504c^{23}d \\
& + 638533632c^{22}d^2 + 1984774144c^{21}d^3 + 4140498944c^{20}d^4 \\
& + 6116081664c^{19}d^5 + 6597459968c^{18}d^6 + 5329403904c^{17}d^7 \\
& + 3354329088c^{16}d^8 + 1782775808c^{15}d^9 + 889044992c^{14}d^{10}
\end{aligned}$$

$$\begin{aligned}
& + 420593664c^{13}d^{11} + 165609472c^{12}d^{12} + 46137344c^{11}d^{13} \\
& + 7716864c^{10}d^{14} + 573440c^9d^{15} + 110313472c^{23} \\
& + 1200029696c^{22}d + 5948878848c^{21}d^2 + 17768452096c^{20}d^3 \\
& + 35640414208c^{19}d^4 + 50721255424c^{18}d^5 + 52941381632c^{17}d^6 \\
& + 41685590016c^{16}d^7 + 25781624832c^{15}d^8 + 13414752256c^{14}d^9 \\
& + 6333427712c^{13}d^{10} + 2711310336c^{12}d^{11} + 946720768c^{11}d^{12} \\
& + 233844736c^{10}d^{13} + 34766848c^9d^{14} + 2293760c^8d^{15} \\
& + 715038720c^{22} + 7472136192c^{21}d + 35560729600c^{20}d^2 \\
& + 101951154176c^{19}d^3 + 196402320384c^{18}d^4 + 268932012032c^{17}d^5 \\
& + 271046500352c^{16}d^6 + 207147681792c^{15}d^7 + 124782520320c^{14}d^8 \\
& + 62687416320c^{13}d^9 + 27708214272c^{12}d^{10} + 10702428160c^{11}d^{11} \\
& + 3310625792c^{10}d^{12} + 722540544c^9d^{13} + 94939136c^8d^{14} \\
& + 5519360c^7d^{15} + 3348508672c^{21} + 33568323072c^{20}d \\
& + 153153029376c^{19}d^2 + 420824112128c^{18}d^3 + 777276012032c^{17}d^4 \\
& + 1021813828352c^{16}d^5 + 991142811136c^{15}d^6 + 731056590080c^{14}d^7 \\
& + 424816826368c^{13}d^8 + 203473523712c^{12}d^9 + 83389062400c^{11}d^{10} \\
& + 28947736064c^{10}d^{11} + 7908616704c^9d^{12} + 1516522240c^8d^{13} \\
& + 174612480c^7d^{14} + 8843520c^6d^{15} + 12063705856c^{20} \\
& + 115827988224c^{19}d + 505719408384c^{18}d^2 + 1329146717888c^{17}d^3 \\
& + 2348207292160c^{16}d^4 + 2954547980416c^{15}d^5 + 2745911350080c^{14}d^6 \\
& + 1941420732096c^{13}d^7 + 1077505604480c^{12}d^8 + 486341860992c^{11}d^9 \\
& + 183132162048c^{10}d^{10} + 56832362176c^9d^{11} + 13639612032c^8d^{12} \\
& + 2278776832c^7d^{13} + 227140032c^6d^{14} + 9856448c^5d^{15} \\
& + 34757500224c^{19} + 318992916160c^{18}d + 1329901769536c^{17}d^2 \\
& + 3334738172928c^{16}d^3 + 5617788388960c^{15}d^4 + 6738060043136c^{14}d^5 \\
& + 5967210490592c^{13}d^6 + 4012998365216c^{12}d^7 + 2105651005920c^{11}d^8 \\
& + 885642090304c^{10}d^9 + 303563771552c^9d^{10} + 83616606592c^8d^{11} \\
& + 17484387200c^7d^{12} + 2514657792c^6d^{13} + 213276224c^5d^{14} \\
& + 7738080c^4d^{15} + 82172148544c^{18} + 719167501184c^{17}d \\
& + 2855207725664c^{16}d^2 + 6808924799744c^{15}d^3 + 10895255233280c^{14}d^4 \\
& + 12395931288736c^{13}d^5 + 10393019269168c^{12}d^6 + 6591759325664c^{11}d^7 \\
& + 3235595196624c^{10}d^8 + 1253867799904c^9d^9 + 387188590192c^8d^{10} \\
& + 93751354624c^7d^{11} + 16877664240c^6d^{12} + 2054114240c^5d^{13} \\
& + 144582144c^4d^{14} + 4225760c^3d^{15} + 162264217600c^{17} \\
& + 1350447273728c^{16}d + 5089087871584c^{15}d^2 + 11497531435744c^{14}d^3 \\
& + 17393253079328c^{13}d^4 + 18661590007232c^{12}d^5 + 14702919357056c^{11}d^6 \\
& + 8712755921072c^{10}d^7 + 3956234318168c^9d^8 + 1395490166728c^8d^9 \\
& + 383510837392c^7d^{10} + 80565022688c^6d^{11} + 12275501208c^5d^{12} \\
& + 1233186824c^4d^{13} + 69368320c^3d^{14} + 1536640c^2d^{15} \\
& + 270991023744c^{16} + 2137540699008c^{15}d + 7616399000160c^{14}d^2 \\
& + 16227183037056c^{13}d^3 + 23079286732736c^{12}d^4 + 23192320189024c^{11}d^5 \\
& + 17025672773024c^{10}d^6 + 9328761911152c^9d^7 + 3870560994464c^8d^8 \\
& + 1225658087864c^7d^9 + 295262028052c^6d^{10} + 52847645828c^5d^{11} \\
& + 6647777696c^4d^{12} + 530865608c^3d^{13} + 22441804c^2d^{14}
\end{aligned}$$

$$\begin{aligned}
& + 336140cd^{15} + 386091343552c^{15} + 2875151418816c^{14}d \\
& + 9642504408736c^{13}d^2 + 19269174106912c^{12}d^3 + 25598541343776c^{11}d^4 \\
& + 23902311721984c^{10}d^5 + 16191097519616c^9d^6 + 8106199104176c^8d^7 \\
& + 3030027593524c^7d^8 + 847207062740c^6d^9 + 175392451892c^5d^{10} \\
& + 26066926604c^4d^{11} + 2606924152c^3d^{12} + 155618638c^2d^{13} \\
& + 4408236cd^{14} + 33614d^{15} + 472008960448c^{14} \\
& + 3303368932608c^{13}d + 10372031109632c^{12}d^2 + 19318101877632c^{11}d^3 \\
& + 23789054194720c^{10}d^4 + 20449228364224c^9d^5 + 12637488505040c^8d^6 \\
& + 5701448372496c^7d^7 + 1887575294508c^6d^8 + 456307151504c^5d^9 \\
& + 79004229674c^4d^{10} + 9381503060c^3d^{11} + 701415612c^2d^{12} \\
& + 27906676cd^{13} + 398566d^{14} + 496881236608c^{13} \\
& + 3250842192384c^{12}d + 9496670597504c^{11}d^2 + 16363221744000c^{10}d^3 \\
& + 18511489595968c^9d^4 + 14489404625216c^8d^5 + 8059274867920c^7d^6 \\
& + 3221451893728c^6d^7 + 924527091536c^5d^8 + 187846799704c^4d^9 \\
& + 26140609338c^3d^{10} + 2330443470c^2d^{11} + 116126108cd^{12} \\
& + 2316720d^{13} + 451077839360c^{12} + 2741873321472c^{11}d \\
& + 7397925399616c^{10}d^2 + 11689141374400c^9d^3 + 12018760591232c^8d^4 \\
& + 8453162677376c^7d^5 + 4161722186064c^6d^6 + 1442478562320c^5d^7 \\
& + 348690666368c^4d^8 + 57172832912c^3d^9 + 6002176242c^2d^{10} \\
& + 357893708cd^{11} + 8940652d^{12} + 353026484928c^{11} \\
& + 1979265227328c^{10}d + 4889695836032c^9d^2 + 7010465311808c^8d^3 \\
& + 6466378178848c^7d^4 + 4019871209536c^6d^5 + 1714674234288c^5d^6 \\
& + 500692487712c^4d^7 + 97835207796c^3d^8 + 12140293068c^2d^9 \\
& + 855767618cd^{10} + 25654442d^{11} + 237680293568c^{10} \\
& + 1218570315136c^9d + 2727773010016c^8d^2 + 3503171364096c^7d^3 \\
& + 2851853000128c^6d^4 + 1534063347296c^5d^5 + 550842524064c^4d^6 \\
& + 130028316688c^3d^7 + 19252056732c^2d^8 + 1607911496cd^9 \\
& + 57154112d^{10} + 137100852992c^9 + 636141654272c^8d \\
& + 1273883848544c^7d^2 + 1441910318816c^6d^3 + 1014343516448c^5d^4 \\
& + 458784481792c^4d^5 + 133069444976c^3d^6 + 23798849584c^2d^7 \\
& + 2373485136cd^8 + 100130304d^9 + 67330064512c^8 \\
& + 279155488640c^7d + 492089925088c^6d^2 + 480706504192c^5d^3 \\
& + 283981964096c^4d^4 + 103637565344c^3d^5 + 22756405904c^2d^6 \\
& + 2740564160cd^7 + 138108744d^8 + 27897313600c^7 \\
& + 101707251008c^6d + 154551998880c^5d^2 + 126634595232c^4d^3 \\
& + 60290574688c^3d^4 + 16638696256c^2d^5 + 2458023600cd^6 \\
& + 149502800d^7 + 9629022272c^6 + 30230602240c^5d \\
& + 38488637248c^4d^2 + 25386199680c^3d^3 + 9129293344c^2d^4 \\
& + 1693209664cd^5 + 126208160d^6 + 2719714944c^5 \\
& + 7146895360c^4d + 7316918720c^3d^2 + 3641256128c^2d^3 \\
& + 879020160cd^4 + 82166016d^5 + 612729856c^4 \\
& + 1293496320c^3d + 998118016c^2d^2 + 333069568cd^3 \\
& + 40476544d^4 + 105957376c^3 + 168428544c^2d \\
& + 87059200cd^2 + 14608640d^3 + 13214720c^2
\end{aligned}$$

$$\begin{aligned}
& + 14057472cd + 3650048d^2 + 1058816c \\
& + 565248d + 40960
\end{aligned}$$

$W_L$

$W_L := -A_1A_0U_K + [(-A_1)^2 - A_2A_0]V_K$ . Every coefficient is strictly positive. 418 terms.

$$\begin{aligned}
W_L(c, d) = & 2097152c^{30} + 31457280c^{29}d + 219676672c^{28}d^2 \\
& + 946864128c^{27}d^3 + 2815426560c^{26}d^4 + 6114246656c^{25}d^5 \\
& + 10019143680c^{24}d^6 + 12636389376c^{23}d^7 + 12445548544c^{22}d^8 \\
& + 9746513920c^{21}d^9 + 6297747456c^{20}d^{10} + 3612344320c^{19}d^{11} \\
& + 2003828736c^{18}d^{12} + 1078984704c^{17}d^{13} + 508559360c^{16}d^{14} \\
& + 185597952c^{15}d^{15} + 47185920c^{14}d^{16} + 7340032c^{13}d^{17} \\
& + 524288c^{12}d^{18} + 54525952c^{29} + 790626304c^{28}d \\
& + 5333057536c^{27}d^2 + 22189178880c^{26}d^3 + 63661932544c^{25}d^4 \\
& + 133403246592c^{24}d^5 + 211093815296c^{23}d^6 + 257624899584c^{22}d^7 \\
& + 246549315584c^{21}d^8 + 188894937088c^{20}d^9 + 120294473728c^{19}d^{10} \\
& + 67867770880c^{18}d^{11} + 36140482560c^{17}d^{12} + 18058313728c^{16}d^{13} \\
& + 7759724544c^{15}d^{14} + 2576089088c^{14}d^{15} + 596901888c^{13}d^{16} \\
& + 84672512c^{12}d^{17} + 5505024c^{11}d^{18} + 684589056c^{28} \\
& + 9587785728c^{27}d + 62417993728c^{26}d^2 + 250488815616c^{25}d^3 \\
& + 692913668096c^{24}d^4 + 1400076500992c^{23}d^5 + 2138020085760c^{22}d^6 \\
& + 2523338375168c^{21}d^7 + 2344322433024c^{20}d^8 + 1753472827392c^{19}d^9 \\
& + 1095119241216c^{18}d^{10} + 602726465536c^{17}d^{11} + 306006884352c^{16}d^{12} \\
& + 141649510400c^{15}d^{13} + 55499063296c^{14}d^{14} + 16743137280c^{13}d^{15} \\
& + 3527573504c^{12}d^{16} + 454950912c^{11}d^{17} + 26836992c^{10}d^{18} \\
& + 5533401088c^{27} + 74787258368c^{26}d + 469496037376c^{25}d^2 \\
& + 1815749754880c^{24}d^3 + 4838859177984c^{23}d^4 + 9420219252736c^{22}d^5 \\
& + 13871887876096c^{21}d^6 + 15818416095232c^{20}d^7 + 14246928875520c^{19}d^8 \\
& + 10373899255808c^{18}d^9 + 6318875475968c^{17}d^{10} + 3365793660928c^{16}d^{11} \\
& + 1618788253696c^{15}d^{12} + 692520714240c^{14}d^{13} + 247153229824c^{13}d^{14} \\
& + 67620536320c^{12}d^{15} + 12911411200c^{11}d^{16} + 1507491840c^{10}d^{17} \\
& + 80281600c^9d^{18} + 32384114688c^{26} + 422005678080c^{25}d \\
& + 2552319645696c^{24}d^2 + 9503934017536c^{23}d^3 + 24377312190464c^{22}d^4 \\
& + 45681748357120c^{21}d^5 + 64802483822592c^{20}d^6 + 71306117885952c^{19}d^7 \\
& + 62134360645632c^{18}d^8 + 43887104667648c^{17}d^9 + 25914805948416c^{16}d^{10} \\
& + 13256481460224c^{15}d^{11} + 6002546139136c^{14}d^{12} + 2365904162816c^{13}d^{13} \\
& + 767500509184c^{12}d^{14} + 189847670784c^{11}d^{15} + 32703578112c^{10}d^{16} \\
& + 3436769280c^9d^{17} + 164075520c^8d^{18} + 146320564224c^{25} \\
& + 1836599492608c^{24}d + 10690663790592c^{23}d^2 + 38288336688128c^{22}d^3 \\
& + 94422663930880c^{21}d^4 + 170125392700416c^{20}d^5 + 232171627516928c^{19}d^6 \\
& + 246083992110080c^{18}d^7 + 206905389347840c^{17}d^8 + 141150597534720c^{16}d^9 \\
& + 80278574617600c^{15}d^{10} + 39140021271552c^{14}d^{11} + 16582963457024c^{13}d^{12} \\
& + 5999058512896c^{12}d^{13} + 1763196042240c^{11}d^{14} + 392600235008c^{10}d^{15} \\
& + 60650642432c^9d^{16} + 5692660736c^8d^{17} + 241296384c^7d^{18} \\
& + 531366158848c^{24} + 6417336542720c^{23}d + 35910867947008c^{22}d^2
\end{aligned}$$

$$\begin{aligned}
& + 123555236854272c^{21}d^3 + 292574986264960c^{20}d^4 + 506097698482688c^{19}d^5 \\
& + 663286367272832c^{18}d^6 + 675611294168576c^{17}d^7 + 546253435668352c^{16}d^8 \\
& + 358119101725184c^{15}d^9 + 194842895658368c^{14}d^{10} + 89861565756928c^{13}d^{11} \\
& + 35399940668544c^{12}d^{12} + 11699062277632c^{11}d^{13} + 3101241386624c^{10}d^{14} \\
& + 617961609728c^9d^{15} + 84943116416c^8d^{16} + 7048523776c^7d^{17} \\
& + 261799552c^6d^{18} + 1594033101056c^{23} + 18500443375360c^{22}d \\
& + 99396304336000c^{21}d^2 + 328070529344256c^{20}d^3 + 744779662867072c^{19}d^4 \\
& + 1234628105108480c^{18}d^5 + 1550422651026944c^{17}d^6 + 1513124395854080c^{16}d^7 \\
& + 1171629292529024c^{15}d^8 + 733971116397568c^{14}d^9 + 379279812996608c^{13}d^{10} \\
& + 164208909833472c^{12}d^{11} + 59743034018176c^{11}d^{12} + 17933912623104c^{10}d^{13} \\
& + 4261825898496c^9d^{14} + 754091785984c^8d^{15} + 91282197888c^7d^{16} \\
& + 6605783296c^6d^{17} + 211134336c^5d^{18} + 4028405974656c^{22} \\
& + 44868415328384c^{21}d + 231095068251200c^{20}d^2 + 730502655523008c^{19}d^3 \\
& + 1586851105080512c^{18}d^4 + 2515187520004032c^{17}d^5 + 3017923173563232c^{16}d^6 \\
& + 2811738201998528c^{15}d^7 + 2074930110349120c^{14}d^8 + 1234269491357888c^{13}d^9 \\
& + 601213416522464c^{12}d^{10} + 242413607844672c^{11}d^{11} + 80850341677888c^{10}d^{12} \\
& + 21889696688192c^9d^{13} + 4626347875616c^8d^{14} + 719468733248c^7d^{15} \\
& + 75648985664c^6d^{16} + 4686422720c^5d^{17} + 125620320c^4d^{18} \\
& + 8701331981952c^{21} + 92858864577664c^{20}d + 457685405060544c^{19}d^2 \\
& + 1382813883788224c^{18}d^3 + 2867637504650560c^{17}d^4 + 4333895612357696c^{16}d^5 \\
& + 4951637651103552c^{15}d^6 + 4384944532379680c^{14}d^7 + 3066792096442176c^{13}d^8 \\
& + 1720261580824224c^{12}d^9 + 783513324973440c^{11}d^{10} + 291714355167808c^{10}d^{11} \\
& + 88436704780416c^9d^{12} + 21405712114112c^8d^{13} + 3981662973184c^7d^{14} \\
& + 536831186656c^6d^{15} + 48122766720c^5d^{16} + 2486118880c^4d^{17} \\
& + 53782400c^3d^{18} + 16239086907008c^{20} + 165739435738496c^{19}d \\
& + 780116912094592c^{18}d^2 + 2247434375593984c^{17}d^3 + 4436927110507744c^{16}d^4 \\
& + 6372416197090336c^{15}d^5 + 6904406012030768c^{14}d^6 + 5781953501634224c^{13}d^7 \\
& + 3808342956506784c^{12}d^8 + 1999062856652480c^{11}d^9 + 843921230392592c^{10}d^{10} \\
& + 287397520381440c^9d^{11} + 78408226743000c^8d^{12} + 16777876090112c^7d^{13} \\
& + 2708818171336c^6d^{14} + 310783039504c^5d^{15} + 23135921928c^4d^{16} \\
& + 958786528c^3d^{17} + 15731352c^2d^{18} + 26399160498816c^{19} \\
& + 257125996055680c^{18}d + 1152942591050880c^{17}d^2 + 3158196608347520c^{16}d^3 \\
& + 5915993337768672c^{15}d^4 + 8042526336042464c^{14}d^5 + 8223795447718288c^{13}d^6 \\
& + 6474296403775984c^{12}d^7 + 3987326092007216c^{11}d^8 + 1942010443487824c^{10}d^9 \\
& + 752497369255528c^9d^{10} + 231862331667424c^8d^{11} + 56236376755032c^7d^{12} \\
& + 10484949358992c^6d^{13} + 1441918261472c^5d^{14} + 137126575600c^4d^{15} \\
& + 8153793256c^3d^{16} + 254621248c^2d^{17} + 2823576cd^{18} \\
& + 37607921314432c^{18} + 348689521850240c^{17}d + 1485195306762816c^{16}d^2 \\
& + 3855449219399232c^{15}d^3 + 6825859420358496c^{14}d^4 + 8742567377435936c^{13}d^5 \\
& + 8389515544768448c^{12}d^6 + 6167108686321216c^{11}d^7 + 3522547271944728c^{10}d^8 \\
& + 1576570890824536c^9d^9 + 554461937237308c^8d^{10} + 152584827405828c^7d^{11} \\
& + 32399127700820c^6d^{12} + 5162234174652c^5d^{13} + 588928289822c^4d^{14} \\
& + 44651384652c^3d^{15} + 1991932712c^2d^{16} + 41748588cd^{17} \\
& + 235298d^{18} + 47150488938112c^{17} + 414940560786048c^{16}d
\end{aligned}$$

$$\begin{aligned}
& + 1673248609159488c^{15}d^2 + 4100353064603712c^{14}d^3 + 6829797783263712c^{13}d^4 \\
& + 8196735797044064c^{12}d^5 + 7333715865946816c^{11}d^6 + 4994483118349120c^{10}d^7 \\
& + 2621039076559448c^9d^8 + 1065995998611640c^8d^9 + 335765584356964c^7d^{10} \\
& + 81212364706516c^6d^{11} + 14795990218364c^5d^{12} + 1961141053660c^4d^{13} \\
& + 178480812328c^3d^{14} + 10132943338c^2d^{15} + 302036196cd^{16} \\
& + 3193330d^{17} + 52171287341184c^{16} + 434317247635328c^{15}d \\
& + 1651662399600640c^{14}d^2 + 3803472696806016c^{13}d^3 + 5928568116023616c^{12}d^4 \\
& + 6624856439196992c^{11}d^5 + 5484603612044768c^{10}d^6 + 342899235505696c^9d^7 \\
& + 1635203756321592c^8d^8 + 596319361629608c^7d^9 + 165486388781016c^6d^{10} \\
& + 34458779515056c^5d^{11} + 5240250953746c^4d^{12} + 555383153798c^3d^{13} \\
& + 37864446221c^2d^{14} + 1433525625cd^{15} + 21455336d^{16} \\
& + 51024276314752c^{15} + 400252445267072c^{14}d + 1428976109938176c^{13}d^2 \\
& + 3076001709380352c^{12}d^3 + 4458912254232768c^{11}d^4 + 4604908181211712c^{10}d^5 \\
& + 3496205674923744c^9d^6 + 1985056799384736c^8d^7 + 848878155342840c^7d^8 \\
& + 273035467614360c^6d^9 + 65367763953096c^5d^{10} + 11393382746744c^4d^{11} \\
& + 1389342538354c^3d^{12} + 110521919186c^2d^{13} + 5025933479cd^{14} \\
& + 95346797d^{15} + 44123684604288c^{14} + 324676062452352c^{13}d \\
& + 1082522482029632c^{12}d^2 + 2164817065925952c^{11}d^3 + 2897052480940800c^{10}d^4 \\
& + 2740944147737600c^9d^5 + 1888235538724256c^8d^6 + 960933021969664c^7d^7 \\
& + 362488957400904c^6d^8 + 100679485693560c^5d^9 + 20211854200612c^4d^{10} \\
& + 2831486141492c^3d^{11} + 259766416442c^2d^{12} + 13804799134cd^{13} \\
& + 314574120d^{14} + 33711386570112c^{13} + 231474580101504c^{12}d \\
& + 716349021272128c^{11}d^2 + 1321246061487168c^{10}d^3 + 1618243269489920c^9d^4 \\
& + 1387932574417152c^8d^5 + 856382893177728c^7d^6 + 384299127903456c^6d^7 \\
& + 125200915332744c^5d^8 + 29182596681000c^4d^9 + 4715676610932c^3d^{10} \\
& + 497957547924c^2d^{11} + 30576446922cd^{12} + 815231522d^{13} \\
& + 22712259818368c^{12} + 144639223222400c^{11}d + 412492104304512c^{10}d^2 \\
& + 695686292458240c^9d^3 + 771736212621664c^8d^4 + 592358393836704c^7d^5 \\
& + 322087710467760c^6d^6 + 124787611838640c^5d^7 + 34121694073128c^4d^8 \\
& + 6405958844920c^3d^9 + 781183563872c^2d^{10} + 55294828952cd^{11} \\
& + 1705937880d^{12} + 13451538872192c^{11} + 78883193726336c^{10}d \\
& + 205545695176832c^9d^2 + 313722224297344c^8d^3 + 311201945644128c^7d^4 \\
& + 210345186952288c^6d^5 + 98690088424464c^5d^6 + 32082011680368c^4d^7 \\
& + 7066177461048c^3d^8 + 1001609220152c^2d^9 + 81947976248cd^{10} \\
& + 2917771200d^{11} + 6972360694656c^{10} + 37331946130048c^9d \\
& + 87967174483904c^8d^2 + 119969178774464c^7d^3 + 104718049531744c^6d^4 \\
& + 61034555914528c^5d^5 + 24014944947904c^4d^6 + 6286402984576c^3d^7 \\
& + 1045393617792c^2d^8 + 99449579280cd^9 + 4095396792d^{10} \\
& + 3143901093760c^9 + 15212302932864c^8d + 32006916537536c^7d^2 \\
& + 38383201345472c^6d^3 + 28870929621984c^5d^4 + 14103019361632c^4d^5 \\
& + 4466153857024c^3d^6 + 882404498752c^2d^7 + 98471545856cd^8 \\
& + 4715752032d^9 + 1223426687872c^8 + 5282461803648c^7d \\
& + 9766691612928c^6d^2 + 10086132965504c^5d^3 + 6354224838400c^4d^4 \\
& + 2496744354304c^3d^5 + 596476481344c^2d^6 + 79055903232cd^7
\end{aligned}$$

$$\begin{aligned}
& + 4440249696d^8 + 406580486784c^7 + 1541711972992c^6d \\
& + 2453140355200c^5d^2 + 2120529907456c^4d^3 + 1073928874880c^3d^4 \\
& + 318152011136c^2d^5 + 50961177600cd^6 + 3398043776d^7 \\
& + 113792371200c^6 + 371128591104c^5d + 494000762240c^4d^2 \\
& + 343067375616c^3d^3 + 130917569408c^2d^4 + 25989174272cd^5 \\
& + 2093211136d^6 + 26318087680c^5 + 71763590144c^4d \\
& + 76698035200c^3d^2 + 40111236096c^2d^3 + 10250892800cd^4 \\
& + 1022625792d^5 + 4898107392c^4 + 10718027776c^3d \\
& + 8621376512c^2d^2 + 3017730048cd^3 + 387325440d^4 \\
& + 705288192c^3 + 1160906752c^2d + 624646144cd^2 \\
& + 109744128d^3 + 73768960c^2 + 81178624cd \\
& + 21911552d^2 + 4988928c + 2752512d \\
& + 163840
\end{aligned}$$

## Appendix A.3.5

The eliminant  $\Omega$  of the appendix, the partial derivatives entering  $\Gamma$ , the polynomial  $\Gamma$  itself, and the two resultants used to locate the zeros of  $\Gamma$  on the curve  $\Omega = 0$ . Polynomials are in  $(m_i, m_j, \rho)$  or  $(m_i, m_j)$  with integer coefficients ordered by decreasing total degree.

$\Omega$

$\text{Res}_\rho(q, P_j) = -A_0 \Omega(m_i, m_j)$ ; irreducible, of degree 14 in each variable. For  $m_j \geq 1$  it has exactly one root  $m_i = \varphi(m_j) > m_j$ , with  $\varphi(1) \approx 1.782$ . 141 terms.

$$\begin{aligned}
\Omega(m_i, m_j) = & 32768m_i^{14}m_j^{11} - 65536m_i^{13}m_j^{12} + 32768m_i^{12}m_j^{13} \\
& - 49152m_i^{14}m_j^{10} + 16384m_i^{13}m_j^{11} + 106496m_i^{12}m_j^{12} \\
& - 131072m_i^{11}m_j^{13} + 24576m_i^{10}m_j^{14} + 49152m_i^{14}m_j^9 \\
& + 57344m_i^{12}m_j^{11} - 36864m_i^{11}m_j^{12} + 139264m_i^{10}m_j^{13} \\
& - 36864m_i^9m_j^{14} - 32768m_i^{14}m_j^8 - 18432m_i^{13}m_j^9 \\
& - 139776m_i^{12}m_j^{10} - 350208m_i^{11}m_j^{11} + 140288m_i^{10}m_j^{12} \\
& - 126976m_i^9m_j^{13} + 38400m_i^8m_j^{14} + 17024m_i^{14}m_j^7 \\
& + 11008m_i^{13}m_j^8 + 187520m_i^{12}m_j^9 + 458240m_i^{11}m_j^{10} \\
& + 363392m_i^{10}m_j^{11} - 224512m_i^9m_j^{12} + 82816m_i^8m_j^{13} \\
& - 25600m_i^7m_j^{14} - 6720m_i^{14}m_j^6 - 5568m_i^{13}m_j^7 \\
& - 138560m_i^{12}m_j^8 - 425664m_i^{11}m_j^9 - 354752m_i^{10}m_j^{10} \\
& - 254016m_i^9m_j^{11} + 251328m_i^8m_j^{12} - 52288m_i^7m_j^{13} \\
& + 12928m_i^6m_j^{14} + 2128m_i^{14}m_j^5 + 1376m_i^{13}m_j^6 \\
& + 79616m_i^{12}m_j^7 + 247456m_i^{11}m_j^8 + 259552m_i^{10}m_j^9 \\
& + 69472m_i^9m_j^{10} + 153216m_i^8m_j^{11} - 175456m_i^7m_j^{12} \\
& + 27216m_i^6m_j^{13} - 4992m_i^5m_j^{14} - 512m_i^{14}m_j^4 \\
& - 288m_i^{13}m_j^5 - 34456m_i^{12}m_j^6 - 110352m_i^{11}m_j^7 \\
& - 90288m_i^{10}m_j^8 + 3696m_i^9m_j^9 + 74096m_i^8m_j^{10} \\
& - 128304m_i^7m_j^{11} + 92144m_i^6m_j^{12} - 11728m_i^5m_j^{13} \\
& + 1448m_i^4m_j^{14} + 96m_i^{14}m_j^3 + 12088m_i^{12}m_j^5 \\
& + 35952m_i^{11}m_j^6 + 18960m_i^{10}m_j^7 - 67072m_i^9m_j^8 \\
& - 62232m_i^8m_j^9 - 61056m_i^7m_j^{10} + 90016m_i^6m_j^{11}
\end{aligned}$$

$$\begin{aligned}
& -37024m_i^5m_j^{12} + 4176m_i^4m_j^{13} - 336m_i^3m_j^{14} \\
& -12m_i^{14}m_j^2 + 4m_i^{13}m_j^3 - 3334m_i^{12}m_j^4 \\
& -9004m_i^{11}m_j^5 + 1902m_i^{10}m_j^6 + 40528m_i^9m_j^7 \\
& + 63304m_i^8m_j^8 + 15632m_i^7m_j^9 + 37356m_i^6m_j^{10} \\
& - 47732m_i^5m_j^{11} + 11198m_i^4m_j^{12} - 1092m_i^3m_j^{13} \\
& + 50m_i^2m_j^{14} + 1m_i^{14}m_j - 2m_i^{13}m_j^2 \\
& + 730m_i^{12}m_j^3 + 1620m_i^{11}m_j^4 - 2119m_i^{10}m_j^5 \\
& - 17330m_i^9m_j^6 - 22372m_i^8m_j^7 - 18824m_i^7m_j^8 \\
& + 2371m_i^6m_j^9 - 19718m_i^5m_j^{10} + 19098m_i^4m_j^{11} \\
& - 2700m_i^3m_j^{12} + 243m_i^2m_j^{13} - 6m_i m_j^{14} \\
& - 121m_i^{12}m_j^2 - 200m_i^{11}m_j^3 + 719m_i^{10}m_j^4 \\
& + 4448m_i^9m_j^5 + 7238m_i^8m_j^6 + 1696m_i^7m_j^7 \\
& + 4942m_i^6m_j^8 - 3648m_i^5m_j^9 + 8115m_i^4m_j^{10} \\
& - 5400m_i^3m_j^{11} + 419m_i^2m_j^{12} - 32m_i m_j^{13} \\
& + 14m_i^{12}m_j + 14m_i^{11}m_j^2 - 120m_i^{10}m_j^3 \\
& - 866m_i^9m_j^4 - 1152m_i^8m_j^5 - 860m_i^7m_j^6 \\
& + 1176m_i^6m_j^7 - 1804m_i^5m_j^8 + 1966m_i^4m_j^9 \\
& - 2706m_i^3m_j^{10} + 1184m_i^2m_j^{11} - 50m_i m_j^{12} \\
& + 4m_j^{13} - 1m_i^{12} + 11m_i^{10}m_j^2 \\
& + 96m_i^9m_j^3 + 170m_i^8m_j^4 - 16m_i^7m_j^5 \\
& + 50m_i^6m_j^6 - 384m_i^5m_j^7 + 407m_i^4m_j^8 \\
& - 432m_i^3m_j^9 + 515m_i^2m_j^{10} - 160m_i m_j^{11} \\
& - 8m_i^9m_j^2 - 32m_i^7m_j^4 + 80m_i^6m_j^5 \\
& - 112m_i^5m_j^6 + 176m_i^4m_j^7 - 160m_i^3m_j^8 \\
& + 112m_i^2m_j^9 - 72m_i m_j^{10} + 16m_j^{11}
\end{aligned}$$

$q_\rho$

$\partial q/\partial \rho = 2A_2\rho + A_1$ . 13 terms.

$$\begin{aligned}
q_\rho(m_i, m_j, \rho) = & -8m_i^3m_j^2 + 8m_i^2m_j^3 - 16m_i^2m_j^2\rho \\
& + 2m_i^3m_j + 4m_i^2m_j\rho - 2m_i m_j^3 \\
& + 8m_i m_j^2\rho - 1m_i^3 + 1m_i^2m_j \\
& - 2m_i^2\rho - 1m_i m_j^2 + 1m_j^3 \\
& - 2m_j^2\rho
\end{aligned}$$

$q_{m_i}$

$\partial q/\partial m_i$ . 15 terms.

$$\begin{aligned}
q_{m_i}(m_i, m_j, \rho) = & -24m_i^2m_j^2\rho + 16m_i m_j^3\rho - 16m_i m_j^2\rho^2 \\
& + 6m_i^2m_j\rho + 4m_i m_j\rho^2 - 2m_j^3\rho \\
& + 4m_j^2\rho^2 - 3m_i^2\rho + 16m_i m_j^2 \\
& + 2m_i m_j\rho - 2m_i\rho^2 - 1m_j^2\rho \\
& - 8m_i m_j - 2m_j^2 + 2m_i
\end{aligned}$$

$q_{m_j}$

$\partial q/\partial m_j$ . 15 terms.

$$\begin{aligned} q_{m_j}(m_i, m_j, \rho) = & -16m_i^3 m_j \rho + 24m_i^2 m_j^2 \rho - 16m_i^2 m_j \rho^2 \\ & + 2m_i^3 \rho + 2m_i^2 \rho^2 - 6m_i m_j^2 \rho \\ & + 8m_i m_j \rho^2 + 16m_i^2 m_j + 1m_i^2 \rho \\ & - 2m_i m_j \rho + 3m_j^2 \rho - 2m_j \rho^2 \\ & - 4m_i^2 - 4m_i m_j + 2m_j \end{aligned}$$

$\omega_{m_i}$

$\omega_{m_i} := \mathcal{D}^2 \partial w_j / \partial m_i$  with  $w_j$  as in the appendix and  $\mathcal{D} = 4m_i^2 m_j^2 (1 + \rho)(m_i + \rho)$ . 13 terms.

$$\begin{aligned} \omega_{m_i}(m_i, m_j, \rho) = & -32m_i^4 m_j^4 \rho - 32m_i^4 m_j^4 + 16m_i^4 m_j^3 \rho \\ & + 16m_i^3 m_j^4 \rho + 8m_i^2 m_j^4 \rho^2 + 16m_i^4 m_j^3 \\ & - 4m_i^4 m_j^2 \rho + 16m_i^3 m_j^4 - 4m_i^2 m_j^4 \rho \\ & - 8m_i m_j^4 \rho^2 - 4m_i^4 m_j^2 - 12m_i^2 m_j^4 \\ & - 8m_i m_j^4 \rho \end{aligned}$$

$\omega_{m_j}$

$\omega_{m_j} := \mathcal{D}^2 \partial w_j / \partial m_j$ . 15 terms.

$$\begin{aligned} \omega_{m_j}(m_i, m_j, \rho) = & -8m_i^6 m_j \rho^2 - 16m_i^5 m_j \rho^3 - 8m_i^4 m_j \rho^4 \\ & - 16m_i^6 m_j \rho + 16m_i^5 m_j^2 \rho - 32m_i^5 m_j \rho^2 \\ & + 16m_i^4 m_j^2 \rho^2 - 16m_i^4 m_j \rho^3 - 8m_i^6 m_j \\ & + 16m_i^5 m_j^2 - 24m_i^5 m_j \rho + 16m_i^4 m_j^2 \rho \\ & - 16m_i^4 m_j \rho^2 - 8m_i^5 m_j - 8m_i^4 m_j \rho \end{aligned}$$

$\omega_\rho$

$\omega_\rho := \mathcal{D}^2 \partial w_j / \partial \rho$ . 13 terms.

$$\begin{aligned} \omega_\rho(m_i, m_j, \rho) = & -32m_i^5 m_j^4 - 64m_i^4 m_j^4 \rho + 16m_i^5 m_j^3 \\ & - 24m_i^4 m_j^4 + 32m_i^4 m_j^3 \rho + 16m_i^3 m_j^4 \rho \\ & - 4m_i^5 m_j^2 + 16m_i^4 m_j^3 - 8m_i^4 m_j^2 \rho \\ & + 4m_i^3 m_j^4 - 8m_i^2 m_j^4 \rho - 4m_i^4 m_j^2 \\ & - 4m_i^2 m_j^4 \end{aligned}$$

$\Omega_{m_i}$

$\partial \Omega / \partial m_i$ , strictly positive at  $m_i = \varphi(m_j)$ . 139 terms.

$$\begin{aligned} \Omega_{m_i}(m_i, m_j) = & 458752m_i^{13} m_j^{11} - 851968m_i^{12} m_j^{12} + 393216m_i^{11} m_j^{13} \\ & - 688128m_i^{13} m_j^{10} + 212992m_i^{12} m_j^{11} + 1277952m_i^{11} m_j^{12} \\ & - 1441792m_i^{10} m_j^{13} + 245760m_i^9 m_j^{14} + 688128m_i^{13} m_j^9 \\ & + 688128m_i^{11} m_j^{11} - 405504m_i^{10} m_j^{12} + 1392640m_i^9 m_j^{13} \\ & - 331776m_i^8 m_j^{14} - 458752m_i^{13} m_j^8 - 239616m_i^{12} m_j^9 \\ & - 1677312m_i^{11} m_j^{10} - 3852288m_i^{10} m_j^{11} + 1402880m_i^9 m_j^{12} \\ & - 1142784m_i^8 m_j^{13} + 307200m_i^7 m_j^{14} + 238336m_i^{13} m_j^7 \\ & + 143104m_i^{12} m_j^8 + 2250240m_i^{11} m_j^9 + 5040640m_i^{10} m_j^{10} \end{aligned}$$

$$\begin{aligned}
& + 3633920m_i^9m_j^{11} - 2020608m_i^8m_j^{12} + 662528m_i^7m_j^{13} \\
& - 179200m_i^6m_j^{14} - 94080m_i^{13}m_j^6 - 72384m_i^{12}m_j^7 \\
& - 1662720m_i^{11}m_j^8 - 4682304m_i^{10}m_j^9 - 3547520m_i^9m_j^{10} \\
& - 2286144m_i^8m_j^{11} + 2010624m_i^7m_j^{12} - 366016m_i^6m_j^{13} \\
& + 77568m_i^5m_j^{14} + 29792m_i^{13}m_j^5 + 17888m_i^{12}m_j^6 \\
& + 955392m_i^{11}m_j^7 + 2722016m_i^{10}m_j^8 + 2595520m_i^9m_j^9 \\
& + 625248m_i^8m_j^{10} + 1225728m_i^7m_j^{11} - 1228192m_i^6m_j^{12} \\
& + 163296m_i^5m_j^{13} - 24960m_i^4m_j^{14} - 7168m_i^{13}m_j^4 \\
& - 3744m_i^{12}m_j^5 - 413472m_i^{11}m_j^6 - 1213872m_i^{10}m_j^7 \\
& - 902880m_i^9m_j^8 + 33264m_i^8m_j^9 + 592768m_i^7m_j^{10} \\
& - 898128m_i^6m_j^{11} + 552864m_i^5m_j^{12} - 58640m_i^4m_j^{13} \\
& + 5792m_i^3m_j^{14} + 1344m_i^{13}m_j^3 + 145056m_i^{11}m_j^5 \\
& + 395472m_i^{10}m_j^6 + 189600m_i^9m_j^7 - 603648m_i^8m_j^8 \\
& - 497856m_i^7m_j^9 - 427392m_i^6m_j^{10} + 540096m_i^5m_j^{11} \\
& - 185120m_i^4m_j^{12} + 16704m_i^3m_j^{13} - 1008m_i^2m_j^{14} \\
& - 168m_i^{13}m_j^2 + 52m_i^{12}m_j^3 - 40008m_i^{11}m_j^4 \\
& - 99044m_i^{10}m_j^5 + 19020m_i^9m_j^6 + 364752m_i^8m_j^7 \\
& + 506432m_i^7m_j^8 + 109424m_i^6m_j^9 + 224136m_i^5m_j^{10} \\
& - 238660m_i^4m_j^{11} + 44792m_i^3m_j^{12} - 3276m_i^2m_j^{13} \\
& + 100m_i m_j^{14} + 14m_i^{13}m_j - 26m_i^{12}m_j^2 \\
& + 8760m_i^{11}m_j^3 + 17820m_i^{10}m_j^4 - 21190m_i^9m_j^5 \\
& - 155970m_i^8m_j^6 - 178976m_i^7m_j^7 - 131768m_i^6m_j^8 \\
& + 14226m_i^5m_j^9 - 98590m_i^4m_j^{10} + 76392m_i^3m_j^{11} \\
& - 8100m_i^2m_j^{12} + 486m_i m_j^{13} - 6m_j^{14} \\
& - 1452m_i^{11}m_j^2 - 2200m_i^{10}m_j^3 + 7190m_i^9m_j^4 \\
& + 40032m_i^8m_j^5 + 57904m_i^7m_j^6 + 11872m_i^6m_j^7 \\
& + 29652m_i^5m_j^8 - 18240m_i^4m_j^9 + 32460m_i^3m_j^{10} \\
& - 16200m_i^2m_j^{11} + 838m_i m_j^{12} - 32m_j^{13} \\
& + 168m_i^{11}m_j + 154m_i^{10}m_j^2 - 1200m_i^9m_j^3 \\
& - 7794m_i^8m_j^4 - 9216m_i^7m_j^5 - 6020m_i^6m_j^6 \\
& + 7056m_i^5m_j^7 - 9020m_i^4m_j^8 + 7864m_i^3m_j^9 \\
& - 8118m_i^2m_j^{10} + 2368m_i m_j^{11} - 50m_j^{12} \\
& - 12m_i^{11} + 110m_i^9m_j^2 + 864m_i^8m_j^3 \\
& + 1360m_i^7m_j^4 - 112m_i^6m_j^5 + 300m_i^5m_j^6 \\
& - 1920m_i^4m_j^7 + 1628m_i^3m_j^8 - 1296m_i^2m_j^9 \\
& + 1030m_i m_j^{10} - 160m_j^{11} - 72m_i^8m_j^2 \\
& - 224m_i^6m_j^4 + 480m_i^5m_j^5 - 560m_i^4m_j^6 \\
& + 704m_i^3m_j^7 - 480m_i^2m_j^8 + 224m_i m_j^9 \\
& - 72m_j^{10}
\end{aligned}$$

$\Omega_{m_j}$

$\partial\Omega/\partial m_j$ . 140 terms.

$$\begin{aligned}
\Omega_{m_j}(m_i, m_j) = & 360448m_i^{14}m_j^{10} - 786432m_i^{13}m_j^{11} + 425984m_i^{12}m_j^{12} \\
& - 491520m_i^{14}m_j^9 + 180224m_i^{13}m_j^{10} + 1277952m_i^{12}m_j^{11}
\end{aligned}$$

$$\begin{aligned}
& -1703936m_i^{11}m_j^{12} + 344064m_i^{10}m_j^{13} + 442368m_i^{14}m_j^8 \\
& + 630784m_i^{12}m_j^{10} - 442368m_i^{11}m_j^{11} + 1810432m_i^{10}m_j^{12} \\
& - 516096m_i^9m_j^{13} - 262144m_i^{14}m_j^7 - 165888m_i^{13}m_j^8 \\
& - 1397760m_i^{12}m_j^9 - 3852288m_i^{11}m_j^{10} + 1683456m_i^{10}m_j^{11} \\
& - 1650688m_i^9m_j^{12} + 537600m_i^8m_j^{13} + 119168m_i^{14}m_j^6 \\
& + 88064m_i^{13}m_j^7 + 1687680m_i^{12}m_j^8 + 4582400m_i^{11}m_j^9 \\
& + 3997312m_i^{10}m_j^{10} - 2694144m_i^9m_j^{11} + 1076608m_i^8m_j^{12} \\
& - 358400m_i^7m_j^{13} - 40320m_i^{14}m_j^5 - 38976m_i^{13}m_j^6 \\
& - 1108480m_i^{12}m_j^7 - 3830976m_i^{11}m_j^8 - 3547520m_i^{10}m_j^9 \\
& - 2794176m_i^9m_j^{10} + 3015936m_i^8m_j^{11} - 679744m_i^7m_j^{12} \\
& + 180992m_i^6m_j^{13} + 10640m_i^{14}m_j^4 + 8256m_i^{13}m_j^5 \\
& + 557312m_i^{12}m_j^6 + 1979648m_i^{11}m_j^7 + 2335968m_i^{10}m_j^8 \\
& + 694720m_i^9m_j^9 + 1685376m_i^8m_j^{10} - 2105472m_i^7m_j^{11} \\
& + 353808m_i^6m_j^{12} - 69888m_i^5m_j^{13} - 2048m_i^{14}m_j^3 \\
& - 1440m_i^{13}m_j^4 - 206736m_i^{12}m_j^5 - 772464m_i^{11}m_j^6 \\
& - 722304m_i^{10}m_j^7 + 33264m_i^9m_j^8 + 740960m_i^8m_j^9 \\
& - 1411344m_i^7m_j^{10} + 1105728m_i^6m_j^{11} - 152464m_i^5m_j^{12} \\
& + 20272m_i^4m_j^{13} + 288m_i^{14}m_j^2 + 60440m_i^{12}m_j^4 \\
& + 215712m_i^{11}m_j^5 + 132720m_i^{10}m_j^6 - 536576m_i^9m_j^7 \\
& - 560088m_i^8m_j^8 - 610560m_i^7m_j^9 + 990176m_i^6m_j^{10} \\
& - 444288m_i^5m_j^{11} + 54288m_i^4m_j^{12} - 4704m_i^3m_j^{13} \\
& - 24m_i^{14}m_j + 12m_i^{13}m_j^2 - 13336m_i^{12}m_j^3 \\
& - 45020m_i^{11}m_j^4 + 11412m_i^{10}m_j^5 + 283696m_i^9m_j^6 \\
& + 506432m_i^8m_j^7 + 140688m_i^7m_j^8 + 373560m_i^6m_j^9 \\
& - 525052m_i^5m_j^{10} + 134376m_i^4m_j^{11} - 14196m_i^3m_j^{12} \\
& + 700m_i^2m_j^{13} + 1m_i^{14} - 4m_i^{13}m_j \\
& + 2190m_i^{12}m_j^2 + 6480m_i^{11}m_j^3 - 10595m_i^{10}m_j^4 \\
& - 103980m_i^9m_j^5 - 156604m_i^8m_j^6 - 150592m_i^7m_j^7 \\
& + 21339m_i^6m_j^8 - 197180m_i^5m_j^9 + 210078m_i^4m_j^{10} \\
& - 32400m_i^3m_j^{11} + 3159m_i^2m_j^{12} - 84m_im_j^{13} \\
& - 242m_i^{12}m_j - 600m_i^{11}m_j^2 + 2876m_i^{10}m_j^3 \\
& + 22240m_i^9m_j^4 + 43428m_i^8m_j^5 + 11872m_i^7m_j^6 \\
& + 39536m_i^6m_j^7 - 32832m_i^5m_j^8 + 81150m_i^4m_j^9 \\
& - 59400m_i^3m_j^{10} + 5028m_i^2m_j^{11} - 416m_im_j^{12} \\
& + 14m_i^{12} + 28m_i^{11}m_j - 360m_i^{10}m_j^2 \\
& - 3464m_i^9m_j^3 - 5760m_i^8m_j^4 - 5160m_i^7m_j^5 \\
& + 8232m_i^6m_j^6 - 14432m_i^5m_j^7 + 17694m_i^4m_j^8 \\
& - 27060m_i^3m_j^9 + 13024m_i^2m_j^{10} - 600m_im_j^{11} \\
& + 52m_j^{12} + 22m_i^{10}m_j + 288m_i^9m_j^2 \\
& + 680m_i^8m_j^3 - 80m_i^7m_j^4 + 300m_i^6m_j^5 \\
& - 2688m_i^5m_j^6 + 3256m_i^4m_j^7 - 3888m_i^3m_j^8 \\
& + 5150m_i^2m_j^9 - 1760m_im_j^{10} - 16m_i^9m_j \\
& - 128m_i^7m_j^3 + 400m_i^6m_j^4 - 672m_i^5m_j^5 \\
& + 1232m_i^4m_j^6 - 1280m_i^3m_j^7 + 1008m_i^2m_j^8
\end{aligned}$$

$$-720m_i m_j^9 + 176m_j^{10}$$

$\Gamma$

$\Gamma := q_\rho(\Omega_{m_i}\omega_{m_j} - \Omega_{m_j}\omega_{m_i}) + \omega_\rho(q_{m_i}\Omega_{m_j} - q_{m_j}\Omega_{m_i})$ , so that  $\mathcal{D}^2\Omega_{m_i}q_\rho dw_j/dm_j = \Gamma$  at  $m_i = \varphi(m_j)$ . 1519 terms.

$$\begin{aligned} \Gamma(m_i, m_j, \rho) = & 29360128m_i^{22}m_j^{14}\rho^2 - 50331648m_i^{21}m_j^{16}\rho - 83886080m_i^{21}m_j^{15}\rho^2 \\ & + 117440512m_i^{21}m_j^{14}\rho^3 + 293601280m_i^{20}m_j^{17}\rho - 71303168m_i^{20}m_j^{16}\rho^2 \\ & - 276824064m_i^{20}m_j^{15}\rho^3 + 146800640m_i^{20}m_j^{14}\rho^4 - 436207616m_i^{19}m_j^{18}\rho \\ & + 411041792m_i^{19}m_j^{17}\rho^2 + 109051904m_i^{19}m_j^{16}\rho^3 - 301989888m_i^{19}m_j^{15}\rho^4 \\ & + 58720256m_i^{19}m_j^{14}\rho^5 + 192937984m_i^{18}m_j^{19}\rho - 452984832m_i^{18}m_j^{18}\rho^2 \\ & + 16777216m_i^{18}m_j^{17}\rho^3 + 180355072m_i^{18}m_j^{16}\rho^4 - 109051904m_i^{18}m_j^{15}\rho^5 \\ & + 167772160m_i^{17}m_j^{19}\rho^2 + 33554432m_i^{17}m_j^{18}\rho^3 - 25165824m_i^{17}m_j^{17}\rho^4 \\ & + 50331648m_i^{17}m_j^{16}\rho^5 + 58720256m_i^{22}m_j^{14}\rho - 51380224m_i^{22}m_j^{13}\rho^2 \\ & - 92274688m_i^{21}m_j^{16} - 117440512m_i^{21}m_j^{15}\rho + 306184192m_i^{21}m_j^{14}\rho^2 \\ & - 205520896m_i^{21}m_j^{13}\rho^3 + 293601280m_i^{20}m_j^{17} - 299892736m_i^{20}m_j^{16}\rho \\ & - 333447168m_i^{20}m_j^{15}\rho^2 + 461373440m_i^{20}m_j^{14}\rho^3 - 256901120m_i^{20}m_j^{13}\rho^4 \\ & - 310378496m_i^{19}m_j^{18} + 264241152m_i^{19}m_j^{17}\rho + 154140672m_i^{19}m_j^{16}\rho^2 \\ & - 159383552m_i^{19}m_j^{15}\rho^3 + 239075328m_i^{19}m_j^{14}\rho^4 - 102760448m_i^{19}m_j^{13}\rho^5 \\ & + 109051904m_i^{18}m_j^{19} + 480247808m_i^{18}m_j^{18}\rho - 663748608m_i^{18}m_j^{17}\rho^2 \\ & + 232783872m_i^{18}m_j^{16}\rho^3 + 262144000m_i^{18}m_j^{15}\rho^4 + 25165824m_i^{18}m_j^{14}\rho^5 \\ & - 687865856m_i^{17}m_j^{19}\rho + 846200832m_i^{17}m_j^{18}\rho^2 - 318767104m_i^{17}m_j^{17}\rho^3 \\ & - 506462208m_i^{17}m_j^{16}\rho^4 + 205520896m_i^{17}m_j^{15}\rho^5 + 100663296m_i^{16}m_j^{20}\rho \\ & - 484442112m_i^{16}m_j^{19}\rho^2 - 161480704m_i^{16}m_j^{18}\rho^3 + 177209344m_i^{16}m_j^{17}\rho^4 \\ & - 209715200m_i^{16}m_j^{16}\rho^5 + 25165824m_i^{15}m_j^{20}\rho^2 + 100663296m_i^{15}m_j^{19}\rho^3 \\ & - 15728640m_i^{15}m_j^{18}\rho^4 + 31457280m_i^{15}m_j^{17}\rho^5 + 29360128m_i^{22}m_j^{14} \\ & - 102760448m_i^{22}m_j^{13}\rho + 58720256m_i^{22}m_j^{12}\rho^2 + 52428800m_i^{21}m_j^{15} \\ & + 260046848m_i^{21}m_j^{14}\rho - 468975616m_i^{21}m_j^{13}\rho^2 + 234881024m_i^{21}m_j^{12}\rho^3 \\ & - 25165824m_i^{20}m_j^{16} + 286785536m_i^{20}m_j^{15}\rho + 508559360m_i^{20}m_j^{14}\rho^2 \\ & - 606076928m_i^{20}m_j^{13}\rho^3 + 293601280m_i^{20}m_j^{12}\rho^4 - 486539264m_i^{19}m_j^{17} \\ & - 4194304m_i^{19}m_j^{16}\rho + 134217728m_i^{19}m_j^{15}\rho^2 + 401604608m_i^{19}m_j^{14}\rho^3 \\ & - 216268800m_i^{19}m_j^{13}\rho^4 + 117440512m_i^{19}m_j^{12}\rho^5 + 947912704m_i^{18}m_j^{18} \\ & - 22020096m_i^{18}m_j^{17}\rho - 713031680m_i^{18}m_j^{16}\rho^2 - 78118912m_i^{18}m_j^{15}\rho^3 \\ & + 170917888m_i^{18}m_j^{14}\rho^4 + 23592960m_i^{18}m_j^{13}\rho^5 - 606076928m_i^{17}m_j^{19} \\ & + 235929600m_i^{17}m_j^{18}\rho + 2450259968m_i^{17}m_j^{17}\rho^2 - 1069023232m_i^{17}m_j^{16}\rho^3 \\ & + 151519232m_i^{17}m_j^{15}\rho^4 + 47185920m_i^{17}m_j^{14}\rho^5 + 88080384m_i^{16}m_j^{20} \\ & + 815267840m_i^{16}m_j^{19}\rho - 1721761792m_i^{16}m_j^{18}\rho^2 + 1563426816m_i^{16}m_j^{17}\rho^3 \\ & + 260046848m_i^{16}m_j^{16}\rho^4 - 101187584m_i^{16}m_j^{15}\rho^5 - 204472320m_i^{15}m_j^{20}\rho \\ & + 923271168m_i^{15}m_j^{19}\rho^2 - 327155712m_i^{15}m_j^{18}\rho^3 - 190054400m_i^{15}m_j^{17}\rho^4 \\ & + 268959744m_i^{15}m_j^{16}\rho^5 - 59768832m_i^{14}m_j^{20}\rho^2 - 132120576m_i^{14}m_j^{19}\rho^3 \\ & + 25165824m_i^{14}m_j^{18}\rho^4 - 58195968m_i^{14}m_j^{17}\rho^5 - 51380224m_i^{22}m_j^{13} \\ & + 117440512m_i^{22}m_j^{12}\rho - 45875200m_i^{22}m_j^{11}\rho^2 - 38797312m_i^{21}m_j^{14} \\ & - 381157376m_i^{21}m_j^{13}\rho + 490995712m_i^{21}m_j^{12}\rho^2 - 183500800m_i^{21}m_j^{11}\rho^3 \\ & - 14680064m_i^{20}m_j^{15} + 21233664m_i^{20}m_j^{14}\rho - 709754880m_i^{20}m_j^{13}\rho^2 \\ & + 558366720m_i^{20}m_j^{12}\rho^3 - 229376000m_i^{20}m_j^{11}\rho^4 - 225443840m_i^{19}m_j^{16} \\ & - 304087040m_i^{19}m_j^{15}\rho + 492109824m_i^{19}m_j^{14}\rho^2 - 649068544m_i^{19}m_j^{13}\rho^3 \end{aligned}$$

$$\begin{aligned}
& + 113508352m_i^{19}m_j^{12}\rho^4 - 91750400m_i^{19}m_j^{11}\rho^5 + 624951296m_i^{18}m_j^{17} \\
& - 2152726528m_i^{18}m_j^{16}\rho + 363593728m_i^{18}m_j^{15}\rho^2 + 61865984m_i^{18}m_j^{14}\rho^3 \\
& - 496369664m_i^{18}m_j^{13}\rho^4 - 71303168m_i^{18}m_j^{12}\rho^5 - 1068498944m_i^{17}m_j^{18} \\
& - 487063552m_i^{17}m_j^{17}\rho - 2232287232m_i^{17}m_j^{16}\rho^2 + 1381761024m_i^{17}m_j^{15}\rho^3 \\
& - 1140523008m_i^{17}m_j^{14}\rho^4 - 227278848m_i^{17}m_j^{13}\rho^5 + 972029952m_i^{16}m_j^{19} \\
& - 612368384m_i^{16}m_j^{18}\rho - 2709127168m_i^{16}m_j^{17}\rho^2 + 259915776m_i^{16}m_j^{16}\rho^3 \\
& + 515375104m_i^{16}m_j^{15}\rho^4 - 543817728m_i^{16}m_j^{14}\rho^5 - 198180864m_i^{15}m_j^{20} \\
& - 694681600m_i^{15}m_j^{19}\rho + 2012020736m_i^{15}m_j^{18}\rho^2 - 1363935232m_i^{15}m_j^{17}\rho^3 \\
& - 145620992m_i^{15}m_j^{16}\rho^4 + 185335808m_i^{15}m_j^{15}\rho^5 + 271319040m_i^{14}m_j^{20}\rho \\
& - 1199308800m_i^{14}m_j^{19}\rho^2 + 747241472m_i^{14}m_j^{18}\rho^3 + 141164544m_i^{14}m_j^{17}\rho^4 \\
& - 247857152m_i^{14}m_j^{16}\rho^5 + 101449728m_i^{13}m_j^{20}\rho^2 + 84934656m_i^{13}m_j^{19}\rho^3 \\
& - 26935296m_i^{13}m_j^{18}\rho^4 + 64487424m_i^{13}m_j^{17}\rho^5 + 58720256m_i^{22}m_j^{12} \\
& - 91750400m_i^{22}m_j^{11}\rho + 28098560m_i^{22}m_j^{10}\rho^2 - 18874368m_i^{21}m_j^{13} \\
& + 360513536m_i^{21}m_j^{12}\rho - 374767616m_i^{21}m_j^{11}\rho^2 + 112394240m_i^{21}m_j^{10}\rho^3 \\
& + 241696768m_i^{20}m_j^{14} - 298090496m_i^{20}m_j^{13}\rho + 650788864m_i^{20}m_j^{12}\rho^2 \\
& - 418938880m_i^{20}m_j^{11}\rho^3 + 140492800m_i^{20}m_j^{10}\rho^4 + 359137280m_i^{19}m_j^{15} \\
& + 448266240m_i^{19}m_j^{14}\rho - 834699264m_i^{19}m_j^{13}\rho^2 + 750813184m_i^{19}m_j^{12}\rho^3 \\
& - 80576512m_i^{19}m_j^{11}\rho^4 + 56197120m_i^{19}m_j^{10}\rho^5 + 753139712m_i^{18}m_j^{16} \\
& + 3711467520m_i^{18}m_j^{15}\rho - 1412513792m_i^{18}m_j^{14}\rho^2 - 92012544m_i^{18}m_j^{13}\rho^3 \\
& + 780877824m_i^{18}m_j^{12}\rho^4 + 55345152m_i^{18}m_j^{11}\rho^5 - 2009071616m_i^{17}m_j^{17} \\
& + 4208984064m_i^{17}m_j^{16}\rho + 4616454144m_i^{17}m_j^{15}\rho^2 - 2462973952m_i^{17}m_j^{14}\rho^3 \\
& + 1506246656m_i^{17}m_j^{13}\rho^4 + 379060224m_i^{17}m_j^{12}\rho^5 + 1586495488m_i^{16}m_j^{18} \\
& - 31424512m_i^{16}m_j^{17}\rho + 1950924800m_i^{16}m_j^{16}\rho^2 + 326664192m_i^{16}m_j^{15}\rho^3 \\
& + 193118208m_i^{16}m_j^{14}\rho^4 + 869335040m_i^{16}m_j^{13}\rho^5 - 1233649664m_i^{15}m_j^{19} \\
& + 897843200m_i^{15}m_j^{18}\rho + 3427696640m_i^{15}m_j^{17}\rho^2 - 1837137920m_i^{15}m_j^{16}\rho^3 \\
& - 704413696m_i^{15}m_j^{15}\rho^4 + 700088320m_i^{15}m_j^{14}\rho^5 + 291766272m_i^{14}m_j^{20} \\
& + 245989376m_i^{14}m_j^{19}\rho - 1862631424m_i^{14}m_j^{18}\rho^2 + 956170240m_i^{14}m_j^{17}\rho^3 \\
& + 51363840m_i^{14}m_j^{16}\rho^4 - 323649536m_i^{14}m_j^{15}\rho^5 - 226557952m_i^{13}m_j^{20}\rho \\
& + 1109065728m_i^{13}m_j^{19}\rho^2 - 916258816m_i^{13}m_j^{18}\rho^3 - 52789248m_i^{13}m_j^{17}\rho^4 \\
& + 170393600m_i^{13}m_j^{16}\rho^5 - 116457472m_i^{12}m_j^{20}\rho^2 + 13500416m_i^{12}m_j^{19}\rho^3 \\
& + 19038208m_i^{12}m_j^{18}\rho^4 - 47906816m_i^{12}m_j^{17}\rho^5 - 45875200m_i^{22}m_j^{11} \\
& + 56197120m_i^{22}m_j^{10}\rho - 13504512m_i^{22}m_j^9\rho^2 + 35160064m_i^{21}m_j^{12} \\
& - 269025280m_i^{21}m_j^{11}\rho + 225640448m_i^{21}m_j^{10}\rho^2 - 54018048m_i^{21}m_j^9\rho^3 \\
& - 275152896m_i^{20}m_j^{13} + 445521920m_i^{20}m_j^{12}\rho - 508870656m_i^{20}m_j^{11}\rho^2 \\
& + 249757696m_i^{20}m_j^{10}\rho^3 - 67522560m_i^{20}m_j^9\rho^4 - 758677504m_i^{19}m_j^{14} \\
& - 521420800m_i^{19}m_j^{13}\rho + 1145774080m_i^{19}m_j^{12}\rho^2 - 652345344m_i^{19}m_j^{11}\rho^3 \\
& + 47382528m_i^{19}m_j^{10}\rho^4 - 27009024m_i^{19}m_j^9\rho^5 - 865304576m_i^{18}m_j^{15} \\
& - 4318896128m_i^{18}m_j^{14}\rho + 1407320064m_i^{18}m_j^{13}\rho^2 + 342646784m_i^{18}m_j^{12}\rho^3 \\
& - 694796288m_i^{18}m_j^{11}\rho^4 - 32931840m_i^{18}m_j^{10}\rho^5 + 83001344m_i^{17}m_j^{16} \\
& - 5075402752m_i^{17}m_j^{15}\rho - 4182433792m_i^{17}m_j^{14}\rho^2 + 2802163712m_i^{17}m_j^{13}\rho^3 \\
& - 1469382656m_i^{17}m_j^{12}\rho^4 - 328171520m_i^{17}m_j^{11}\rho^5 + 2636021760m_i^{16}m_j^{17} \\
& - 5026340864m_i^{16}m_j^{16}\rho - 3954401280m_i^{16}m_j^{15}\rho^2 + 883392512m_i^{16}m_j^{14}\rho^3 \\
& + 24977408m_i^{16}m_j^{13}\rho^4 - 970121216m_i^{16}m_j^{12}\rho^5 - 1825538048m_i^{15}m_j^{18} \\
& + 878952448m_i^{15}m_j^{17}\rho - 2015223808m_i^{15}m_j^{16}\rho^2 + 749879296m_i^{15}m_j^{15}\rho^3
\end{aligned}$$

$$\begin{aligned}
& + 475201536m_i^{15}m_j^{14}\rho^4 - 897359872m_i^{15}m_j^{13}\rho^5 + 1127841792m_i^{14}m_j^{19} \\
& - 565600256m_i^{14}m_j^{18}\rho - 3802705920m_i^{14}m_j^{17}\rho^2 + 3304022016m_i^{14}m_j^{16}\rho^3 \\
& + 514555904m_i^{14}m_j^{15}\rho^4 - 540459008m_i^{14}m_j^{14}\rho^5 - 291307520m_i^{13}m_j^{20} \\
& + 38731776m_i^{13}m_j^{19}\rho + 1538383872m_i^{13}m_j^{18}\rho^2 - 654778368m_i^{13}m_j^{17}\rho^3 \\
& - 67207168m_i^{13}m_j^{16}\rho^4 + 392839168m_i^{13}m_j^{15}\rho^5 + 138100736m_i^{12}m_j^{20}\rho \\
& - 782901248m_i^{12}m_j^{19}\rho^2 + 728776704m_i^{12}m_j^{18}\rho^3 + 7401472m_i^{12}m_j^{17}\rho^4 \\
& - 101801984m_i^{12}m_j^{16}\rho^5 + 105824256m_i^{11}m_j^{20}\rho^2 - 78118912m_i^{11}m_j^{19}\rho^3 \\
& - 10289152m_i^{11}m_j^{18}\rho^4 + 26312704m_i^{11}m_j^{17}\rho^5 + 28098560m_i^{22}m_j^{10} \\
& - 27009024m_i^{22}m_j^9\rho + 5318656m_i^{22}m_j^8\rho^2 - 38887424m_i^{21}m_j^{11} \\
& + 157229056m_i^{21}m_j^{10}\rho - 108401664m_i^{21}m_j^9\rho^2 + 21274624m_i^{21}m_j^8\rho^3 \\
& + 278003712m_i^{20}m_j^{12} - 406181888m_i^{20}m_j^{11}\rho + 311291904m_i^{20}m_j^{10}\rho^2 \\
& - 122859520m_i^{20}m_j^9\rho^3 + 26593280m_i^{20}m_j^8\rho^4 + 611745792m_i^{19}m_j^{13} \\
& + 557490176m_i^{19}m_j^{12}\rho - 1014425600m_i^{19}m_j^{11}\rho^2 + 440107008m_i^{19}m_j^{10}\rho^3 \\
& - 28550144m_i^{19}m_j^9\rho^4 + 10637312m_i^{19}m_j^8\rho^5 + 1260650496m_i^{18}m_j^{14} \\
& + 3613237248m_i^{18}m_j^{13}\rho - 1173794816m_i^{18}m_j^{12}\rho^2 - 460990464m_i^{18}m_j^{11}\rho^3 \\
& + 477892608m_i^{18}m_j^{10}\rho^4 + 12916736m_i^{18}m_j^9\rho^5 - 35864576m_i^{17}m_j^{15} \\
& + 4379656192m_i^{17}m_j^{14}\rho + 4126722048m_i^{17}m_j^{13}\rho^2 - 2647003136m_i^{17}m_j^{12}\rho^3 \\
& + 927252480m_i^{17}m_j^{11}\rho^4 + 219947008m_i^{17}m_j^{10}\rho^5 - 42958848m_i^{16}m_j^{16} \\
& + 4611092480m_i^{16}m_j^{15}\rho + 2902081536m_i^{16}m_j^{14}\rho^2 - 816699392m_i^{16}m_j^{13}\rho^3 \\
& - 233455616m_i^{16}m_j^{12}\rho^4 + 687605760m_i^{16}m_j^{11}\rho^5 - 3084648448m_i^{15}m_j^{17} \\
& + 5118468096m_i^{15}m_j^{16}\rho + 2632470528m_i^{15}m_j^{15}\rho^2 - 1476075520m_i^{15}m_j^{14}\rho^3 \\
& - 1124968448m_i^{15}m_j^{13}\rho^4 + 839561216m_i^{15}m_j^{12}\rho^5 + 2107064320m_i^{14}m_j^{18} \\
& - 2174312448m_i^{14}m_j^{17}\rho + 2107082752m_i^{14}m_j^{16}\rho^2 - 2017949696m_i^{14}m_j^{15}\rho^3 \\
& - 505987072m_i^{14}m_j^{14}\rho^4 + 428550144m_i^{14}m_j^{13}\rho^5 - 898752512m_i^{13}m_j^{19} \\
& + 438853632m_i^{13}m_j^{18}\rho + 3541847040m_i^{13}m_j^{17}\rho^2 - 3649048576m_i^{13}m_j^{16}\rho^3 \\
& - 72810496m_i^{13}m_j^{15}\rho^4 + 307609600m_i^{13}m_j^{14}\rho^5 + 225935360m_i^{12}m_j^{20} \\
& - 178599936m_i^{12}m_j^{19}\rho - 1233686528m_i^{12}m_j^{18}\rho^2 + 820348928m_i^{12}m_j^{17}\rho^3 \\
& + 44169216m_i^{12}m_j^{16}\rho^4 - 309372928m_i^{12}m_j^{15}\rho^5 - 51412992m_i^{11}m_j^{20}\rho \\
& + 415094784m_i^{11}m_j^{19}\rho^2 - 454770688m_i^{11}m_j^{18}\rho^3 + 5934080m_i^{11}m_j^{17}\rho^4 \\
& + 52445184m_i^{11}m_j^{16}\rho^5 - 71655424m_i^{10}m_j^{20}\rho^2 + 81739776m_i^{10}m_j^{19}\rho^3 \\
& + 4272128m_i^{10}m_j^{18}\rho^4 - 11026432m_i^{10}m_j^{17}\rho^5 - 13504512m_i^{22}m_j^9 \\
& + 10637312m_i^{22}m_j^8\rho - 1688064m_i^{22}m_j^7\rho^2 + 25993216m_i^{21}m_j^{10} \\
& - 75442176m_i^{21}m_j^9\rho + 42655744m_i^{21}m_j^8\rho^2 - 6752256m_i^{21}m_j^7\rho^3 \\
& - 200749056m_i^{20}m_j^{11} + 276074496m_i^{20}m_j^{10}\rho - 159737856m_i^{20}m_j^9\rho^2 \\
& + 49283072m_i^{20}m_j^8\rho^3 - 8440320m_i^{20}m_j^7\rho^4 - 406790144m_i^{19}m_j^{12} \\
& - 491272192m_i^{19}m_j^{11}\rho + 711960576m_i^{19}m_j^{10}\rho^2 - 239224832m_i^{19}m_j^9\rho^3 \\
& + 13148160m_i^{19}m_j^8\rho^4 - 3376128m_i^{19}m_j^7\rho^5 - 755122176m_i^{18}m_j^{13} \\
& - 2447472640m_i^{18}m_j^{12}\rho + 575804416m_i^{18}m_j^{11}\rho^2 + 425716736m_i^{18}m_j^{10}\rho^3 \\
& - 254175232m_i^{18}m_j^9\rho^4 - 4116480m_i^{18}m_j^8\rho^5 - 600178688m_i^{17}m_j^{14} \\
& - 2163920896m_i^{17}m_j^{13}\rho - 3165307904m_i^{17}m_j^{12}\rho^2 + 1827774464m_i^{17}m_j^{11}\rho^3 \\
& - 437981184m_i^{17}m_j^{10}\rho^4 - 112750592m_i^{17}m_j^9\rho^5 + 217333760m_i^{16}m_j^{15} \\
& - 2917320704m_i^{16}m_j^{14}\rho - 2665839616m_i^{16}m_j^{13}\rho^2 + 600210432m_i^{16}m_j^{12}\rho^3 \\
& + 452832256m_i^{16}m_j^{11}\rho^4 - 379700224m_i^{16}m_j^{10}\rho^5 + 26181632m_i^{15}m_j^{16} \\
& - 3717570560m_i^{15}m_j^{15}\rho - 603148288m_i^{15}m_j^{14}\rho^2 + 155648000m_i^{15}m_j^{13}\rho^3
\end{aligned}$$

$$\begin{aligned}
& + 1171801088m_i^{15}m_j^{12}\rho^4 - 456198144m_i^{15}m_j^{11}\rho^5 + 2624098304m_i^{14}m_j^{17} \\
& - 4211537920m_i^{14}m_j^{16}\rho - 2155402752m_i^{14}m_j^{15}\rho^2 + 2455964672m_i^{14}m_j^{14}\rho^3 \\
& + 902367232m_i^{14}m_j^{13}\rho^4 - 293358592m_i^{14}m_j^{12}\rho^5 - 1886855168m_i^{13}m_j^{18} \\
& + 2515392512m_i^{13}m_j^{17}\rho - 2147636224m_i^{13}m_j^{16}\rho^2 + 2064713728m_i^{13}m_j^{15}\rho^3 \\
& + 147456000m_i^{13}m_j^{14}\rho^4 - 27955200m_i^{13}m_j^{13}\rho^5 + 598290432m_i^{12}m_j^{19} \\
& - 351979520m_i^{12}m_j^{18}\rho - 2348878848m_i^{12}m_j^{17}\rho^2 + 2464754688m_i^{12}m_j^{16}\rho^3 \\
& - 115847680m_i^{12}m_j^{15}\rho^4 - 196539392m_i^{12}m_j^{14}\rho^5 - 137682944m_i^{11}m_j^{20} \\
& + 155547648m_i^{11}m_j^{19}\rho + 937705472m_i^{11}m_j^{18}\rho^2 - 897005568m_i^{11}m_j^{17}\rho^3 \\
& - 19160064m_i^{11}m_j^{16}\rho^4 + 177556480m_i^{11}m_j^{15}\rho^5 + 7675904m_i^{10}m_j^{20}\rho \\
& - 185719808m_i^{10}m_j^{19}\rho^2 + 251797504m_i^{10}m_j^{18}\rho^3 - 4608000m_i^{10}m_j^{17}\rho^4 \\
& - 23014400m_i^{10}m_j^{16}\rho^5 + 39440384m_i^9m_j^{20}\rho^2 - 55214080m_i^9m_j^{19}\rho^3 \\
& - 1390592m_i^9m_j^{18}\rho^4 + 3579904m_i^9m_j^{17}\rho^5 + 5318656m_i^{22}m_j^8 \\
& - 3376128m_i^{22}m_j^7\rho + 439040m_i^{22}m_j^6\rho^2 - 14160896m_i^{21}m_j^9 \\
& + 29453312m_i^{21}m_j^8\rho - 13625856m_i^{21}m_j^7\rho^2 + 1756160m_i^{21}m_j^6\rho^3 \\
& + 123367424m_i^{20}m_j^{10} - 147671296m_i^{20}m_j^9\rho + 66206208m_i^{20}m_j^8\rho^2 \\
& - 16258560m_i^{20}m_j^7\rho^3 + 2195200m_i^{20}m_j^6\rho^4 + 171786240m_i^{19}m_j^{11} \\
& + 355213312m_i^{19}m_j^{10}\rho - 390242560m_i^{19}m_j^9\rho^2 + 105162240m_i^{19}m_j^8\rho^3 \\
& - 5144064m_i^{19}m_j^7\rho^4 + 878080m_i^{19}m_j^6\rho^5 + 345671680m_i^{18}m_j^{12} \\
& + 1305029888m_i^{18}m_j^{11}\rho - 166457856m_i^{18}m_j^{10}\rho^2 - 280577024m_i^{18}m_j^9\rho^3 \\
& + 110891520m_i^{18}m_j^8\rho^4 + 864768m_i^{18}m_j^7\rho^5 + 184985600m_i^{17}m_j^{13} \\
& + 557170688m_i^{17}m_j^{12}\rho + 2143353600m_i^{17}m_j^{11}\rho^2 - 987210752m_i^{17}m_j^{10}\rho^3 \\
& + 143455232m_i^{17}m_j^9\rho^4 + 47800832m_i^{17}m_j^8\rho^5 + 534386688m_i^{16}m_j^{14} \\
& + 189714944m_i^{16}m_j^{13}\rho + 1636125696m_i^{16}m_j^{12}\rho^2 - 89073152m_i^{16}m_j^{11}\rho^3 \\
& - 382930944m_i^{16}m_j^{10}\rho^4 + 159764992m_i^{16}m_j^9\rho^5 - 385671168m_i^{15}m_j^{15} \\
& + 2277374976m_i^{15}m_j^{14}\rho - 59276800m_i^{15}m_j^{13}\rho^2 + 422492160m_i^{15}m_j^{12}\rho^3 \\
& - 918729216m_i^{15}m_j^{11}\rho^4 + 187663360m_i^{15}m_j^{10}\rho^5 + 374583296m_i^{14}m_j^{16} \\
& + 2390981120m_i^{14}m_j^{15}\rho + 524534016m_i^{14}m_j^{14}\rho^2 - 735034880m_i^{14}m_j^{13}\rho^3 \\
& - 647451136m_i^{14}m_j^{12}\rho^4 + 33009152m_i^{14}m_j^{11}\rho^5 - 2150315008m_i^{13}m_j^{17} \\
& + 3190092800m_i^{13}m_j^{16}\rho + 2235600384m_i^{13}m_j^{15}\rho^2 - 2161319936m_i^{13}m_j^{14}\rho^3 \\
& - 218992640m_i^{13}m_j^{13}\rho^4 - 23683072m_i^{13}m_j^{12}\rho^5 + 1430927360m_i^{12}m_j^{18} \\
& - 2166188288m_i^{12}m_j^{17}\rho + 1027597824m_i^{12}m_j^{16}\rho^2 - 640752640m_i^{12}m_j^{15}\rho^3 \\
& - 52294912m_i^{12}m_j^{14}\rho^4 - 73550336m_i^{12}m_j^{13}\rho^5 - 347734016m_i^{11}m_j^{19} \\
& + 365919232m_i^{11}m_j^{18}\rho + 1096735488m_i^{11}m_j^{17}\rho^2 - 1287990784m_i^{11}m_j^{16}\rho^3 \\
& + 119624704m_i^{11}m_j^{15}\rho^4 + 131145216m_i^{11}m_j^{14}\rho^5 + 68354048m_i^{10}m_j^{20} \\
& - 106251008m_i^{10}m_j^{19}\rho - 563380736m_i^{10}m_j^{18}\rho^2 + 707230208m_i^{10}m_j^{17}\rho^3 \\
& + 3609600m_i^{10}m_j^{16}\rho^4 - 77252608m_i^{10}m_j^{15}\rho^5 + 6305792m_i^9m_j^{20}\rho \\
& + 66888960m_i^9m_j^{19}\rho^2 - 125177856m_i^9m_j^{18}\rho^3 + 1783808m_i^9m_j^{17}\rho^4 \\
& + 8318464m_i^9m_j^{16}\rho^5 - 17144832m_i^8m_j^{20}\rho^2 + 27199488m_i^8m_j^{19}\rho^3 \\
& + 356864m_i^8m_j^{18}\rho^4 - 899072m_i^8m_j^{17}\rho^5 - 1688064m_i^{22}m_j^7 \\
& + 878080m_i^{22}m_j^6\rho - 89600m_i^{22}m_j^5\rho^2 + 5868544m_i^{21}m_j^8 \\
& - 9467904m_i^{21}m_j^7\rho + 3553792m_i^{21}m_j^6\rho^2 - 358400m_i^{21}m_j^5\rho^3 \\
& - 60975616m_i^{20}m_j^9 + 63651712m_i^{20}m_j^8\rho - 22813440m_i^{20}m_j^7\rho^2 \\
& + 4334080m_i^{20}m_j^6\rho^3 - 448000m_i^{20}m_j^5\rho^4 - 48004096m_i^{19}m_j^{10} \\
& - 204992256m_i^{19}m_j^9\rho + 174936320m_i^{19}m_j^8\rho^2 - 37995520m_i^{19}m_j^7\rho^3
\end{aligned}$$

$$\begin{aligned}
& + 1519104m_i^{19}m_j^6\rho^4 - 179200m_i^{19}m_j^5\rho^5 - 57440768m_i^{18}m_j^{11} \\
& - 554485632m_i^{18}m_j^{10}\rho - 19811840m_i^{18}m_j^9\rho^2 + 144274176m_i^{18}m_j^8\rho^3 \\
& - 39454976m_i^{18}m_j^7\rho^4 - 139264m_i^{18}m_j^6\rho^5 - 5188608m_i^{17}m_j^{12} \\
& + 212683264m_i^{17}m_j^{11}\rho - 1150906496m_i^{17}m_j^{10}\rho^2 + 404928768m_i^{17}m_j^9\rho^3 \\
& - 29076352m_i^{17}m_j^8\rho^4 - 16493056m_i^{17}m_j^7\rho^5 - 220695552m_i^{16}m_j^{13} \\
& + 837495296m_i^{16}m_j^{12}\rho - 834280832m_i^{16}m_j^{11}\rho^2 - 125221632m_i^{16}m_j^{10}\rho^3 \\
& + 225574528m_i^{16}m_j^9\rho^4 - 54098176m_i^{16}m_j^8\rho^5 - 466723840m_i^{15}m_j^{14} \\
& - 227190272m_i^{15}m_j^{13}\rho + 690987520m_i^{15}m_j^{12}\rho^2 - 671625216m_i^{15}m_j^{11}\rho^3 \\
& + 484487168m_i^{15}m_j^{10}\rho^4 - 50004480m_i^{15}m_j^9\rho^5 + 415126016m_i^{14}m_j^{15} \\
& - 1781639936m_i^{14}m_j^{14}\rho - 17223680m_i^{14}m_j^{13}\rho^2 + 226067456m_i^{14}m_j^{12}\rho^3 \\
& + 321901824m_i^{14}m_j^{11}\rho^4 + 34980864m_i^{14}m_j^{10}\rho^5 - 322890752m_i^{13}m_j^{16} \\
& - 861662208m_i^{13}m_j^{15}\rho - 1234849792m_i^{13}m_j^{14}\rho^2 + 680963584m_i^{13}m_j^{13}\rho^3 \\
& + 11382784m_i^{13}m_j^{12}\rho^4 + 114426368m_i^{13}m_j^{11}\rho^5 + 1517815296m_i^{12}m_j^{17} \\
& - 2193759872m_i^{12}m_j^{16}\rho - 907436032m_i^{12}m_j^{15}\rho^2 + 639376896m_i^{12}m_j^{14}\rho^3 \\
& - 26103296m_i^{12}m_j^{13}\rho^4 + 45707776m_i^{12}m_j^{12}\rho^5 - 883705856m_i^{11}m_j^{18} \\
& + 1390021888m_i^{11}m_j^{17}\rho - 95269888m_i^{11}m_j^{16}\rho^2 - 240004096m_i^{11}m_j^{15}\rho^3 \\
& + 62977792m_i^{11}m_j^{14}\rho^4 + 57089536m_i^{11}m_j^{13}\rho^5 + 173410816m_i^{10}m_j^{19} \\
& - 285365120m_i^{10}m_j^{18}\rho - 418420736m_i^{10}m_j^{17}\rho^2 + 649992448m_i^{10}m_j^{16}\rho^3 \\
& - 63896576m_i^{10}m_j^{15}\rho^4 - 74499072m_i^{10}m_j^{14}\rho^5 - 27819008m_i^9m_j^{20} \\
& + 56377856m_i^9m_j^{19}\rho + 282472064m_i^9m_j^{18}\rho^2 - 427843328m_i^9m_j^{17}\rho^3 \\
& + 887680m_i^9m_j^{16}\rho^4 + 25985024m_i^9m_j^{15}\rho^5 - 5788416m_i^8m_j^{20}\rho \\
& - 22488960m_i^8m_j^{19}\rho^2 + 57629440m_i^8m_j^{18}\rho^3 - 459392m_i^8m_j^{17}\rho^4 \\
& - 2394368m_i^8m_j^{16}\rho^5 + 6128896m_i^7m_j^{20}\rho^2 - 10465280m_i^7m_j^{19}\rho^3 \\
& - 68864m_i^7m_j^{18}\rho^4 + 169984m_i^7m_j^{17}\rho^5 + 439040m_i^{22}m_j^6 \\
& - 179200m_i^{22}m_j^5\rho + 14336m_i^{22}m_j^4\rho^2 - 2012800m_i^{21}m_j^7 \\
& + 2458496m_i^{21}m_j^6\rho - 730944m_i^{21}m_j^5\rho^2 + 57344m_i^{21}m_j^4\rho^3 \\
& + 25433856m_i^{20}m_j^8\rho - 22417472m_i^{20}m_j^7\rho + 6311680m_i^{20}m_j^6\rho^2 \\
& - 918528m_i^{20}m_j^5\rho^3 + 71680m_i^{20}m_j^4\rho^4 - 464384m_i^{19}m_j^9 \\
& + 96224128m_i^{19}m_j^8\rho - 63714304m_i^{19}m_j^7\rho^2 + 11125248m_i^{19}m_j^6\rho^3 \\
& - 361024m_i^{19}m_j^5\rho^4 + 28672m_i^{19}m_j^4\rho^5 - 22977536m_i^{18}m_j^{10} \\
& + 176560256m_i^{18}m_j^9\rho + 49144768m_i^{18}m_j^8\rho^2 - 58270848m_i^{18}m_j^7\rho^3 \\
& + 11573632m_i^{18}m_j^6\rho^4 + 5760m_i^{18}m_j^5\rho^5 - 91286656m_i^{17}m_j^{11} \\
& - 309647744m_i^{17}m_j^{10}\rho + 501220160m_i^{17}m_j^9\rho^2 - 126536960m_i^{17}m_j^8\rho^3 \\
& - 151296m_i^{17}m_j^7\rho^4 + 4740608m_i^{17}m_j^6\rho^5 + 176682496m_i^{16}m_j^{12} \\
& - 846796096m_i^{16}m_j^{11}\rho + 267108416m_i^{16}m_j^{10}\rho^2 + 140117376m_i^{16}m_j^9\rho^3 \\
& - 95379264m_i^{16}m_j^8\rho^4 + 14338432m_i^{16}m_j^7\rho^5 + 241911296m_i^{15}m_j^{13} \\
& - 127176192m_i^{15}m_j^{12}\rho - 654916864m_i^{15}m_j^{11}\rho^2 + 443301248m_i^{15}m_j^{10}\rho^3 \\
& - 192766912m_i^{15}m_j^9\rho^4 + 8372352m_i^{15}m_j^8\rho^5 + 148696832m_i^{14}m_j^{14} \\
& + 572942720m_i^{14}m_j^{13}\rho - 336527744m_i^{14}m_j^{12}\rho^2 + 39489792m_i^{14}m_j^{11}\rho^3 \\
& - 74891840m_i^{14}m_j^{10}\rho^4 - 36184320m_i^{14}m_j^9\rho^5 - 606968704m_i^{13}m_j^{15} \\
& + 686811776m_i^{13}m_j^{14}\rho + 703134912m_i^{13}m_j^{13}\rho^2 - 426245632m_i^{13}m_j^{12}\rho^3 \\
& + 91624704m_i^{13}m_j^{11}\rho^4 - 67390976m_i^{13}m_j^{10}\rho^5 + 354736384m_i^{12}m_j^{16} \\
& - 180187968m_i^{12}m_j^{15}\rho + 355977728m_i^{12}m_j^{14}\rho^2 + 116809984m_i^{12}m_j^{13}\rho^3 \\
& + 90504704m_i^{12}m_j^{12}\rho^4 - 69276160m_i^{12}m_j^{11}\rho^5 - 962526208m_i^{11}m_j^{17}
\end{aligned}$$

$$\begin{aligned}
& + 1482248576m_i^{11}m_j^{16}\rho - 18938240m_i^{11}m_j^{15}\rho^2 + 88132096m_i^{11}m_j^{14}\rho^3 \\
& + 49035584m_i^{11}m_j^{13}\rho^4 - 11678720m_i^{11}m_j^{12}\rho^5 + 450790912m_i^{10}m_j^{18} \\
& - 773563904m_i^{10}m_j^{17}\rho - 113633216m_i^{10}m_j^{16}\rho^2 + 362907264m_i^{10}m_j^{15}\rho^3 \\
& - 60519552m_i^{10}m_j^{14}\rho^4 - 29127296m_i^{10}m_j^{13}\rho^5 - 73526656m_i^9m_j^{19} \\
& + 179901312m_i^9m_j^{18}\rho + 118000448m_i^9m_j^{17}\rho^2 - 344574720m_i^9m_j^{16}\rho^3 \\
& + 23299584m_i^9m_j^{15}\rho^4 + 32527872m_i^9m_j^{14}\rho^5 + 9293312m_i^8m_j^{20} \\
& - 26113856m_i^8m_j^{19}\rho - 109165888m_i^8m_j^{18}\rho^2 + 199859072m_i^8m_j^{17}\rho^3 \\
& - 829632m_i^8m_j^{16}\rho^4 - 6776704m_i^8m_j^{15}\rho^5 + 2883840m_i^7m_j^{20}\rho \\
& + 6656640m_i^7m_j^{19}\rho^2 - 22893440m_i^7m_j^{18}\rho^3 + 67776m_i^7m_j^{17}\rho^4 \\
& + 535936m_i^7m_j^{16}\rho^5 - 1759872m_i^6m_j^{20}\rho^2 + 3161856m_i^6m_j^{19}\rho^3 \\
& + 10048m_i^6m_j^{18}\rho^4 - 23296m_i^6m_j^{17}\rho^5 - 89600m_i^5m_j^{22} \\
& + 28672m_i^{22}m_j^4\rho - 1568m_i^{22}m_j^3\rho^2 + 535488m_i^{21}m_j^6 \\
& - 508800m_i^{21}m_j^5\rho + 116864m_i^{21}m_j^4\rho^2 - 6272m_i^{21}m_j^3\rho^3 \\
& - 8723648m_i^{20}m_j^7 + 6405120m_i^{20}m_j^6\rho - 1414080m_i^{20}m_j^5\rho^2 \\
& + 148480m_i^{20}m_j^4\rho^3 - 7840m_i^{20}m_j^3\rho^4 + 7897472m_i^{19}m_j^8 \\
& - 36500096m_i^{19}m_j^7\rho + 19000928m_i^{19}m_j^6\rho^2 - 2645376m_i^{19}m_j^5\rho^3 \\
& + 61056m_i^{19}m_j^4\rho^4 - 3136m_i^{19}m_j^3\rho^5 + 26436352m_i^{18}m_j^9 \\
& - 37657984m_i^{18}m_j^8\rho - 30554432m_i^{18}m_j^7\rho^2 + 18895168m_i^{18}m_j^6\rho^3 \\
& - 2757824m_i^{18}m_j^5\rho^4 + 768m_i^{18}m_j^4\rho^5 + 60357184m_i^{17}m_j^{10} \\
& + 192630784m_i^{17}m_j^9\rho - 170471072m_i^{17}m_j^8\rho^2 + 27922944m_i^{17}m_j^7\rho^3 \\
& + 2684576m_i^{17}m_j^6\rho^4 - 1107328m_i^{17}m_j^5\rho^5 - 87188544m_i^{16}m_j^{11} \\
& + 456516544m_i^{16}m_j^{10}\rho - 43865312m_i^{16}m_j^9\rho^2 - 76268608m_i^{16}m_j^8\rho^3 \\
& + 30625280m_i^{16}m_j^7\rho^4 - 2996352m_i^{16}m_j^6\rho^5 - 227252352m_i^{15}m_j^{12} \\
& + 79009280m_i^{15}m_j^{11}\rho + 433665920m_i^{15}m_j^{10}\rho^2 - 208419328m_i^{15}m_j^9\rho^3 \\
& + 55366048m_i^{15}m_j^8\rho^4 + 122112m_i^{15}m_j^7\rho^5 + 123207168m_i^{14}m_j^{13} \\
& - 495234688m_i^{14}m_j^{12}\rho + 253044576m_i^{14}m_j^{11}\rho^2 + 8931200m_i^{14}m_j^{10}\rho^3 \\
& - 3565792m_i^{14}m_j^9\rho^4 + 15722624m_i^{14}m_j^8\rho^5 + 172941120m_i^{13}m_j^{14} \\
& - 62093952m_i^{13}m_j^{13}\rho - 304832640m_i^{13}m_j^{12}\rho^2 + 186795136m_i^{13}m_j^{11}\rho^3 \\
& - 88970496m_i^{13}m_j^{10}\rho^4 + 31528256m_i^{13}m_j^9\rho^5 + 349233856m_i^{12}m_j^{15} \\
& + 52366016m_i^{12}m_j^{14}\rho - 92212992m_i^{12}m_j^{13}\rho^2 + 18916480m_i^{12}m_j^{12}\rho^3 \\
& - 91529568m_i^{12}m_j^{11}\rho^4 + 24777472m_i^{12}m_j^{10}\rho^5 - 304342144m_i^{11}m_j^{16} \\
& + 430750720m_i^{11}m_j^{15}\rho + 164660320m_i^{11}m_j^{14}\rho^2 - 286936192m_i^{11}m_j^{13}\rho^3 \\
& - 38147968m_i^{11}m_j^{12}\rho^4 + 23462464m_i^{11}m_j^{11}\rho^5 + 531362048m_i^{10}m_j^{17} \\
& - 861657408m_i^{10}m_j^{16}\rho + 169835392m_i^{10}m_j^{15}\rho^2 - 139406784m_i^{10}m_j^{14}\rho^3 \\
& - 28173248m_i^{10}m_j^{13}\rho^4 - 2158080m_i^{10}m_j^{12}\rho^5 - 192251200m_i^9m_j^{18} \\
& + 372877568m_i^9m_j^{17}\rho + 74087264m_i^9m_j^{16}\rho^2 - 233287936m_i^9m_j^{15}\rho^3 \\
& + 35257824m_i^9m_j^{14}\rho^4 + 12322944m_i^9m_j^{13}\rho^5 + 26658368m_i^8m_j^{19} \\
& - 88031104m_i^8m_j^{18}\rho - 35172448m_i^8m_j^{17}\rho^2 + 185695680m_i^8m_j^{16}\rho^3 \\
& - 6297088m_i^8m_j^{15}\rho^4 - 10574464m_i^8m_j^{14}\rho^5 - 2558080m_i^7m_j^{20} \\
& + 10206592m_i^7m_j^{19}\rho + 34379200m_i^7m_j^{18}\rho^2 - 75312640m_i^7m_j^{17}\rho^3 \\
& + 326496m_i^7m_j^{16}\rho^4 + 1328384m_i^7m_j^{15}\rho^5 - 1019328m_i^6m_j^{20}\rho \\
& - 1933184m_i^6m_j^{19}\rho^2 + 7740160m_i^6m_j^{18}\rho^3 - 5792m_i^6m_j^{17}\rho^4 \\
& - 87424m_i^6m_j^{16}\rho^5 + 405312m_i^5m_j^{20}\rho^2 - 750336m_i^5m_j^{19}\rho^3 \\
& - 896m_i^5m_j^{18}\rho^4 + 1984m_i^5m_j^{17}\rho^5 + 14336m_i^{22}m_j^4
\end{aligned}$$

$$\begin{aligned}
& -3136m_i^{22}m_j^3\rho + 112m_i^{22}m_j^2\rho^2 - 114816m_i^{21}m_j^5 \\
& + 80128m_i^{21}m_j^4\rho - 12864m_i^{21}m_j^3\rho^2 + 448m_i^{21}m_j^2\rho^3 \\
& + 2491264m_i^{20}m_j^6 - 1481648m_i^{20}m_j^5\rho + 241600m_i^{20}m_j^4\rho^2 \\
& - 16736m_i^{20}m_j^3\rho^3 + 560m_i^{20}m_j^2\rho^4 - 4934848m_i^{19}m_j^7 \\
& + 11259072m_i^{19}m_j^6\rho - 4573872m_i^{19}m_j^5\rho^2 + 490080m_i^{19}m_j^4\rho^3 \\
& - 7424m_i^{19}m_j^3\rho^4 + 224m_i^{19}m_j^2\rho^5 - 12048448m_i^{18}m_j^8 \\
& + 2088304m_i^{18}m_j^7\rho + 12258048m_i^{18}m_j^6\rho^2 - 4858368m_i^{18}m_j^5\rho^3 \\
& + 522624m_i^{18}m_j^4\rho^4 - 416m_i^{18}m_j^3\rho^5 - 27588160m_i^{17}m_j^9 \\
& - 80120192m_i^{17}m_j^8\rho + 44732912m_i^{17}m_j^7\rho^2 - 3438240m_i^{17}m_j^6\rho^3 \\
& - 1172704m_i^{17}m_j^5\rho^4 + 208352m_i^{17}m_j^4\rho^5 + 56169856m_i^{16}m_j^{10} \\
& - 175627568m_i^{16}m_j^9\rho - 11000192m_i^{16}m_j^8\rho^2 + 28471264m_i^{16}m_j^7\rho^3 \\
& - 7342944m_i^{16}m_j^6\rho^4 + 467744m_i^{16}m_j^5\rho^5 + 107280832m_i^{15}m_j^{11} \\
& + 39719584m_i^{15}m_j^{10}\rho - 197533488m_i^{15}m_j^9\rho^2 + 68095712m_i^{15}m_j^8\rho^3 \\
& - 11757696m_i^{15}m_j^7\rho^4 - 441760m_i^{15}m_j^6\rho^5 - 78241856m_i^{14}m_j^{12} \\
& + 335516880m_i^{14}m_j^{11}\rho - 122340480m_i^{14}m_j^{10}\rho^2 - 18831552m_i^{14}m_j^9\rho^3 \\
& + 12452128m_i^{14}m_j^8\rho^4 - 4949216m_i^{14}m_j^7\rho^5 - 318997760m_i^{13}m_j^{13} \\
& + 195143712m_i^{13}m_j^{12}\rho + 145081136m_i^{13}m_j^{11}\rho^2 - 121233632m_i^{13}m_j^{10}\rho^3 \\
& + 44090784m_i^{13}m_j^9\rho^4 - 8797280m_i^{13}m_j^8\rho^5 + 3681152m_i^{12}m_j^{14} \\
& - 279279504m_i^{12}m_j^{13}\rho + 21405280m_i^{12}m_j^{12}\rho^2 - 3340000m_i^{12}m_j^{11}\rho^3 \\
& + 44944736m_i^{12}m_j^{10}\rho^4 - 9460128m_i^{12}m_j^9\rho^5 - 128215104m_i^{11}m_j^{15} \\
& - 138807616m_i^{11}m_j^{14}\rho - 169309584m_i^{11}m_j^{13}\rho^2 + 68123424m_i^{11}m_j^{12}\rho^3 \\
& + 33544320m_i^{11}m_j^{11}\rho^4 - 3441504m_i^{11}m_j^{10}\rho^5 + 207293376m_i^{10}m_j^{16} \\
& - 373813360m_i^{10}m_j^{15}\rho - 141137984m_i^{10}m_j^{14}\rho^2 + 148023424m_i^{10}m_j^{13}\rho^3 \\
& + 4457536m_i^{10}m_j^{12}\rho^4 - 6154336m_i^{10}m_j^{11}\rho^5 - 244104768m_i^9m_j^{17} \\
& + 434845824m_i^9m_j^{16}\rho - 116718384m_i^9m_j^{15}\rho^2 + 55237280m_i^9m_j^{14}\rho^3 \\
& + 13401568m_i^9m_j^{13}\rho^4 + 2585120m_i^9m_j^{12}\rho^5 + 66857600m_i^8m_j^{18} \\
& - 159928464m_i^8m_j^{17}\rho + 1951552m_i^8m_j^{16}\rho^2 + 105514912m_i^8m_j^{15}\rho^3 \\
& - 13417056m_i^8m_j^{14}\rho^4 - 4305184m_i^8m_j^{13}\rho^5 - 7962304m_i^7m_j^{19} \\
& + 34285280m_i^7m_j^{18}\rho + 9796464m_i^7m_j^{17}\rho^2 - 84882528m_i^7m_j^{16}\rho^3 \\
& + 1186048m_i^7m_j^{15}\rho^4 + 2543136m_i^7m_j^{14}\rho^5 + 559552m_i^6m_j^{20} \\
& - 3327120m_i^6m_j^{19}\rho - 8271024m_i^6m_j^{18}\rho^2 + 22641728m_i^6m_j^{17}\rho^3 \\
& - 69856m_i^6m_j^{16}\rho^4 - 188704m_i^6m_j^{15}\rho^5 + 266976m_i^5m_j^{20}\rho \\
& + 509008m_i^5m_j^{19}\rho^2 - 2139360m_i^5m_j^{18}\rho^3 - 672m_i^5m_j^{17}\rho^4 \\
& + 9824m_i^5m_j^{16}\rho^5 - 73184m_i^4m_j^{20}\rho^2 + 136896m_i^4m_j^{19}\rho^3 \\
& + 48m_i^4m_j^{18}\rho^4 - 96m_i^4m_j^{17}\rho^5 - 1568m_i^{22}m_j^3 \\
& + 224m_i^{22}m_j^2\rho + 17824m_i^{21}m_j^4 - 8832m_i^{21}m_j^3\rho \\
& + 896m_i^{21}m_j^2\rho^2 - 578240m_i^{20}m_j^5 + 267544m_i^{20}m_j^4\rho \\
& - 30784m_i^{20}m_j^3\rho^2 + 1120m_i^{20}m_j^2\rho^3 + 1895968m_i^{19}m_j^6 \\
& - 2780240m_i^{19}m_j^5\rho + 869928m_i^{19}m_j^4\rho^2 - 69568m_i^{19}m_j^3\rho^3 \\
& + 448m_i^{19}m_j^2\rho^4 + 3705504m_i^{18}m_j^7 + 2121048m_i^{18}m_j^6\rho \\
& - 3582168m_i^{18}m_j^5\rho^2 + 979312m_i^{18}m_j^4\rho^3 - 76192m_i^{18}m_j^3\rho^4 \\
& + 8388256m_i^{17}m_j^8 + 24562432m_i^{17}m_j^7\rho - 8452296m_i^{17}m_j^6\rho^2 \\
& - 214368m_i^{17}m_j^5\rho^3 + 307264m_i^{17}m_j^4\rho^4 - 30144m_i^{17}m_j^3\rho^5 \\
& - 25999200m_i^{16}m_j^9 + 48510056m_i^{16}m_j^8\rho + 9465816m_i^{16}m_j^7\rho^2
\end{aligned}$$

$$\begin{aligned}
& -7553328m_i^{16}m_j^6\rho^3 + 1296784m_i^{16}m_j^5\rho^4 - 50880m_i^{16}m_j^4\rho^5 \\
& -49575776m_i^{15}m_j^{10} - 46309520m_i^{15}m_j^9\rho + 68567272m_i^{15}m_j^8\rho^2 \\
& -16734208m_i^{15}m_j^7\rho^3 + 1632880m_i^{15}m_j^6\rho^4 + 134432m_i^{15}m_j^5\rho^5 \\
& + 78530336m_i^{14}m_j^{11} - 200317528m_i^{14}m_j^{10}\rho + 34208200m_i^{14}m_j^9\rho^2 \\
& + 16020528m_i^{14}m_j^8\rho^3 - 5501712m_i^{14}m_j^7\rho^4 + 1042112m_i^{14}m_j^6\rho^5 \\
& + 179004096m_i^{13}m_j^{12} - 123812000m_i^{13}m_j^{11}\rho - 69528936m_i^{13}m_j^{10}\rho^2 \\
& + 48579360m_i^{13}m_j^9\rho^3 - 14625488m_i^{13}m_j^8\rho^4 + 2009312m_i^{13}m_j^7\rho^5 \\
& + 99198208m_i^{12}m_j^{13} + 52415592m_i^{12}m_j^{12}\rho - 9122632m_i^{12}m_j^{11}\rho^2 \\
& + 21619984m_i^{12}m_j^{10}\rho^3 - 15671456m_i^{12}m_j^9\rho^4 + 1598592m_i^{12}m_j^8\rho^5 \\
& - 88991968m_i^{11}m_j^{14} + 198091152m_i^{11}m_j^{13}\rho + 72300312m_i^{11}m_j^{12}\rho^2 \\
& - 28536640m_i^{11}m_j^{11}\rho^3 - 8749664m_i^{11}m_j^{10}\rho^4 + 1598784m_i^{11}m_j^9\rho^5 \\
& + 31729632m_i^{10}m_j^{15} + 69136616m_i^{10}m_j^{14}\rho + 96860856m_i^{10}m_j^{13}\rho^2 \\
& - 24993200m_i^{10}m_j^{12}\rho^3 - 8491072m_i^{10}m_j^{11}\rho^4 - 510592m_i^{10}m_j^{10}\rho^5 \\
& - 120498336m_i^9m_j^{16} + 221048128m_i^9m_j^{15}\rho + 60487144m_i^9m_j^{14}\rho^2 \\
& - 47251104m_i^9m_j^{13}\rho^3 + 2275360m_i^9m_j^{12}\rho^4 + 1527808m_i^9m_j^{11}\rho^5 \\
& + 95654944m_i^8m_j^{17} - 186233256m_i^8m_j^{16}\rho + 41612360m_i^8m_j^{15}\rho^2 \\
& - 3711504m_i^8m_j^{14}\rho^3 - 5614896m_i^8m_j^{13}\rho^4 - 960448m_i^8m_j^{12}\rho^5 \\
& - 19505632m_i^7m_j^{18} + 60183632m_i^7m_j^{17}\rho - 16331848m_i^7m_j^{16}\rho^2 \\
& - 42974976m_i^7m_j^{15}\rho^3 + 3624496m_i^7m_j^{14}\rho^4 + 1108384m_i^7m_j^{13}\rho^5 \\
& + 2011520m_i^6m_j^{19} - 10820552m_i^6m_j^{18}\rho - 3305896m_i^6m_j^{17}\rho^2 \\
& + 31563280m_i^6m_j^{16}\rho^3 - 159504m_i^6m_j^{15}\rho^4 - 430144m_i^6m_j^{14}\rho^5 \\
& - 97888m_i^5m_j^{20} + 917984m_i^5m_j^{19}\rho + 1482440m_i^5m_j^{18}\rho^2 \\
& - 5410080m_i^5m_j^{17}\rho^3 + 10864m_i^5m_j^{16}\rho^4 + 16608m_i^5m_j^{15}\rho^5 \\
& - 54592m_i^4m_j^{20}\rho - 115352m_i^4m_j^{19}\rho^2 + 469968m_i^4m_j^{18}\rho^3 \\
& + 64m_i^4m_j^{17}\rho^4 - 512m_i^4m_j^{16}\rho^5 + 9248m_i^3m_j^{20}\rho^2 \\
& - 17408m_i^3m_j^{19}\rho^3 + 112m_i^{22}m_j^2 - 1992m_i^{21}m_j^3 \\
& + 568m_i^{21}m_j^2\rho + 107344m_i^{20}m_j^4 - 36884m_i^{20}m_j^3\rho \\
& + 2456m_i^{20}m_j^2\rho^2 - 530640m_i^{19}m_j^5 + 540824m_i^{19}m_j^4\rho \\
& - 126288m_i^{19}m_j^3\rho^2 + 6608m_i^{19}m_j^2\rho^3 - 764000m_i^{18}m_j^6 \\
& - 1035416m_i^{18}m_j^5\rho + 782688m_i^{18}m_j^4\rho^2 - 148608m_i^{18}m_j^3\rho^3 \\
& + 7680m_i^{18}m_j^2\rho^4 - 2048752m_i^{17}m_j^7 - 5566192m_i^{17}m_j^6\rho \\
& + 978232m_i^{17}m_j^5\rho^2 + 215456m_i^{17}m_j^4\rho^3 - 53936m_i^{17}m_j^3\rho^4 \\
& + 3072m_i^{17}m_j^2\rho^5 + 10196480m_i^{16}m_j^8 - 9874896m_i^{16}m_j^7\rho \\
& - 3368160m_i^{16}m_j^6\rho^2 + 1422112m_i^{16}m_j^5\rho^3 - 153264m_i^{16}m_j^4\rho^4 \\
& + 3232m_i^{16}m_j^3\rho^5 + 14813744m_i^{15}m_j^9 + 24893776m_i^{15}m_j^8\rho \\
& - 17598856m_i^{15}m_j^7\rho^2 + 2881888m_i^{15}m_j^6\rho^3 - 134720m_i^{15}m_j^5\rho^4 \\
& - 20032m_i^{15}m_j^4\rho^5 - 38341472m_i^{14}m_j^{10} + 79915448m_i^{14}m_j^9\rho \\
& - 4792368m_i^{14}m_j^8\rho^2 - 6161632m_i^{14}m_j^7\rho^3 + 1515680m_i^{14}m_j^6\rho^4 \\
& - 166144m_i^{14}m_j^5\rho^5 - 102887840m_i^{13}m_j^{11} + 67059728m_i^{13}m_j^{10}\rho \\
& + 27748760m_i^{13}m_j^9\rho^2 - 16484832m_i^{13}m_j^8\rho^3 + 3326416m_i^{13}m_j^7\rho^4 \\
& - 265472m_i^{13}m_j^6\rho^5 - 30667904m_i^{12}m_j^{12} - 24180872m_i^{12}m_j^{11}\rho \\
& + 11424912m_i^{12}m_j^{10}\rho^2 - 7000288m_i^{12}m_j^9\rho^3 + 3401328m_i^{12}m_j^8\rho^4 \\
& - 333152m_i^{12}m_j^7\rho^5 - 1799856m_i^{11}m_j^{13} - 39436912m_i^{11}m_j^{12}\rho \\
& - 28641160m_i^{11}m_j^{11}\rho^2 + 3387648m_i^{11}m_j^{10}\rho^3 + 2516736m_i^{11}m_j^9\rho^4
\end{aligned}$$

$$\begin{aligned}
& + 24448m_i^{11}m_j^8\rho^5 + 64519840m_i^{10}m_j^{14} - 83796008m_i^{10}m_j^{13}\rho \\
& - 25908208m_i^{10}m_j^{12}\rho^2 + 8453344m_i^{10}m_j^{11}\rho^3 + 77376m_i^{10}m_j^{10}\rho^4 \\
& - 270080m_i^{10}m_j^9\rho^5 + 2468208m_i^9m_j^{15} - 10080496m_i^9m_j^{14}\rho \\
& - 37226712m_i^9m_j^{13}\rho^2 + 4339936m_i^9m_j^{12}\rho^3 + 1916656m_i^9m_j^{11}\rho^4 \\
& + 399616m_i^9m_j^{10}\rho^5 + 51346752m_i^8m_j^{16} - 101804592m_i^8m_j^{15}\rho \\
& - 5833696m_i^8m_j^{14}\rho^2 + 7754336m_i^8m_j^{13}\rho^3 - 1162576m_i^8m_j^{12}\rho^4 \\
& - 398368m_i^8m_j^{11}\rho^5 - 29831088m_i^7m_j^{17} + 66442960m_i^7m_j^{16}\rho \\
& - 17401368m_i^7m_j^{15}\rho^2 - 3381344m_i^7m_j^{14}\rho^3 + 1631040m_i^7m_j^{13}\rho^4 \\
& + 242368m_i^7m_j^{12}\rho^5 + 4453552m_i^6m_j^{18} - 18876920m_i^6m_j^{17}\rho \\
& + 10783728m_i^6m_j^{16}\rho^2 + 14999264m_i^6m_j^{15}\rho^3 - 658656m_i^6m_j^{14}\rho^4 \\
& - 200704m_i^6m_j^{13}\rho^5 - 395288m_i^5m_j^{19} + 2640536m_i^5m_j^{18}\rho \\
& + 1023496m_i^5m_j^{17}\rho^2 - 9180832m_i^5m_j^{16}\rho^3 + 9712m_i^5m_j^{15}\rho^4 \\
& + 48128m_i^5m_j^{14}\rho^5 + 11760m_i^4m_j^{20} - 195172m_i^4m_j^{19}\rho \\
& - 171880m_i^4m_j^{18}\rho^2 + 996192m_i^4m_j^{17}\rho^3 - 624m_i^4m_j^{16}\rho^4 \\
& - 800m_i^4m_j^{15}\rho^5 + 7352m_i^3m_j^{20}\rho + 23672m_i^3m_j^{19}\rho^2 \\
& - 81360m_i^3m_j^{18}\rho^3 - 816m_i^2m_j^{20}\rho^2 + 1440m_i^2m_j^{19}\rho^3 \\
& + 108m_i^{21}m_j^2 - 15340m_i^{20}m_j^3 + 3412m_i^{20}m_j^2\rho \\
& - 96m_i^{20}m_j\rho^2 + 111152m_i^{19}m_j^4 - 79408m_i^{19}m_j^3\rho \\
& + 12616m_i^{19}m_j^2\rho^2 - 384m_i^{19}m_j\rho^3 + 99352m_i^{18}m_j^5 \\
& + 276768m_i^{18}m_j^4\rho - 123776m_i^{18}m_j^3\rho^2 + 15552m_i^{18}m_j^2\rho^3 \\
& - 480m_i^{18}m_j\rho^4 + 382648m_i^{17}m_j^6 + 898448m_i^{17}m_j^5\rho \\
& - 6680m_i^{17}m_j^4\rho^2 - 50816m_i^{17}m_j^3\rho^3 + 6240m_i^{17}m_j^2\rho^4 \\
& - 192m_i^{17}m_j\rho^5 - 2922736m_i^{16}m_j^7 + 1374640m_i^{16}m_j^6\rho \\
& + 705872m_i^{16}m_j^5\rho^2 - 176832m_i^{16}m_j^4\rho^3 + 10384m_i^{16}m_j^3\rho^4 \\
& - 3524832m_i^{15}m_j^8 - 8231632m_i^{15}m_j^7\rho + 3430440m_i^{15}m_j^6\rho^2 \\
& - 361920m_i^{15}m_j^5\rho^3 - 528m_i^{15}m_j^4\rho^4 + 1568m_i^{15}m_j^3\rho^5 \\
& + 15961512m_i^{14}m_j^9 - 24226104m_i^{14}m_j^8\rho - 663616m_i^{14}m_j^7\rho^2 \\
& + 1743232m_i^{14}m_j^6\rho^3 - 271184m_i^{14}m_j^5\rho^4 + 16128m_i^{14}m_j^4\rho^5 \\
& + 38048624m_i^{13}m_j^{10} - 21108688m_i^{13}m_j^9\rho - 8635144m_i^{13}m_j^8\rho^2 \\
& + 3495712m_i^{13}m_j^7\rho^3 - 495856m_i^{13}m_j^6\rho^4 + 28128m_i^{13}m_j^5\rho^5 \\
& + 16207112m_i^{12}m_j^{11} + 5986936m_i^{12}m_j^{10}\rho - 4436256m_i^{12}m_j^9\rho^2 \\
& + 2433152m_i^{12}m_j^8\rho^3 - 615648m_i^{12}m_j^7\rho^4 + 19200m_i^{12}m_j^6\rho^5 \\
& - 13005120m_i^{11}m_j^{12} + 19442192m_i^{11}m_j^{11}\rho + 5577032m_i^{11}m_j^{10}\rho^2 \\
& - 1602400m_i^{11}m_j^9\rho^3 - 65888m_i^{11}m_j^8\rho^4 + 25536m_i^{11}m_j^7\rho^5 \\
& - 2706520m_i^{10}m_j^{13} + 7521960m_i^{10}m_j^{12}\rho + 7975440m_i^{10}m_j^{11}\rho^2 \\
& - 792960m_i^{10}m_j^{10}\rho^3 - 423744m_i^{10}m_j^9\rho^4 - 45312m_i^{10}m_j^8\rho^5 \\
& - 32958008m_i^9m_j^{14} + 24828400m_i^9m_j^{13}\rho + 7385880m_i^9m_j^{12}\rho^2 \\
& - 2084160m_i^9m_j^{11}\rho^3 + 594496m_i^9m_j^{10}\rho^4 + 44160m_i^9m_j^9\rho^5 \\
& - 617040m_i^8m_j^{15} - 7639760m_i^8m_j^{14}\rho + 5310352m_i^8m_j^{13}\rho^2 \\
& + 2001216m_i^8m_j^{12}\rho^3 - 609904m_i^8m_j^{11}\rho^4 - 81152m_i^8m_j^{10}\rho^5 \\
& - 18717728m_i^7m_j^{16} + 38909200m_i^7m_j^{15}\rho - 1286376m_i^7m_j^{14}\rho^2 \\
& - 1881984m_i^7m_j^{13}\rho^3 + 374640m_i^7m_j^{12}\rho^4 + 66080m_i^7m_j^{11}\rho^5 \\
& + 7791000m_i^6m_j^{17} - 20223912m_i^6m_j^{16}\rho + 6899168m_i^6m_j^{15}\rho^2 \\
& + 2199936m_i^6m_j^{14}\rho^3 - 335760m_i^6m_j^{13}\rho^4 - 35328m_i^6m_j^{12}\rho^5
\end{aligned}$$

$$\begin{aligned}
& -834652m_i^5m_j^{18} + 5001296m_i^5m_j^{17}\rho - 4099576m_i^5m_j^{16}\rho^2 \\
& -4351840m_i^5m_j^{15}\rho^3 + 80464m_i^5m_j^{14}\rho^4 + 21088m_i^5m_j^{13}\rho^5 \\
& + 64292m_i^4m_j^{19} - 518828m_i^4m_j^{18}\rho - 264032m_i^4m_j^{17}\rho^2 \\
& + 2051328m_i^4m_j^{16}\rho^3 - 320m_i^4m_j^{15}\rho^4 - 2560m_i^4m_j^{14}\rho^5 \\
& - 1008m_i^3m_j^{20} + 35296m_i^3m_j^{19}\rho - 1488m_i^3m_j^{18}\rho^2 \\
& - 127456m_i^3m_j^{17}\rho^3 - 744m_i^2m_j^{20}\rho - 3120m_i^2m_j^{19}\rho^2 \\
& + 9664m_i^2m_j^{18}\rho^3 + 1528m_i^{20}m_j^2 - 192m_i^{20}m_j\rho \\
& - 16888m_i^{19}m_j^3 + 8064m_i^{19}m_j^2\rho - 768m_i^{19}m_j\rho^2 \\
& - 4072m_i^{18}m_j^4 - 47872m_i^{18}m_j^3\rho + 13296m_i^{18}m_j^2\rho^2 \\
& - 960m_i^{18}m_j\rho^3 - 66112m_i^{17}m_j^5 - 93064m_i^{17}m_j^4\rho \\
& - 19632m_i^{17}m_j^3\rho^2 + 6752m_i^{17}m_j^2\rho^3 - 384m_i^{17}m_j\rho^4 \\
& + 661264m_i^{16}m_j^6 - 125616m_i^{16}m_j^5\rho - 94192m_i^{16}m_j^4\rho^2 \\
& + 11680m_i^{16}m_j^3\rho^3 + 452024m_i^{15}m_j^7 + 1974352m_i^{15}m_j^6\rho \\
& - 481632m_i^{15}m_j^5\rho^2 + 27264m_i^{15}m_j^4\rho^3 + 256m_i^{15}m_j^3\rho^4 \\
& - 4617496m_i^{14}m_j^8 + 5156624m_i^{14}m_j^7\rho + 537984m_i^{14}m_j^6\rho^2 \\
& - 311008m_i^{14}m_j^5\rho^3 + 32832m_i^{14}m_j^4\rho^4 - 1152m_i^{14}m_j^3\rho^5 \\
& - 11680512m_i^{13}m_j^9 + 5311032m_i^{13}m_j^8\rho + 1753296m_i^{13}m_j^7\rho^2 \\
& - 547872m_i^{13}m_j^6\rho^3 + 44416m_i^{13}m_j^5\rho^4 - 2321136m_i^{12}m_j^{10} \\
& - 3158800m_i^{12}m_j^9\rho + 1643792m_i^{12}m_j^8\rho^2 - 434400m_i^{12}m_j^7\rho^3 \\
& + 59968m_i^{12}m_j^6\rho^4 - 4736m_i^{12}m_j^5\rho^5 + 3955240m_i^{11}m_j^{11} \\
& - 5562128m_i^{11}m_j^{10}\rho - 1873152m_i^{11}m_j^9\rho^2 + 406528m_i^{11}m_j^8\rho^3 \\
& + 15872m_i^{11}m_j^7\rho^4 + 7680m_i^{11}m_j^6\rho^5 + 5259368m_i^{10}m_j^{12} \\
& - 4857296m_i^{10}m_j^{11}\rho - 222752m_i^{10}m_j^{10}\rho^2 + 24224m_i^{10}m_j^9\rho^3 \\
& - 36992m_i^{10}m_j^8\rho^4 - 12544m_i^{10}m_j^7\rho^5 + 2316288m_i^9m_j^{13} \\
& + 1506728m_i^9m_j^{12}\rho - 2953328m_i^9m_j^{11}\rho^2 + 693984m_i^9m_j^{10}\rho^3 \\
& + 37248m_i^9m_j^9\rho^4 + 18944m_i^9m_j^8\rho^5 + 7487184m_i^8m_j^{14} \\
& - 4500944m_i^8m_j^{13}\rho + 1206960m_i^8m_j^{12}\rho^2 - 402976m_i^8m_j^{11}\rho^3 \\
& - 116864m_i^8m_j^{10}\rho^4 - 16640m_i^8m_j^9\rho^5 + 1354664m_i^7m_j^{15} \\
& + 4518320m_i^7m_j^{14}\rho - 1483360m_i^7m_j^{13}\rho^2 - 563840m_i^7m_j^{12}\rho^3 \\
& + 102656m_i^7m_j^{11}\rho^4 + 14848m_i^7m_j^{10}\rho^5 + 4862360m_i^6m_j^{16} \\
& - 11268240m_i^6m_j^{15}\rho + 1310976m_i^6m_j^{14}\rho^2 + 502624m_i^6m_j^{13}\rho^3 \\
& - 57280m_i^6m_j^{12}\rho^4 - 8832m_i^6m_j^{11}\rho^5 - 1521088m_i^5m_j^{17} \\
& + 4831656m_i^5m_j^{16}\rho - 2442576m_i^5m_j^{15}\rho^2 - 689632m_i^5m_j^{14}\rho^3 \\
& + 37504m_i^5m_j^{13}\rho^4 + 3584m_i^5m_j^{12}\rho^5 + 106008m_i^4m_j^{18} \\
& - 1014192m_i^4m_j^{17}\rho + 1091632m_i^4m_j^{16}\rho^2 + 958816m_i^4m_j^{15}\rho^3 \\
& - 4544m_i^4m_j^{14}\rho^4 - 1152m_i^4m_j^{13}\rho^5 - 6800m_i^3m_j^{19} \\
& + 67472m_i^3m_j^{18}\rho + 64448m_i^3m_j^{17}\rho^2 - 346112m_i^3m_j^{16}\rho^3 \\
& - 4016m_i^2m_j^{19}\rho + 2992m_i^2m_j^{18}\rho^2 + 10400m_i^2m_j^{17}\rho^3 \\
& + 416m_i^{19}m_j^2 - 832m_i^{18}m_j^3 - 96m_i^{20}m_j \\
& + 1760m_i^{19}m_j^2 - 480m_i^{19}m_j\rho - 936m_i^{18}m_j^3 \\
& + 5384m_i^{18}m_j^2\rho - 768m_i^{18}m_j\rho^2 + 9608m_i^{17}m_j^4 \\
& + 4048m_i^{17}m_j^3\rho + 3568m_i^{17}m_j^2\rho^2 - 384m_i^{17}m_j\rho^3 \\
& - 107168m_i^{16}m_j^5 + 5528m_i^{16}m_j^4\rho + 5552m_i^{16}m_j^3\rho^2 \\
& - 8720m_i^{15}m_j^6 - 327536m_i^{15}m_j^5\rho + 48176m_i^{15}m_j^4\rho^2
\end{aligned}$$

$$\begin{aligned}
& -2080m_i^{15}m_j^3\rho^3 + 1051736m_i^{14}m_j^7 - 794336m_i^{14}m_j^6\rho \\
& -138688m_i^{14}m_j^5\rho^2 + 40704m_i^{14}m_j^4\rho^3 - 2304m_i^{14}m_j^3\rho^4 \\
& + 2575736m_i^{13}m_j^8 - 962896m_i^{13}m_j^7\rho - 247360m_i^{13}m_j^6\rho^2 \\
& + 38048m_i^{13}m_j^5\rho^3 + 148640m_i^{12}m_j^9 + 909672m_i^{12}m_j^8\rho \\
& - 304896m_i^{12}m_j^7\rho^2 + 78848m_i^{12}m_j^6\rho^3 - 9472m_i^{12}m_j^5\rho^4 \\
& - 2809792m_i^{11}m_j^{10} + 1680272m_i^{11}m_j^9\rho + 346112m_i^{11}m_j^8\rho^2 \\
& - 100352m_i^{11}m_j^7\rho^3 + 15360m_i^{11}m_j^6\rho^4 - 321848m_i^{10}m_j^{11} \\
& + 1082272m_i^{10}m_j^{10}\rho - 10096m_i^{10}m_j^9\rho^2 + 31232m_i^{10}m_j^8\rho^3 \\
& - 25088m_i^{10}m_j^7\rho^4 - 2769224m_i^9m_j^{12} + 1040464m_i^9m_j^{11}\rho \\
& + 453872m_i^9m_j^{10}\rho^2 - 136896m_i^9m_j^9\rho^3 + 37888m_i^9m_j^8\rho^4 \\
& + 2052832m_i^8m_j^{13} - 2450712m_i^8m_j^{12}\rho - 67216m_i^8m_j^{11}\rho^2 \\
& + 55040m_i^8m_j^{10}\rho^3 - 33280m_i^8m_j^9\rho^4 - 2007824m_i^7m_j^{14} \\
& + 1585200m_i^7m_j^{13}\rho - 431888m_i^7m_j^{12}\rho^2 - 1952m_i^7m_j^{11}\rho^3 \\
& + 29696m_i^7m_j^{10}\rho^4 - 259512m_i^6m_j^{15} - 2026560m_i^6m_j^{14}\rho \\
& + 442016m_i^6m_j^{13}\rho^2 + 162304m_i^6m_j^{12}\rho^3 - 17664m_i^6m_j^{11}\rho^4 \\
& - 1062840m_i^5m_j^{16} + 2713488m_i^5m_j^{15}\rho - 468832m_i^5m_j^{14}\rho^2 \\
& - 145504m_i^5m_j^{13}\rho^3 + 7168m_i^5m_j^{12}\rho^4 + 240512m_i^4m_j^{17} \\
& - 947688m_i^4m_j^{16}\rho + 649696m_i^4m_j^{15}\rho^2 + 139520m_i^4m_j^{14}\rho^3 \\
& - 2304m_i^4m_j^{13}\rho^4 - 10240m_i^3m_j^{18} + 167312m_i^3m_j^{17}\rho \\
& - 221984m_i^3m_j^{16}\rho^2 - 140352m_i^3m_j^{15}\rho^3 + 576m_i^2m_j^{19} \\
& - 6536m_i^2m_j^{18}\rho - 9328m_i^2m_j^{17}\rho^2 + 39168m_i^2m_j^{16}\rho^3 \\
& + 416m_i m_j^{19}\rho - 832m_i m_j^{18}\rho^2 - 96m_i^{19}m_j \\
& + 176m_i^{18}m_j^2 - 288m_i^{18}m_j\rho - 1304m_i^{17}m_j^3 \\
& + 208m_i^{17}m_j^2\rho - 192m_i^{17}m_j\rho^2 + 12520m_i^{16}m_j^4 \\
& - 536m_i^{16}m_j^3\rho - 12640m_i^{15}m_j^5 + 38072m_i^{15}m_j^4\rho \\
& - 3152m_i^{15}m_j^3\rho^2 - 175456m_i^{14}m_j^6 + 76688m_i^{14}m_j^5\rho \\
& + 21760m_i^{14}m_j^4\rho^2 - 2560m_i^{14}m_j^3\rho^3 - 440312m_i^{13}m_j^7 \\
& + 136160m_i^{13}m_j^6\rho + 10272m_i^{13}m_j^5\rho^2 + 118264m_i^{12}m_j^8 \\
& - 159016m_i^{12}m_j^7\rho + 63168m_i^{12}m_j^6\rho^2 - 10368m_i^{12}m_j^5\rho^3 \\
& + 832896m_i^{11}m_j^9 - 345112m_i^{11}m_j^8\rho - 98992m_i^{11}m_j^7\rho^2 \\
& + 21760m_i^{11}m_j^6\rho^3 + 326992m_i^{10}m_j^{10} - 199456m_i^{10}m_j^9\rho \\
& + 78656m_i^{10}m_j^8\rho^2 - 30848m_i^{10}m_j^7\rho^3 + 580664m_i^9m_j^{11} \\
& - 323632m_i^9m_j^{10}\rho - 144512m_i^9m_j^9\rho^2 + 49920m_i^9m_j^8\rho^3 \\
& - 439720m_i^8m_j^{12} + 271768m_i^8m_j^{11}\rho + 95872m_i^8m_j^{10}\rho^2 \\
& - 39168m_i^8m_j^9\rho^3 - 309856m_i^7m_j^{13} + 378888m_i^7m_j^{12}\rho \\
& - 45488m_i^7m_j^{11}\rho^2 + 20480m_i^7m_j^{10}\rho^3 + 53440m_i^6m_j^{14} \\
& - 173808m_i^6m_j^{13}\rho + 128640m_i^6m_j^{12}\rho^2 - 1792m_i^6m_j^{11}\rho^3 \\
& + 193240m_i^5m_j^{15} + 380800m_i^5m_j^{14}\rho - 120928m_i^5m_j^{13}\rho^2 \\
& - 24576m_i^5m_j^{12}\rho^3 + 142536m_i^4m_j^{16} - 426584m_i^4m_j^{15}\rho \\
& + 99008m_i^4m_j^{14}\rho^2 + 27008m_i^4m_j^{13}\rho^3 - 23392m_i^3m_j^{17} \\
& + 121688m_i^3m_j^{16}\rho - 106320m_i^3m_j^{15}\rho^2 - 19712m_i^3m_j^{14}\rho^3 \\
& + 160m_i^2m_j^{18} - 17504m_i^2m_j^{17}\rho + 28992m_i^2m_j^{16}\rho^2 \\
& + 12672m_i^2m_j^{15}\rho^3 + 1408m_i m_j^{17}\rho^2 - 2816m_i m_j^{16}\rho^3 \\
& + 80m_i^{17}m_j^2 - 816m_i^{16}m_j^3 + 2672m_i^{15}m_j^4
\end{aligned}$$

$$\begin{aligned}
& -2336m_i^{15}m_j^3\rho + 21696m_i^{14}m_j^5 - 3936m_i^{14}m_j^4\rho \\
& -1280m_i^{14}m_j^3\rho^2 + 53024m_i^{13}m_j^6 - 14784m_i^{13}m_j^5\rho \\
& -21696m_i^{12}m_j^7 + 21632m_i^{12}m_j^6\rho - 5056m_i^{12}m_j^5\rho^2 \\
& -205696m_i^{11}m_j^8 + 28320m_i^{11}m_j^7\rho + 10880m_i^{11}m_j^6\rho^2 \\
& + 6576m_i^{10}m_j^9 + 30112m_i^{10}m_j^8\rho - 14272m_i^{10}m_j^7\rho^2 \\
& -304304m_i^9m_j^{10} + 69376m_i^9m_j^9\rho + 21760m_i^9m_j^8\rho^2 \\
& + 266640m_i^8m_j^{11} - 72704m_i^8m_j^{10}\rho - 13184m_i^8m_j^9\rho^2 \\
& -50640m_i^7m_j^{12} + 55840m_i^7m_j^{11}\rho - 2816m_i^7m_j^{10}\rho^2 \\
& + 176352m_i^6m_j^{13} - 140832m_i^6m_j^{12}\rho + 14720m_i^6m_j^{11}\rho^2 \\
& -17600m_i^5m_j^{14} + 78144m_i^5m_j^{13}\rho - 30208m_i^5m_j^{12}\rho^2 \\
& -37792m_i^4m_j^{15} - 74880m_i^4m_j^{14}\rho + 29504m_i^4m_j^{13}\rho^2 \\
& -15136m_i^3m_j^{16} + 52320m_i^3m_j^{15}\rho - 19328m_i^3m_j^{14}\rho^2 \\
& + 1712m_i^2m_j^{17} - 11296m_i^2m_j^{16}\rho + 12096m_i^2m_j^{15}\rho^2 \\
& + 1408m_i m_j^{17}\rho - 2816m_i m_j^{16}\rho^2 - 144m_i^{15}m_j^3 \\
& -2176m_i^{14}m_j^4 + 256m_i^{14}m_j^3\rho - 3488m_i^{13}m_j^5 \\
& + 384m_i^{12}m_j^6 + 1152m_i^{12}m_j^5\rho + 28560m_i^{11}m_j^7 \\
& -6400m_i^{11}m_j^6\rho - 6528m_i^{10}m_j^8 + 8064m_i^{10}m_j^7\rho \\
& + 50496m_i^9m_j^9 - 18432m_i^9m_j^8\rho - 34816m_i^8m_j^{10} \\
& + 18688m_i^8m_j^9\rho + 11920m_i^7m_j^{11} - 16896m_i^7m_j^{10}\rho \\
& -37504m_i^6m_j^{12} + 15360m_i^6m_j^{11}\rho + 3168m_i^5m_j^{13} \\
& -4096m_i^5m_j^{12}\rho - 11648m_i^4m_j^{14} + 2688m_i^4m_j^{13}\rho \\
& + 9840m_i^3m_j^{15} + 768m_i^3m_j^{14}\rho + 128m_i^2m_j^{16} \\
& -1152m_i^2m_j^{15}\rho + 128m_i^{14}m_j^3 + 576m_i^{12}m_j^5 \\
& -3200m_i^{11}m_j^6 + 4032m_i^{10}m_j^7 - 9216m_i^9m_j^8 \\
& + 9344m_i^8m_j^9 - 8448m_i^7m_j^{10} + 7680m_i^6m_j^{11} \\
& -2048m_i^5m_j^{12} + 1344m_i^4m_j^{13} + 384m_i^3m_j^{14} \\
& -576m_i^2m_j^{15}
\end{aligned}$$

$R$

$\text{Res}_\rho(q, \Gamma) = A_0^2 A_2 m_i^2 m_j^2 R(m_i, m_j)$ ;  $R$  is of degree 46 in  $m_i$  and 44 in  $m_j$ . 1387 terms.

$$\begin{aligned}
R(m_i, m_j) = & -10836786603360256m_i^{45}m_j^{34} + 96405179523399680m_i^{44}m_j^{35} - 437834326273425408m_i^{43}m_j^{36} \\
& + 1221179186459181056m_i^{42}m_j^{37} - 2153987259278295040m_i^{41}m_j^{38} + 2393522464459063296m_i^{40}m_j^{39} \\
& - 1621999553295155200m_i^{39}m_j^{40} + 612348811834032128m_i^{38}m_j^{41} - 98797716825440256m_i^{37}m_j^{42} \\
& + 3448068464705536m_i^{46}m_j^{32} + 10836786603360256m_i^{45}m_j^{33} - 158787071236898816m_i^{44}m_j^{34} \\
& + 685180462057914368m_i^{43}m_j^{35} - 1294538602264395776m_i^{42}m_j^{36} + 38034306228027392m_i^{41}m_j^{37} \\
& + 4694334108464054272m_i^{40}m_j^{38} - 9835263451851653120m_i^{39}m_j^{39} + 10031029298153914368m_i^{38}m_j^{40} \\
& - 5556069749663727616m_i^{37}m_j^{41} + 1523342573958070272m_i^{36}m_j^{42} - 141546728913371136m_i^{35}m_j^{43} \\
& - 13792273858822144m_i^{46}m_j^{31} + 20934701392855040m_i^{45}m_j^{32} + 134210787332849664m_i^{44}m_j^{33} \\
& - 1062638405826904064m_i^{43}m_j^{34} + 2915491420489908224m_i^{42}m_j^{35} - 4365641704410185728m_i^{41}m_j^{36} \\
& + 5328162979458318336m_i^{40}m_j^{37} - 9550006155141447680m_i^{39}m_j^{38} + 18530658801141415936m_i^{38}m_j^{39} \\
& - 23911316863756271616m_i^{37}m_j^{40} + 17804770020576722944m_i^{36}m_j^{41} - 6874621685878751232m_i^{35}m_j^{42} \\
& + 1093662225916231680m_i^{34}m_j^{43} - 49873847435919360m_i^{33}m_j^{44} + 31463624740438016m_i^{46}m_j^{30} \\
& - 114648276451459072m_i^{45}m_j^{31} + 200709250580742144m_i^{44}m_j^{32} + 646310526992777216m_i^{43}m_j^{33} \\
& - 2267672363543822336m_i^{42}m_j^{34} + 2126253177979273216m_i^{41}m_j^{35} + 6357099154870632448m_i^{40}m_j^{36}
\end{aligned}$$

$$\begin{aligned}
& -26549814916930338816m_i^{39}m_j^{37} + 46291770494020485120m_i^{38}m_j^{38} - 52019474008780046336m_i^{37}m_j^{39} \\
& + 47092021444994924544m_i^{36}m_j^{40} - 34894703551471157248m_i^{35}m_j^{41} + 16447853924645339136m_i^{34}m_j^{42} \\
& - 3411964900895883264m_i^{33}m_j^{43} + 226926005833433088m_i^{32}m_j^{44} - 51074514133450752m_i^{46}m_j^{29} \\
& + 261310483213058048m_i^{45}m_j^{30} - 830795933749870592m_i^{44}m_j^{31} + 401929224312586240m_i^{43}m_j^{32} \\
& + 1281467058277580800m_i^{42}m_j^{33} - 6271045877322743808m_i^{41}m_j^{34} + 5617054162098323456m_i^{40}m_j^{35} \\
& - 1985944499158777856m_i^{39}m_j^{36} + 27602987627153719296m_i^{38}m_j^{37} - 89698866129698553856m_i^{37}m_j^{38} \\
& + 124262352798418468864m_i^{36}m_j^{39} - 99348699144549040128m_i^{35}m_j^{40} + 58679534246835519488m_i^{34}m_j^{41} \\
& - 28646572595982893056m_i^{33}m_j^{42} + 7024744605488775168m_i^{32}m_j^{43} - 578328822559014912m_i^{31}m_j^{44} \\
& + 65122699323637760m_i^{46}m_j^{28} - 423831495937884160m_i^{45}m_j^{29} + 1658466093380927488m_i^{44}m_j^{30} \\
& - 2465956091473690624m_i^{43}m_j^{31} + 2501485160457830400m_i^{42}m_j^{32} + 5660360351703629824m_i^{41}m_j^{33} \\
& + 3734580991377801216m_i^{40}m_j^{34} - 31414969701641486336m_i^{39}m_j^{35} + 63719666961258381312m_i^{38}m_j^{36} \\
& - 84259254452625604608m_i^{37}m_j^{37} + 140719306693612142592m_i^{36}m_j^{38} - 202070625196324159488m_i^{35}m_j^{39} \\
& + 178374613050251542528m_i^{34}m_j^{40} - 89609406840005722112m_i^{33}m_j^{41} + 39923020226747170816m_i^{32}m_j^{42} \\
& - 10804596751189671936m_i^{31}m_j^{43} + 1035176728533663744m_i^{30}m_j^{44} - 68516891718582272m_i^{46}m_j^{27} \\
& + 545102199642914816m_i^{45}m_j^{28} - 2404048089271762944m_i^{44}m_j^{29} + 4986124771985981440m_i^{43}m_j^{30} \\
& - 6929609568053886976m_i^{42}m_j^{31} - 6678019042508275712m_i^{41}m_j^{32} - 482577783734665216m_i^{40}m_j^{33} \\
& - 8373277200136273920m_i^{39}m_j^{34} + 8005162631302217728m_i^{38}m_j^{35} - 104627127502231306240m_i^{37}m_j^{36} \\
& + 214008302764797460480m_i^{36}m_j^{37} - 281430664399191801856m_i^{35}m_j^{38} + 280772264192095813632m_i^{34}m_j^{39} \\
& - 266061801916813279232m_i^{33}m_j^{40} + 126663642144296665088m_i^{32}m_j^{41} - 47944258542681194496m_i^{31}m_j^{42} \\
& + 13331271997917757440m_i^{30}m_j^{43} - 1433678294066135040m_i^{29}m_j^{44} + 61381679729606656m_i^{46}m_j^{26} \\
& - 586384738097299456m_i^{45}m_j^{27} + 2844343277186973696m_i^{44}m_j^{28} - 7453730577577934848m_i^{43}m_j^{29} \\
& + 11866127433411330048m_i^{42}m_j^{30} + 1396052662566256640m_i^{41}m_j^{31} + 6773020283121958912m_i^{40}m_j^{32} \\
& + 27668821555881705472m_i^{39}m_j^{33} + 78308165668711694336m_i^{38}m_j^{34} - 59414307235971465216m_i^{37}m_j^{35} \\
& + 155979262308741283840m_i^{36}m_j^{36} - 293172611305734930432m_i^{35}m_j^{37} + 509991545719916331008m_i^{34}m_j^{38} \\
& - 366660178386622611456m_i^{33}m_j^{39} + 327359549945277317120m_i^{32}m_j^{40} - 158527433746526765056m_i^{31}m_j^{41} \\
& + 51261171041312964608m_i^{30}m_j^{42} - 13702706902889136128m_i^{29}m_j^{43} + 1612586651429634048m_i^{28}m_j^{44} \\
& - 47781339369308160m_i^{46}m_j^{25} + 542191914770759680m_i^{45}m_j^{26} - 2864840091631616000m_i^{44}m_j^{27} \\
& + 9072788986660913152m_i^{43}m_j^{28} - 15510736077630472192m_i^{42}m_j^{29} + 6701601915509145600m_i^{41}m_j^{30} \\
& - 24195197898326016m_i^{40}m_j^{31} - 76714689291729502208m_i^{39}m_j^{32} - 82812190431272501248m_i^{38}m_j^{33} \\
& - 129617790424376672256m_i^{37}m_j^{34} + 125251551779641884672m_i^{36}m_j^{35} - 413311996987646672896m_i^{35}m_j^{36} \\
& + 309726526572274384896m_i^{34}m_j^{37} - 748856238771997245440m_i^{33}m_j^{38} + 480225020619366858752m_i^{32}m_j^{39} \\
& - 348747614670567768064m_i^{31}m_j^{40} + 172613603397698846720m_i^{30}m_j^{41} - 49536268611386081280m_i^{29}m_j^{42} \\
& + 12022997974276636672m_i^{28}m_j^{43} - 1516037697876000768m_i^{27}m_j^{44} + 32790059891556352m_i^{46}m_j^{24} \\
& - 438972881301405696m_i^{45}m_j^{25} + 2522971823667150848m_i^{44}m_j^{26} - 9457742405349408768m_i^{43}m_j^{27} \\
& + 17482374199103193088m_i^{42}m_j^{28} - 17415490941145317376m_i^{41}m_j^{29} - 8402594549725134848m_i^{40}m_j^{30} \\
& + 83672800676687118336m_i^{39}m_j^{31} + 119022730175167070208m_i^{38}m_j^{32} + 154697510702974763008m_i^{37}m_j^{33} \\
& + 306027745973165359104m_i^{36}m_j^{34} - 8798293201346625536m_i^{35}m_j^{35} + 870390685790241816576m_i^{34}m_j^{36} \\
& - 362071621063973797888m_i^{33}m_j^{37} + 863444784135628914688m_i^{32}m_j^{38} - 583077280481154170880m_i^{31}m_j^{39} \\
& + 331577740958334189568m_i^{30}m_j^{40} - 162084652773315444736m_i^{29}m_j^{41} + 43113315229772021760m_i^{28}m_j^{42} \\
& - 9125822139562196992m_i^{27}m_j^{43} + 1211418513575510016m_i^{26}m_j^{44} - 20042003177799680m_i^{46}m_j^{23} \\
& + 315001985344471040m_i^{45}m_j^{24} - 1971473595502690304m_i^{44}m_j^{25} + 8608762621060972544m_i^{43}m_j^{26} \\
& - 17377964558689239040m_i^{42}m_j^{27} + 25873056305919295488m_i^{41}m_j^{28} + 19441671467564007424m_i^{40}m_j^{29} \\
& - 70787901616386211840m_i^{39}m_j^{30} - 38508700075596709888m_i^{38}m_j^{31} - 246527213748464123904m_i^{37}m_j^{32} \\
& - 409728883086023720960m_i^{36}m_j^{33} - 750057078234588119040m_i^{35}m_j^{34} - 236167025221697536000m_i^{34}m_j^{35}
\end{aligned}$$

$$\begin{aligned}
& -1278208931596551061504m_i^{33}m_j^{36} + 583571853517451689984m_i^{32}m_j^{37} - 836731880777980051456m_i^{31}m_j^{38} \\
& + 621571564198911016960m_i^{30}m_j^{39} - 287594881431480827904m_i^{29}m_j^{40} + 131310396009934225408m_i^{28}m_j^{41} \\
& - 33418770295942021120m_i^{27}m_j^{42} + 6017853753237962752m_i^{26}m_j^{43} - 829686551213506560m_i^{25}m_j^{44} \\
& + 10994092062277632m_i^{46}m_j^{22} - 202179897580322816m_i^{45}m_j^{23} + 1381753726045257728m_i^{44}m_j^{24} \\
& - 6946494939249246208m_i^{43}m_j^{25} + 15581886494648827904m_i^{42}m_j^{26} - 30114538614297722880m_i^{41}m_j^{27} \\
& - 23471172304370663424m_i^{40}m_j^{28} + 35662237755766734848m_i^{39}m_j^{29} - 73335571961947881472m_i^{38}m_j^{30} \\
& + 108437798706395217920m_i^{37}m_j^{31} + 579481898795589959680m_i^{36}m_j^{32} + 789497237742603468800m_i^{35}m_j^{33} \\
& + 1327069925897492496384m_i^{34}m_j^{34} + 255600930049568538624m_i^{33}m_j^{35} + 1349277065394555715584m_i^{32}m_j^{36} \\
& - 878521736752768483328m_i^{31}m_j^{37} + 719898246711877828608m_i^{30}m_j^{38} - 562696716866327412736m_i^{29}m_j^{39} \\
& + 226237897135150333952m_i^{28}m_j^{40} - 91384534527163498496m_i^{27}m_j^{41} + 22672242902552281088m_i^{26}m_j^{42} \\
& - 3421100438873702400m_i^{25}m_j^{43} + 487113318994018304m_i^{24}m_j^{44} - 5442183461076992m_i^{46}m_j^{21} \\
& + 116827338234134528m_i^{45}m_j^{22} - 874651836613656576m_i^{44}m_j^{23} + 5012405613745930240m_i^{43}m_j^{24} \\
& - 12656604222377164800m_i^{42}m_j^{25} + 29283503081535832064m_i^{41}m_j^{26} + 20608035292077621248m_i^{40}m_j^{27} \\
& - 6711617828098670592m_i^{39}m_j^{28} + 191925918771705806848m_i^{38}m_j^{29} + 71148773993800007680m_i^{37}m_j^{30} \\
& - 284216547900263497728m_i^{36}m_j^{31} - 795322231884475269120m_i^{35}m_j^{32} - 1080512491040224051200m_i^{34}m_j^{33} \\
& - 1573834145452118769664m_i^{33}m_j^{34} + 210527823229154754560m_i^{32}m_j^{35} - 1149383361681799774208m_i^{31}m_j^{36} \\
& + 1059549521516901171200m_i^{30}m_j^{37} - 578528846133107621888m_i^{29}m_j^{38} + 427456710056360280064m_i^{28}m_j^{39} \\
& - 157166455566836432896m_i^{27}m_j^{40} + 53657046347082104832m_i^{26}m_j^{41} - 13102527959808344064m_i^{25}m_j^{42} \\
& + 1627309845877096448m_i^{24}m_j^{43} - 242356368139354112m_i^{23}m_j^{44} + 2440630722625536m_i^{46}m_j^{20} \\
& - 61079921090363392m_i^{45}m_j^{21} + 502698639775236096m_i^{44}m_j^{22} - 3258077798656376832m_i^{43}m_j^{23} \\
& + 9356803726418903040m_i^{42}m_j^{24} - 24830476205414154240m_i^{41}m_j^{25} - 12618851075200385024m_i^{40}m_j^{26} \\
& - 10349892730460569600m_i^{39}m_j^{27} - 246831916054684368896m_i^{38}m_j^{28} - 244644420199464828928m_i^{37}m_j^{29} \\
& - 79783349365948547072m_i^{36}m_j^{30} + 31018670867087360000m_i^{35}m_j^{31} + 734280793777915297792m_i^{34}m_j^{32} \\
& + 872538408007661256704m_i^{33}m_j^{33} + 1246800274751276187648m_i^{32}m_j^{34} - 898491068460990201856m_i^{31}m_j^{35} \\
& + 906412361274410139648m_i^{30}m_j^{36} - 1004601742024618541056m_i^{29}m_j^{37} + 430735924540725002240m_i^{28}m_j^{38} \\
& - 265406342805678194688m_i^{27}m_j^{39} + 91102968002574811136m_i^{26}m_j^{40} - 25020980807585497088m_i^{25}m_j^{41} \\
& + 6091617039173025792m_i^{24}m_j^{42} - 589612394158751744m_i^{23}m_j^{43} + 98610373120229376m_i^{22}m_j^{44} \\
& - 994212061380608m_i^{46}m_j^{19} + 28996160497647616m_i^{45}m_j^{20} - 263259374743453696m_i^{44}m_j^{21} \\
& + 1917172767028936704m_i^{43}m_j^{22} - 6299562692568940544m_i^{42}m_j^{23} + 18698712132988436480m_i^{41}m_j^{24} \\
& + 4236351130218528768m_i^{40}m_j^{25} + 14146903695986524160m_i^{39}m_j^{26} + 238745998737633443840m_i^{38}m_j^{27} \\
& + 295786014105053167616m_i^{37}m_j^{28} + 399990194632495464448m_i^{36}m_j^{29} + 684266370221775781888m_i^{35}m_j^{30} \\
& + 551312373385726525440m_i^{34}m_j^{31} - 248204458805436612608m_i^{33}m_j^{32} - 106605342502191890432m_i^{32}m_j^{33} \\
& - 690045874320359555072m_i^{31}m_j^{34} + 1359638531553980579840m_i^{30}m_j^{35} - 766907082681344851968m_i^{29}m_j^{36} \\
& + 762543152706797699072m_i^{28}m_j^{37} - 278026683762231738368m_i^{27}m_j^{38} + 123058405590876291072m_i^{26}m_j^{39} \\
& - 38140841900169494528m_i^{25}m_j^{40} + 7195124641520680960m_i^{24}m_j^{41} - 1887004575581339648m_i^{23}m_j^{42} \\
& + 98286014899945472m_i^{22}m_j^{43} - 29292594773098496m_i^{21}m_j^{44} + 368445017292800m_i^{46}m_j^{18} \\
& - 12530742037839872m_i^{45}m_j^{19} + 125954114093056000m_i^{44}m_j^{20} - 1025474046868324352m_i^{43}m_j^{21} \\
& + 3868117135409020928m_i^{42}m_j^{22} - 12684832859165294592m_i^{41}m_j^{23} + 1713248441789579264m_i^{40}m_j^{24} \\
& - 11712487346422677504m_i^{39}m_j^{25} - 186750702111262507008m_i^{38}m_j^{26} - 251148288188613656576m_i^{37}m_j^{27} \\
& - 482173321047759650816m_i^{36}m_j^{28} - 1139169780049202118656m_i^{35}m_j^{29} - 1409819007908576231424m_i^{34}m_j^{30} \\
& - 1218756957875510181888m_i^{33}m_j^{31} - 520885519249840799744m_i^{32}m_j^{32} - 728593882518823895040m_i^{31}m_j^{33} \\
& + 391321334727067041792m_i^{30}m_j^{34} - 1380897717657353060352m_i^{29}m_j^{35} + 680442838454382100480m_i^{28}m_j^{36} \\
& - 458424661313878228992m_i^{27}m_j^{37} + 130939579092122992640m_i^{26}m_j^{38} - 24131634683327807488m_i^{25}m_j^{39} \\
& + 3675193258327932928m_i^{24}m_j^{40} + 1603417747125436416m_i^{23}m_j^{41} - 87476668445753344m_i^{22}m_j^{42}
\end{aligned}$$

$$\begin{aligned}
& + 73825948513861632m_i^{21}m_j^{43} + 2974038408298496m_i^{20}m_j^{44} - 124276507672576m_i^{46}m_j^{17} \\
& + 4937101721337856m_i^{45}m_j^{18} - 55142870011084800m_i^{44}m_j^{19} + 499997308616310784m_i^{43}m_j^{20} \\
& - 2167373107452706816m_i^{42}m_j^{21} + 7809062303304908800m_i^{41}m_j^{22} - 4359310104626462720m_i^{40}m_j^{23} \\
& + 7774724170987208704m_i^{39}m_j^{24} + 124392963798079635456m_i^{38}m_j^{25} + 158266601947885207552m_i^{37}m_j^{26} \\
& + 398982379014271533056m_i^{36}m_j^{27} + 1108297067389037838336m_i^{35}m_j^{28} + 1685760303719056932864m_i^{34}m_j^{29} \\
& + 1747437252947185238016m_i^{33}m_j^{30} + 1762281665768279506944m_i^{32}m_j^{31} + 1112301033246825381888m_i^{31}m_j^{32} \\
& + 1090472914689643773952m_i^{30}m_j^{33} - 472058167200342474752m_i^{29}m_j^{34} + 1102867389711488909312m_i^{28}m_j^{35} \\
& - 555626956354896134144m_i^{27}m_j^{36} + 196530666219945328640m_i^{26}m_j^{37} - 12153544618019258368m_i^{25}m_j^{38} \\
& - 29517399051346116608m_i^{24}m_j^{39} + 13082830339268542464m_i^{23}m_j^{40} - 4485040747179409408m_i^{22}m_j^{41} \\
& + 692269923104456704m_i^{21}m_j^{42} - 97257618990170112m_i^{20}m_j^{43} + 3631982192885760m_i^{19}m_j^{44} \\
& + 38134855041024m_i^{16}m_j^{17} - 1774764127682560m_i^{45}m_j^{17} + 22112400308928512m_i^{44}m_j^{18} \\
& - 222697481769779200m_i^{43}m_j^{19} + 1109317096382857216m_i^{42}m_j^{20} - 4386878226582929408m_i^{41}m_j^{21} \\
& + 4538454109265723392m_i^{40}m_j^{22} - 4829369940445822976m_i^{39}m_j^{23} - 71790808200439791616m_i^{38}m_j^{24} \\
& - 73833559183012724736m_i^{37}m_j^{25} - 246361198860698550272m_i^{36}m_j^{26} - 815793285350278692864m_i^{35}m_j^{27} \\
& - 1299485408426945019904m_i^{34}m_j^{28} - 1539328193962509205504m_i^{33}m_j^{29} - 1708600476612828332032m_i^{32}m_j^{30} \\
& - 1949552367747812425728m_i^{31}m_j^{31} - 1122317474007360929792m_i^{30}m_j^{32} - 928751031835998683136m_i^{29}m_j^{33} \\
& + 674018877950379589632m_i^{28}m_j^{34} - 754968126680202149888m_i^{27}m_j^{35} + 371198760849836802048m_i^{26}m_j^{36} \\
& - 19013850191590260736m_i^{25}m_j^{37} - 61422381922053652480m_i^{24}m_j^{38} + 48450562040658591744m_i^{23}m_j^{39} \\
& - 17170682576884465664m_i^{22}m_j^{40} + 4331614142099816448m_i^{21}m_j^{41} - 660780836526456832m_i^{20}m_j^{42} \\
& + 71439955802456064m_i^{19}m_j^{43} - 3459881677357056m_i^{18}m_j^{44} - 10629264572416m_i^{46}m_j^{15} \\
& + 581980550316032m_i^{45}m_j^{16} - 8123226682245120m_i^{44}m_j^{17} + 90717578111533056m_i^{43}m_j^{18} \\
& - 518937439279087616m_i^{42}m_j^{19} + 2255083983852535808m_i^{41}m_j^{20} - 3505800297574203392m_i^{40}m_j^{21} \\
& + 3036998972429631488m_i^{39}m_j^{22} + 36424234276176920576m_i^{38}m_j^{23} + 19868564937599795200m_i^{37}m_j^{24} \\
& + 117852721729851703296m_i^{36}m_j^{25} + 469427910372263116800m_i^{35}m_j^{26} + 726258741295333998592m_i^{34}m_j^{27} \\
& + 783860963141174689792m_i^{33}m_j^{28} + 1084446341126163202048m_i^{32}m_j^{29} + 1428177972527367716864m_i^{31}m_j^{30} \\
& + 1601116742695953367040m_i^{30}m_j^{31} + 670184396383365185536m_i^{29}m_j^{32} + 586340204023015882752m_i^{28}m_j^{33} \\
& - 745535913163024613376m_i^{27}m_j^{34} + 468911768067202842624m_i^{26}m_j^{35} - 176855500926117871616m_i^{25}m_j^{36} \\
& - 76395352341302509568m_i^{24}m_j^{37} + 89483089764915281920m_i^{23}m_j^{38} - 46573728688638001152m_i^{22}m_j^{39} \\
& + 14609492437664808960m_i^{21}m_j^{40} - 3119531850031644672m_i^{20}m_j^{41} + 444620032707018752m_i^{19}m_j^{42} \\
& - 41238321225367552m_i^{18}m_j^{43} + 2012202639949824m_i^{17}m_j^{44} + 2684104507392m_i^{46}m_j^{14} \\
& - 173921283063808m_i^{45}m_j^{15} + 2732705793318912m_i^{44}m_j^{16} - 33818606014414848m_i^{43}m_j^{17} \\
& + 222007143073587200m_i^{42}m_j^{18} - 1062746037248065536m_i^{41}m_j^{19} + 2248314712784371712m_i^{40}m_j^{20} \\
& - 1978100938306027520m_i^{39}m_j^{21} - 16228659240253112320m_i^{38}m_j^{22} + 4175925683305824256m_i^{37}m_j^{23} \\
& - 40816139983058526208m_i^{36}m_j^{24} - 216613922249189572608m_i^{35}m_j^{25} - 255334448890789142528m_i^{34}m_j^{26} \\
& - 129356679179799298048m_i^{33}m_j^{27} - 235442096994730573824m_i^{32}m_j^{28} - 698216084045143474176m_i^{31}m_j^{29} \\
& - 896451345709054427136m_i^{30}m_j^{30} - 947318429689567657984m_i^{29}m_j^{31} - 182367449539989643264m_i^{28}m_j^{32} \\
& - 343230798362460438528m_i^{27}m_j^{33} + 638274897485513678848m_i^{26}m_j^{34} - 264814684993297842176m_i^{25}m_j^{35} \\
& + 27694117371677704192m_i^{24}m_j^{36} + 110203946308228677632m_i^{23}m_j^{37} - 85169746640470589440m_i^{22}m_j^{38} \\
& + 35688485088817201152m_i^{21}m_j^{39} - 10018849310678261760m_i^{20}m_j^{40} + 1884394767905144832m_i^{19}m_j^{41} \\
& - 246021530658693120m_i^{18}m_j^{42} + 20262993153949696m_i^{17}m_j^{43} - 927034156253184m_i^{16}m_j^{44} \\
& - 611633397760m_i^{46}m_j^{13} + 47274534436864m_i^{45}m_j^{14} - 840853779550208m_i^{44}m_j^{15} \\
& + 11534944485343232m_i^{43}m_j^{16} - 86866399652192256m_i^{42}m_j^{17} + 459411456587194368m_i^{41}m_j^{18} \\
& - 1248186463949438976m_i^{40}m_j^{19} + 1271683160945491968m_i^{39}m_j^{20} + 6274404654230159360m_i^{38}m_j^{21} \\
& - 9990651633068408832m_i^{37}m_j^{22} + 7685406085746642944m_i^{36}m_j^{23} + 77188110627786473472m_i^{35}m_j^{24}
\end{aligned}$$

$$\begin{aligned}
& + 7069842141790642176m_i^{34}m_j^{25} - 219750750834730033152m_i^{33}m_j^{26} - 241779604870510977024m_i^{32}m_j^{27} \\
& + 25022563292907044864m_i^{31}m_j^{28} + 334623157154945662976m_i^{30}m_j^{29} + 307707544789069922304m_i^{29}m_j^{30} \\
& + 392671335335864733696m_i^{28}m_j^{31} - 79002653718582198272m_i^{27}m_j^{32} + 240856189769236127744m_i^{26}m_j^{33} \\
& - 444714094210472009728m_i^{25}m_j^{34} + 124935506621440163840m_i^{24}m_j^{35} + 54847728927736758272m_i^{23}m_j^{36} \\
& - 105768805744248143872m_i^{22}m_j^{37} + 65374807617561821184m_i^{21}m_j^{38} - 23478058539062513664m_i^{20}m_j^{39} \\
& + 5899618295421550592m_i^{19}m_j^{40} - 994486819875037184m_i^{18}m_j^{41} + 117730938948837376m_i^{17}m_j^{42} \\
& - 8740656329957376m_i^{16}m_j^{43} + 362238858297344m_i^{15}m_j^{44} + 125084151808m_i^{14}m_j^{45} \\
& - 11653586165760m_i^{13}m_j^{46} + 236220650231808m_i^{12}m_j^{47} - 3596662790856704m_i^{11}m_j^{48} \\
& + 31080347579705344m_i^{10}m_j^{49} - 182178989766690816m_i^9m_j^{50} + 612704220569612288m_i^8m_j^{51} \\
& - 767512469172500480m_i^7m_j^{52} - 2008707746923909120m_i^6m_j^{53} + 8152541073323499520m_i^5m_j^{54} \\
& + 1688812864371595264m_i^4m_j^{55} - 19095365014285254656m_i^3m_j^{56} + 73016031615108999168m_i^2m_j^{57} \\
& + 282848546051928537088m_i^1m_j^{58} + 359401605614576398336m_i^0m_j^{59} + 201561123839386003456m_i^{59}m_j^0 \\
& + 56501126510483828736m_i^{58}m_j^1 - 20064215500338667520m_i^{57}m_j^2 + 70248943568022718464m_i^{56}m_j^3 \\
& - 117140950821576630272m_i^{55}m_j^4 + 132029806459447654400m_i^{54}m_j^5 - 198367026633370523648m_i^{53}m_j^6 \\
& + 260703778616484646912m_i^{52}m_j^7 - 34952505723364304896m_i^{51}m_j^8 - 81105323863671877632m_i^{50}m_j^9 \\
& + 83096015692328476672m_i^{49}m_j^{10} - 43108945152789149696m_i^{48}m_j^{11} + 13643067848496676864m_i^{47}m_j^{12} \\
& - 3065901149166766080m_i^{46}m_j^{13} + 467041593583593472m_i^{45}m_j^{14} - 49838889973542912m_i^{44}m_j^{15} \\
& + 3357866755586048m_i^{43}m_j^{16} - 123339484749824m_i^{42}m_j^{17} - 22783918080m_i^{41}m_j^{18} \\
& + 2594244043776m_i^{40}m_j^{19} - 60421695337472m_i^{39}m_j^{20} + 1023439695145984m_i^{38}m_j^{21} \\
& - 10160604407687168m_i^{37}m_j^{22} + 66221377458271232m_i^{36}m_j^{23} - 269000804830361600m_i^{35}m_j^{24} \\
& + 421292754352970752m_i^{34}m_j^{25} + 454841421606773760m_i^{33}m_j^{26} - 4843358183298491392m_i^{32}m_j^{27} \\
& - 2124419044610980864m_i^{31}m_j^{28} + 1367741399117827072m_i^{30}m_j^{29} - 69400540083066855424m_i^{29}m_j^{30} \\
& - 214036722190927133696m_i^{28}m_j^{31} - 267490244957297123328m_i^{27}m_j^{32} - 178163457278025413632m_i^{26}m_j^{33} \\
& - 90776108281355564032m_i^{25}m_j^{34} - 126821006937301629952m_i^{24}m_j^{35} - 130734310178508663808m_i^{23}m_j^{36} \\
& - 178129712376486181888m_i^{22}m_j^{37} + 42625319685053446144m_i^{21}m_j^{38} - 99328966671413216256m_i^{20}m_j^{39} \\
& + 156094534933589272576m_i^{19}m_j^{40} - 129211613103600847872m_i^{18}m_j^{41} - 13834447388356461568m_i^{17}m_j^{42} \\
& + 74110419883890504704m_i^{16}m_j^{43} - 56513542997140300800m_i^{15}m_j^{44} + 25115114253253133312m_i^{14}m_j^{45} \\
& - 7097647824235311104m_i^{13}m_j^{46} + 1426181350314970112m_i^{12}m_j^{47} - 197017521785458688m_i^{11}m_j^{48} \\
& + 18882856266654720m_i^{10}m_j^{49} - 1156908413103104m_i^9m_j^{50} + 37035758290944m_i^8m_j^{51} \\
& + 3658633216m_i^7m_j^{52} - 518574543872m_i^6m_j^{53} + 14020649275904m_i^5m_j^{54} \\
& - 265101746363392m_i^4m_j^{55} + 3030997185936896m_i^3m_j^{56} - 22038715437006848m_i^2m_j^{57} \\
& + 106360595884813824m_i^1m_j^{58} - 207803562295894016m_i^0m_j^{59} - 5854319211054592m_i^{59}m_j^0 \\
& + 2350189543474164736m_i^{58}m_j^1 + 883210556612537856m_i^{57}m_j^2 + 1532491582050465792m_i^{56}m_j^3 \\
& + 43939205260353183232m_i^{55}m_j^4 + 123438309069774838784m_i^{54}m_j^5 + 143194902607507703296m_i^{53}m_j^6 \\
& + 73779413914770363392m_i^{52}m_j^7 + 27252402975874851328m_i^{51}m_j^8 + 57955354001617000448m_i^{50}m_j^9 \\
& + 139442648861791984128m_i^{49}m_j^{10} + 123183957812608001024m_i^{48}m_j^{11} + 140256040699851512320m_i^{47}m_j^{12} \\
& - 37005859872786184192m_i^{46}m_j^{13} + 60244097208423545344m_i^{45}m_j^{14} - 105534654457089640448m_i^{44}m_j^{15} \\
& + 51373722175668370944m_i^{43}m_j^{16} + 31653047808881886208m_i^{42}m_j^{17} - 54369643329293841920m_i^{41}m_j^{18} \\
& + 34022050525237772288m_i^{40}m_j^{19} - 13116770180426790400m_i^{39}m_j^{20} + 3329182161159937024m_i^{38}m_j^{21} \\
& - 598540713464638976m_i^{37}m_j^{22} + 75019206216193024m_i^{36}m_j^{23} - 6442938927917568m_i^{35}m_j^{24} \\
& + 358536869807104m_i^{34}m_j^{25} - 9852744803328m_i^{33}m_j^{26} - 510490624m_i^{32}m_j^{27} \\
& + 92357138944m_i^{31}m_j^{28} - 2936965054720m_i^{30}m_j^{29} + 62284563945984m_i^{29}m_j^{30} \\
& - 823339335746304m_i^{28}m_j^{31} + 6702642152846848m_i^{27}m_j^{32} - 38015794325525248m_i^{26}m_j^{33} \\
& + 91783817541353984m_i^{25}m_j^{34} - 68729616906788096m_i^{24}m_j^{35} - 966955743659486208m_i^{23}m_j^{36}
\end{aligned}$$

$$\begin{aligned}
& -90092668157560576m_i^{36}m_j^{19} - 995466671021917184m_i^{35}m_j^{20} - 22368311836790297344m_i^{34}m_j^{21} \\
& - 58936645991231333888m_i^{33}m_j^{22} - 57125143359441452800m_i^{32}m_j^{23} - 4827830029254075904m_i^{31}m_j^{24} \\
& + 31508429171262157568m_i^{30}m_j^{25} + 2941754372576197632m_i^{29}m_j^{26} - 45839870360507007232m_i^{28}m_j^{27} \\
& - 92891947896847167488m_i^{27}m_j^{28} - 60464750494839038208m_i^{26}m_j^{29} - 78418655787766537728m_i^{25}m_j^{30} \\
& + 34137300652083101440m_i^{24}m_j^{31} - 35272602917994398208m_i^{23}m_j^{32} + 59723651814013903104m_i^{22}m_j^{33} \\
& - 12386431389294330880m_i^{21}m_j^{34} - 31012121155442380032m_i^{20}m_j^{35} + 34361092163576527872m_i^{19}m_j^{36} \\
& - 18333594440127400192m_i^{18}m_j^{37} + 6190264836796526080m_i^{17}m_j^{38} - 1413401086552632576m_i^{16}m_j^{39} \\
& + 227609074242733568m_i^{15}m_j^{40} - 25843216317333248m_i^{14}m_j^{41} + 1985652173453824m_i^{13}m_j^{42} \\
& - 99972238797312m_i^{12}m_j^{43} + 2322160565760m_i^{11}m_j^{44} + 60650240m_i^{10}m_j^{45} \\
& - 14503559168m_i^9m_j^{46} + 551870001792m_i^8m_j^{47} - 13209809229056m_i^7m_j^{48} \\
& + 203081970767104m_i^6m_j^{49} - 1858279739331584m_i^5m_j^{50} + 12305781212635392m_i^4m_j^{51} \\
& - 36319479565762048m_i^3m_j^{52} + 50340911496550912m_i^2m_j^{53} + 340013358041037568m_i^1m_j^{54} \\
& - 147570098605729280m_i^0m_j^{55} + 360552620231759616m_i^{56}m_j^{57} + 9813907290037231616m_i^{58}m_j^{59} \\
& + 24055958714380555776m_i^{60}m_j^{61} + 16070617954717726464m_i^{62}m_j^{63} - 19272644045018309632m_i^{64}m_j^{65} \\
& - 43069247856747293696m_i^{66}m_j^{67} - 30895146919789891328m_i^{68}m_j^{69} + 3410353206339716864m_i^{70}m_j^{71} \\
& + 21989625883069130496m_i^{72}m_j^{73} + 40510130085165979648m_i^{74}m_j^{75} + 14216745593130606592m_i^{76}m_j^{77} \\
& + 38171199721456739072m_i^{78}m_j^{79} - 24112590495768100352m_i^{80}m_j^{81} + 20635227783986810368m_i^{82}m_j^{83} \\
& - 27851821376624091904m_i^{84}m_j^{85} - 3225587674975298048m_i^{86}m_j^{87} + 22883571801923309824m_i^{88}m_j^{89} \\
& - 19238759229952861184m_i^{90}m_j^{91} + 8899558016777575936m_i^{92}m_j^{93} - 2651129091264031488m_i^{94}m_j^{95} \\
& + 544210578770116608m_i^{96}m_j^{97} - 78585401105731328m_i^{98}m_j^{99} + 8058309345277696m_i^{100}m_j^{101} \\
& - 553152227213696m_i^{102}m_j^{103} + 25033249908224m_i^{104}m_j^{105} - 483044034816m_i^{106}m_j^{107} \\
& - 5945856m_i^{108}m_j^{109} + 1979109440m_i^{110}m_j^{111} - 92232433600m_i^{112}m_j^{113} \\
& + 2512817674304m_i^{114}m_j^{115} - 45307577546624m_i^{116}m_j^{117} + 468105370556288m_i^{118}m_j^{119} \\
& - 3608281347155200m_i^{120}m_j^{121} + 12888894263811456m_i^{122}m_j^{123} - 25323702334787328m_i^{124}m_j^{125} \\
& - 100970224224203520m_i^{126}m_j^{127} + 136283593254697984m_i^{128}m_j^{129} - 89452679455637632m_i^{130}m_j^{131} \\
& - 3842899552963986432m_i^{132}m_j^{133} - 8585322689909761792m_i^{134}m_j^{135} - 1796959765342438144m_i^{136}m_j^{137} \\
& + 18087673216072491392m_i^{138}m_j^{139} + 31618004826883052032m_i^{140}m_j^{141} + 25249706850974775680m_i^{142}m_j^{143} \\
& + 10490584849073156992m_i^{144}m_j^{145} - 2620109696100144640m_i^{146}m_j^{147} - 2141525119789929728m_i^{148}m_j^{149} \\
& - 11295523683632500224m_i^{150}m_j^{151} + 3754064923165446400m_i^{152}m_j^{153} - 18530067196641647488m_i^{154}m_j^{155} \\
& + 13551858251030393088m_i^{156}m_j^{157} - 11309352303108813568m_i^{158}m_j^{159} + 10101697787477263360m_i^{160}m_j^{161} \\
& + 6938604261812505728m_i^{162}m_j^{163} - 14132228454883190784m_i^{164}m_j^{165} + 9665415831246584576m_i^{166}m_j^{167} \\
& - 3906308087719633152m_i^{168}m_j^{169} + 1032335378307570816m_i^{170}m_j^{171} - 190166699984978432m_i^{172}m_j^{173} \\
& + 24642943309298752m_i^{174}m_j^{175} - 2272238708980672m_i^{176}m_j^{177} + 139137821733312m_i^{178}m_j^{179} \\
& - 5607018381184m_i^{180}m_j^{181} + 88016749184m_i^{182}m_j^{183} + 457856m_i^{184}m_j^{185} \\
& - 229941824m_i^{186}m_j^{187} + 13559525984m_i^{188}m_j^{189} - 425135406144m_i^{190}m_j^{191} \\
& + 9096775799136m_i^{192}m_j^{193} - 106688966823296m_i^{194}m_j^{195} + 957346052020352m_i^{196}m_j^{197} \\
& - 4105187803983872m_i^{198}m_j^{199} + 10417835149762368m_i^{200}m_j^{201} + 24176218707586816m_i^{202}m_j^{203} \\
& - 75592230657972544m_i^{204}m_j^{205} + 15506285150787520m_i^{206}m_j^{207} + 1377434726829396800m_i^{208}m_j^{209} \\
& + 2710100262790591168m_i^{210}m_j^{211} - 1222959273818922176m_i^{212}m_j^{213} - 10747798879595956736m_i^{214}m_j^{215} \\
& - 16669459530548070592m_i^{216}m_j^{217} - 13382130901846841472m_i^{218}m_j^{219} - 5969098685746198912m_i^{220}m_j^{221} \\
& - 2611044462607214784m_i^{222}m_j^{223} - 934433460597786112m_i^{224}m_j^{225} - 5102682933740447040m_i^{226}m_j^{227} \\
& + 1627592144422758464m_i^{228}m_j^{229} - 6206570356962716160m_i^{230}m_j^{231} + 9319983361701540288m_i^{232}m_j^{233} \\
& - 6394076399857470720m_i^{234}m_j^{235} + 5365505041355413824m_i^{236}m_j^{237} - 2168515499614951104m_i^{238}m_j^{239} \\
& - 5872914112901319488m_i^{240}m_j^{241} + 7599318482044740160m_i^{242}m_j^{243} - 4382784385167317056m_i^{244}m_j^{245}
\end{aligned}$$

$$\begin{aligned}
& + 1553654995051750400m_i^{15}m_j^{37} - 365603429519013056m_i^{14}m_j^{38} + 60282951673813184m_i^{13}m_j^{39} \\
& - 7011046226807008m_i^{12}m_j^{40} + 578090077840384m_i^{11}m_j^{41} - 31515816469344m_i^{10}m_j^{42} \\
& + 1116375670208m_i^9m_j^{43} - 13878377408m_i^8m_j^{44} - 25088m_i^{46}m_j^5 \\
& + 22040256m_i^{45}m_j^6 - 1726660784m_i^{44}m_j^7 + 63243910592m_i^{43}m_j^8 \\
& - 1632732608832m_i^{42}m_j^9 + 21880615871840m_i^{41}m_j^{10} - 229277791510912m_i^{40}m_j^{11} \\
& + 1173203194819328m_i^{39}m_j^{12} - 3686430878360928m_i^{38}m_j^{13} - 3976604324635168m_i^{37}m_j^{14} \\
& + 32361038604480896m_i^{36}m_j^{15} - 2951682133158496m_i^{35}m_j^{16} - 458964738619588128m_i^{34}m_j^{17} \\
& - 764170125394232544m_i^{33}m_j^{18} + 1063173215738589056m_i^{32}m_j^{19} + 4954067042578244320m_i^{31}m_j^{20} \\
& + 6999579070267721056m_i^{30}m_j^{21} + 4863420299378208352m_i^{29}m_j^{22} + 1562887215020019808m_i^{28}m_j^{23} \\
& - 30225644731214816m_i^{27}m_j^{24} + 1421404642092217760m_i^{26}m_j^{25} + 1794655145457861600m_i^{25}m_j^{26} \\
& + 4695761324562516928m_i^{24}m_j^{27} - 90659423230905376m_i^{23}m_j^{28} + 4244652329098849376m_i^{22}m_j^{29} \\
& - 4594772844530996064m_i^{21}m_j^{30} + 2583022182983267456m_i^{20}m_j^{31} - 2019178261582040608m_i^{19}m_j^{32} \\
& - 480454556786097376m_i^{18}m_j^{33} + 3732358000537415520m_i^{17}m_j^{34} - 3625353706424450304m_i^{16}m_j^{35} \\
& + 1798077888069510560m_i^{15}m_j^{36} - 560363202929575520m_i^{14}m_j^{37} + 117648337571086432m_i^{13}m_j^{38} \\
& - 17311217705798064m_i^{12}m_j^{39} + 1805449162571424m_i^{11}m_j^{40} - 132217268397408m_i^{10}m_j^{41} \\
& + 6399918761024m_i^9m_j^{42} - 195794471104m_i^8m_j^{43} + 1858795680m_i^7m_j^{44} \\
& + 784m_i^{46}m_j^4 - 1659168m_i^{45}m_j^5 + 186486568m_i^{44}m_j^6 \\
& - 8145872064m_i^{43}m_j^7 + 259713676136m_i^{42}m_j^8 - 4009803163624m_i^{41}m_j^9 \\
& + 49381485073248m_i^{40}m_j^{10} - 300421550280440m_i^{39}m_j^{11} + 1144533020720320m_i^{38}m_j^{12} \\
& + 31232227052608m_i^{37}m_j^{13} - 11449917397511104m_i^{36}m_j^{14} + 1677385215310368m_i^{35}m_j^{15} \\
& + 143380593638329744m_i^{34}m_j^{16} + 192724331458943968m_i^{33}m_j^{17} - 523894956439475904m_i^{32}m_j^{18} \\
& - 1903011951067980416m_i^{31}m_j^{19} - 2391395802339722896m_i^{30}m_j^{20} - 1176252976947181056m_i^{29}m_j^{21} \\
& + 452682374812044944m_i^{28}m_j^{22} + 922887348212604064m_i^{27}m_j^{23} + 461160785971245600m_i^{26}m_j^{24} \\
& - 1028673819383337136m_i^{25}m_j^{25} - 1008134595700734016m_i^{24}m_j^{26} - 2613971867573720752m_i^{23}m_j^{27} \\
& + 222734019396556880m_i^{22}m_j^{28} - 2244930242790904384m_i^{21}m_j^{29} + 2076727098691467136m_i^{20}m_j^{30} \\
& - 864681864351913632m_i^{19}m_j^{31} + 48325253646553352m_i^{18}m_j^{32} + 895596858278500704m_i^{17}m_j^{33} \\
& - 1982680132181877440m_i^{16}m_j^{34} + 1549033411789225152m_i^{15}m_j^{35} - 667713324205003328m_i^{14}m_j^{36} \\
& + 183213092987466016m_i^{13}m_j^{37} - 34329665785049464m_i^{12}m_j^{38} + 4492247474245792m_i^{11}m_j^{39} \\
& - 419265215070600m_i^{10}m_j^{40} + 27044222623576m_i^9m_j^{41} - 1157669132704m_i^8m_j^{42} \\
& + 29852540776m_i^7m_j^{43} - 205331280m_i^6m_j^{44} + 88844m_i^{45}m_j^4 \\
& - 16534140m_i^{44}m_j^5 + 888318132m_i^{43}m_j^6 - 36189825928m_i^{42}m_j^7 \\
& + 650597831268m_i^{41}m_j^8 - 9514395336464m_i^{40}m_j^9 + 68732934774796m_i^{39}m_j^{10} \\
& - 314253152808420m_i^{38}m_j^{11} + 296846699736016m_i^{37}m_j^{12} + 3439089155339312m_i^{36}m_j^{13} \\
& - 1126364022951680m_i^{35}m_j^{14} - 41968360214325808m_i^{34}m_j^{15} - 43217350976555648m_i^{33}m_j^{16} \\
& + 201118362322914224m_i^{32}m_j^{17} + 626682750822073024m_i^{31}m_j^{18} + 673089092949090096m_i^{30}m_j^{19} \\
& + 64424010401775928m_i^{29}m_j^{20} - 635662675060804424m_i^{28}m_j^{21} - 892999699299925240m_i^{27}m_j^{22} \\
& - 570423006514988576m_i^{26}m_j^{23} - 192640742218743576m_i^{25}m_j^{24} + 536995261787172336m_i^{24}m_j^{25} \\
& + 280962595665058488m_i^{23}m_j^{26} + 1177182528313714344m_i^{22}m_j^{27} - 287057090241870144m_i^{21}m_j^{28} \\
& + 1026653480886042864m_i^{20}m_j^{29} - 819866772310624480m_i^{19}m_j^{30} + 199098764972100080m_i^{18}m_j^{31} \\
& + 41637432964214560m_i^{17}m_j^{32} - 645344107869015472m_i^{16}m_j^{33} + 913863210973152032m_i^{15}m_j^{34} \\
& - 595452109435744848m_i^{14}m_j^{35} + 224238307804207452m_i^{13}m_j^{36} - 54222719969624924m_i^{12}m_j^{37} \\
& + 9055507416187940m_i^{11}m_j^{38} - 1049606694834712m_i^{10}m_j^{39} + 87339992961044m_i^9m_j^{40} \\
& - 4912408561888m_i^8m_j^{41} + 184881522524m_i^7m_j^{42} - 3882500516m_i^6m_j^{43} \\
& + 17829840m_i^5m_j^{44} - 2716m_i^{45}m_j^3 + 1145378m_i^{44}m_j^4
\end{aligned}$$

$$\begin{aligned}
& -79369516m_i^{43}m_j^5 + 4349452994m_i^{42}m_j^6 - 92331488952m_i^{41}m_j^7 \\
& + 1628297746830m_i^{40}m_j^8 - 13987266575768m_i^{39}m_j^9 + 76469505418614m_i^{38}m_j^{10} \\
& - 145814657223880m_i^{37}m_j^{11} - 884960681327816m_i^{36}m_j^{12} + 557591710130728m_i^{35}m_j^{13} \\
& + 11436820601739384m_i^{34}m_j^{14} + 8374193204510568m_i^{33}m_j^{15} - 65019027105376136m_i^{32}m_j^{16} \\
& - 180359800568805448m_i^{31}m_j^{17} - 151708178097974360m_i^{30}m_j^{18} + 101088354511391552m_i^{29}m_j^{19} \\
& + 378041335641076636m_i^{28}m_j^{20} + 467812807805396048m_i^{27}m_j^{21} + 395771438238790956m_i^{26}m_j^{22} \\
& + 182395431152667352m_i^{25}m_j^{23} + 90056773103645988m_i^{24}m_j^{24} - 215551909229602200m_i^{23}m_j^{25} \\
& - 298904954280924m_i^{22}m_j^{26} - 479324628253032184m_i^{21}m_j^{27} + 205393277419806632m_i^{20}m_j^{28} \\
& - 416359346072261352m_i^{19}m_j^{29} + 270005497976479928m_i^{18}m_j^{30} + 6750358813714552m_i^{17}m_j^{31} \\
& - 132207923130160488m_i^{16}m_j^{32} + 346619550673114344m_i^{15}m_j^{33} - 371568599137295256m_i^{14}m_j^{34} \\
& + 206215955956505820m_i^{13}m_j^{35} - 67970705574701870m_i^{12}m_j^{36} + 14488857596427676m_i^{11}m_j^{37} \\
& - 2150059857398558m_i^{10}m_j^{38} + 219710851289088m_i^9m_j^{39} - 16204410869698m_i^8m_j^{40} \\
& + 784920390288m_i^7m_j^{41} - 25740380074m_i^6m_j^{42} + 418144832m_i^5m_j^{43} \\
& - 1105776m_i^4m_j^{44} - 56043m_i^3m_j^{45} + 5498128m_i^2m_j^{46} \\
& - 441048523m_i^1m_j^{47} + 11272209900m_i^0m_j^{48} - 245193123753m_i^{-1}m_j^{49} \\
& + 2515384124136m_i^{39}m_j^8 - 16464763435773m_i^{38}m_j^9 + 48000791058100m_i^{37}m_j^{10} \\
& + 194866444524748m_i^{36}m_j^{11} - 210770404196696m_i^{35}m_j^{12} - 2871400520933320m_i^{34}m_j^{13} \\
& - 1297564683734776m_i^{33}m_j^{14} + 18301873334043920m_i^{32}m_j^{15} + 45756965480398600m_i^{31}m_j^{16} \\
& + 25347124949675168m_i^{30}m_j^{17} - 64632998023767592m_i^{29}m_j^{18} - 156434470442570886m_i^{28}m_j^{19} \\
& - 189147038717383560m_i^{27}m_j^{20} - 159559170661881622m_i^{26}m_j^{21} - 120755855585835584m_i^{25}m_j^{22} \\
& - 46457108323302474m_i^{24}m_j^{23} - 53595414724842920m_i^{23}m_j^{24} + 84030809362022270m_i^{22}m_j^{25} \\
& - 48139981576469984m_i^{21}m_j^{26} + 184488808934908488m_i^{20}m_j^{27} - 107492359054133784m_i^{19}m_j^{28} \\
& + 147519551700534960m_i^{18}m_j^{29} - 66523167998359496m_i^{17}m_j^{30} - 42312462513165304m_i^{16}m_j^{31} \\
& + 94872970572315928m_i^{15}m_j^{32} - 154574925164906472m_i^{14}m_j^{33} + 134285050217633512m_i^{13}m_j^{34} \\
& - 64293005913728151m_i^{12}m_j^{35} + 18539899809996888m_i^{11}m_j^{36} - 3482421585795079m_i^{10}m_j^{37} \\
& + 456867970053908m_i^9m_j^{38} - 40932688966437m_i^8m_j^{39} + 2651811305120m_i^7m_j^{40} \\
& - 108964528953m_i^6m_j^{41} + 3070261868m_i^5m_j^{42} - 35587340m_i^4m_j^{43} \\
& + 39312m_i^3m_j^{44} + 1568m_i^2m_j^{45} - 266882m_i^1m_j^{46} \\
& + 36535859m_i^0m_j^{47} - 1156257374m_i^{-1}m_j^{48} + 32078209698m_i^{-2}m_j^{49} \\
& - 396121826370m_i^{39}m_j^7 + 3122559646613m_i^{38}m_j^8 - 12575091614394m_i^{37}m_j^9 \\
& - 36300991193134m_i^{36}m_j^{10} + 63708070940432m_i^{35}m_j^{11} + 657064143439364m_i^{34}m_j^{12} \\
& + 115772706378888m_i^{33}m_j^{13} - 4546091077750532m_i^{32}m_j^{14} - 10311389765493392m_i^{31}m_j^{15} \\
& - 2005775130900700m_i^{30}m_j^{16} + 25228634046272696m_i^{29}m_j^{17} + 51897926156763108m_i^{28}m_j^{18} \\
& + 59205837647721916m_i^{27}m_j^{19} + 52731186882805026m_i^{26}m_j^{20} + 37691261617201724m_i^{25}m_j^{21} \\
& + 34366577886429392m_i^{24}m_j^{22} + 10234201565829916m_i^{23}m_j^{23} + 26332180701913046m_i^{22}m_j^{24} \\
& - 35247997856367980m_i^{21}m_j^{25} + 34006438427800248m_i^{20}m_j^{26} - 67443421800570656m_i^{19}m_j^{27} \\
& + 45165990307789476m_i^{18}m_j^{28} - 43906259737613848m_i^{17}m_j^{29} + 6723414243527332m_i^{16}m_j^{30} \\
& + 30970949881355136m_i^{15}m_j^{31} - 48279029679705900m_i^{14}m_j^{32} + 59350471330899992m_i^{13}m_j^{33} \\
& - 43261215476717764m_i^{12}m_j^{34} + 18002523581287766m_i^{11}m_j^{35} - 4531298564891645m_i^{10}m_j^{36} \\
& + 749013392437170m_i^9m_j^{37} - 86221553305362m_i^8m_j^{38} + 6727986216502m_i^7m_j^{39} \\
& - 377633663843m_i^6m_j^{40} + 12920566102m_i^5m_j^{41} - 305174474m_i^4m_j^{42} \\
& + 2175360m_i^3m_j^{43} + 6948m_i^2m_j^{44} - 2347592m_i^1m_j^{45} \\
& + 96194368m_i^0m_j^{46} - 3584088500m_i^{-1}m_j^{47} + 53941892766m_i^{-2}m_j^{48} \\
& - 517719619400m_i^{-3}m_j^{49} + 2749881730102m_i^{-4}m_j^{50} + 5571675416632m_i^{-5}m_j^{51}
\end{aligned}$$

$$\begin{aligned}
& -15801236910960m_i^{35}m_j^{10} - 135635366230216m_i^{34}m_j^{11} + 14954058599692m_i^{33}m_j^{12} \\
& + 1001721991197056m_i^{32}m_j^{13} + 2069019425924780m_i^{31}m_j^{14} - 500815599552176m_i^{30}m_j^{15} \\
& - 7637179271882716m_i^{29}m_j^{16} - 14159283297850384m_i^{28}m_j^{17} - 15461905910938900m_i^{27}m_j^{18} \\
& - 12804210412763712m_i^{26}m_j^{19} - 10685651663316260m_i^{25}m_j^{20} - 7564561258302280m_i^{24}m_j^{21} \\
& - 10671938269639128m_i^{23}m_j^{22} + 98820166102656m_i^{22}m_j^{23} - 11080646816933712m_i^{21}m_j^{24} \\
& + 14974899385221408m_i^{20}m_j^{25} - 16529164572295884m_i^{19}m_j^{26} + 22845019112648064m_i^{18}m_j^{27} \\
& - 15453773249601436m_i^{17}m_j^{28} + 9909474344672224m_i^{16}m_j^{29} + 4488850445294964m_i^{15}m_j^{30} \\
& - 15777254779739856m_i^{14}m_j^{31} + 19921138063935260m_i^{13}m_j^{32} - 19904750793733360m_i^{12}m_j^{33} \\
& + 12420000569499552m_i^{11}m_j^{34} - 4509563171191992m_i^{10}m_j^{35} + 986677572604020m_i^9m_j^{36} \\
& - 143190679106148m_i^8m_j^{37} + 14306077568282m_i^7m_j^{38} - 964709879224m_i^6m_j^{39} \\
& + 45928717274m_i^5m_j^{40} - 1281005144m_i^4m_j^{41} + 24265356m_i^3m_j^{42} \\
& - 77688m_i^2m_j^{43} + 106149m_i^{42}m_j^2 - 6122520m_i^{41}m_j^3 \\
& + 333818990m_i^{40}m_j^4 - 6240068428m_i^{39}m_j^5 + 74221556279m_i^{38}m_j^6 \\
& - 509567544256m_i^{37}m_j^7 - 662990219870m_i^{36}m_j^8 + 3247832937500m_i^{35}m_j^9 \\
& + 25036851003580m_i^{34}m_j^{10} - 10644324568276m_i^{33}m_j^{11} - 195226368597848m_i^{32}m_j^{12} \\
& - 371075517254972m_i^{31}m_j^{13} + 285711475971968m_i^{30}m_j^{14} + 1892731552698412m_i^{29}m_j^{15} \\
& + 3286581533242640m_i^{28}m_j^{16} + 3278596480102980m_i^{27}m_j^{17} + 2600887846699682m_i^{26}m_j^{18} \\
& + 1772390626629484m_i^{25}m_j^{19} + 2174353146676932m_i^{24}m_j^{20} + 1439799193021660m_i^{23}m_j^{21} \\
& + 3120021199800214m_i^{22}m_j^{22} - 1610075013894244m_i^{21}m_j^{23} + 4370714906935252m_i^{20}m_j^{24} \\
& - 5955034959297364m_i^{19}m_j^{25} + 6567239038159168m_i^{18}m_j^{26} - 6987182575874364m_i^{17}m_j^{27} \\
& + 4181865573861704m_i^{16}m_j^{28} - 1034134150972052m_i^{15}m_j^{29} - 3675019465245832m_i^{14}m_j^{30} \\
& + 6476171256120068m_i^{13}m_j^{31} - 6953499663525216m_i^{12}m_j^{32} + 5860181813015212m_i^{11}m_j^{33} \\
& - 3168847405937167m_i^{10}m_j^{34} + 1005050199834044m_i^9m_j^{35} - 189938914300450m_i^8m_j^{36} \\
& + 24114099010368m_i^7m_j^{37} - 2057854478789m_i^6m_j^{38} + 118839950836m_i^5m_j^{39} \\
& - 4633358374m_i^4m_j^{40} + 102853960m_i^3m_j^{41} - 1408052m_i^2m_j^{42} \\
& - 2688m_i^{42}m_j + 267372m_i^{41}m_j^2 - 25018068m_i^{40}m_j^3 \\
& + 597722182m_i^{39}m_j^4 - 9057698792m_i^{38}m_j^5 + 80071952062m_i^{37}m_j^6 \\
& + 50585383132m_i^{36}m_j^7 - 552544521036m_i^{35}m_j^8 - 4099778590292m_i^{34}m_j^9 \\
& + 3209575098816m_i^{33}m_j^{10} + 33380666679844m_i^{32}m_j^{11} + 59589507248464m_i^{31}m_j^{12} \\
& - 82475676728804m_i^{30}m_j^{13} - 394869897713256m_i^{29}m_j^{14} - 648425508750412m_i^{28}m_j^{15} \\
& - 583408227307704m_i^{27}m_j^{16} - 373977921306020m_i^{26}m_j^{17} - 250765007575064m_i^{25}m_j^{18} \\
& - 209507552777908m_i^{24}m_j^{19} - 594953240347924m_i^{23}m_j^{20} - 65688295415076m_i^{22}m_j^{21} \\
& - 899598369466940m_i^{21}m_j^{22} + 995924802671676m_i^{20}m_j^{23} - 1638025069770864m_i^{19}m_j^{24} \\
& + 2102598548049316m_i^{18}m_j^{25} - 2182409159643856m_i^{17}m_j^{26} + 1851888522168972m_i^{16}m_j^{27} \\
& - 792174807841120m_i^{15}m_j^{28} - 427653758170540m_i^{14}m_j^{29} + 1712632352286904m_i^{13}m_j^{30} \\
& - 2232675243279556m_i^{12}m_j^{31} + 2084943601337144m_i^{11}m_j^{32} - 1513839411070668m_i^{10}m_j^{33} \\
& + 714907667448396m_i^9m_j^{34} - 197851239046184m_i^8m_j^{35} + 31993747038174m_i^7m_j^{36} \\
& - 3536219395172m_i^6m_j^{37} + 251687712302m_i^5m_j^{38} - 12335609608m_i^4m_j^{39} \\
& + 370899516m_i^3m_j^{40} - 6496096m_i^2m_j^{41} + 50544m_i^{42}m_j \\
& - 5976m_i^{41}m_j + 1425004m_i^{40}m_j^2 - 45584732m_i^{39}m_j^3 \\
& + 919893912m_i^{38}m_j^4 - 10575202332m_i^{37}m_j^5 + 290155488m_i^{36}m_j^6 \\
& + 76891564840m_i^{35}m_j^7 + 590935766516m_i^{34}m_j^8 - 705671378304m_i^{33}m_j^9 \\
& - 4929562630672m_i^{32}m_j^{10} - 8629373542592m_i^{31}m_j^{11} + 17835816615724m_i^{30}m_j^{12} \\
& + 69283863476512m_i^{29}m_j^{13} + 110588771510408m_i^{28}m_j^{14} + 82028966834880m_i^{27}m_j^{15}
\end{aligned}$$

$$\begin{aligned}
& + 41278741749988m_i^{26}m_j^{16} - 2856688046608m_i^{25}m_j^{17} + 36897118184184m_i^{24}m_j^{18} \\
& + 37124202499848m_i^{23}m_j^{19} + 155484708650332m_i^{22}m_j^{20} - 105734991918104m_i^{21}m_j^{21} \\
& + 288657983222264m_i^{20}m_j^{22} - 422178723962064m_i^{19}m_j^{23} + 575161694522444m_i^{18}m_j^{24} \\
& - 657946139799552m_i^{17}m_j^{25} + 605378435758992m_i^{16}m_j^{26} - 394662909423200m_i^{15}m_j^{27} \\
& + 38294908016932m_i^{14}m_j^{28} + 329099862991168m_i^{13}m_j^{29} - 610912285330568m_i^{12}m_j^{30} \\
& + 655540152766656m_i^{11}m_j^{31} - 538989897633492m_i^{10}m_j^{32} + 341573572689128m_i^9m_j^{33} \\
& - 141540494400996m_i^8m_j^{34} + 34071697918324m_i^7m_j^{35} - 4648516990276m_i^6m_j^{36} \\
& + 444525715924m_i^5m_j^{37} - 25399877336m_i^4m_j^{38} + 1047901736m_i^3m_j^{39} \\
& - 21441616m_i^2m_j^{40} + 299520m_im_j^{41} - 55680m_im_j^{40} \\
& + 2594208m_i^{39}m_j^2 - 75159612m_i^{38}m_j^3 + 1152468464m_i^{37}m_j^4 \\
& - 817771820m_i^{36}m_j^5 - 8537574700m_i^{35}m_j^6 - 74286394032m_i^{34}m_j^7 \\
& + 123443757724m_i^{33}m_j^8 + 613841747848m_i^{32}m_j^9 + 1137050823684m_i^{31}m_j^{10} \\
& - 3120924481384m_i^{30}m_j^{11} - 10171820748980m_i^{29}m_j^{12} - 16215448330264m_i^{28}m_j^{13} \\
& - 8973973456460m_i^{27}m_j^{14} - 1543711681544m_i^{26}m_j^{15} + 5756363371772m_i^{25}m_j^{16} \\
& + 3263386843144m_i^{24}m_j^{17} - 16017121825644m_i^{23}m_j^{18} + 1074045832704m_i^{22}m_j^{19} \\
& - 38749980280612m_i^{21}m_j^{20} + 57734373609712m_i^{20}m_j^{21} - 95599099621300m_i^{19}m_j^{22} \\
& + 140636914846040m_i^{18}m_j^{23} - 175885700557116m_i^{17}m_j^{24} + 178476492727192m_i^{16}m_j^{25} \\
& - 137439924300916m_i^{15}m_j^{26} + 56906592169128m_i^{14}m_j^{27} + 44222683331684m_i^{13}m_j^{28} \\
& - 132190816355432m_i^{12}m_j^{29} + 177659681476156m_i^{11}m_j^{30} - 164084246001240m_i^{10}m_j^{31} \\
& + 119696003665620m_i^9m_j^{32} - 66758395090280m_i^8m_j^{33} + 24337073889596m_i^7m_j^{34} \\
& - 5069006399748m_i^6m_j^{35} + 571166474884m_i^5m_j^{36} - 46937438532m_i^4m_j^{37} \\
& + 2019955184m_i^3m_j^{38} - 70442568m_i^2m_j^{39} + 742864m_im_j^{40} \\
& - 10816m_i^{41} + 1152m_i^{40} - 97200m_i^{39}m_j \\
& + 4677028m_i^{38}m_j^2 - 100419328m_i^{37}m_j^3 + 145124136m_i^{36}m_j^4 \\
& + 723070292m_i^{35}m_j^5 + 8030035704m_i^{34}m_j^6 - 17492229580m_i^{33}m_j^7 \\
& - 61939871056m_i^{32}m_j^8 - 138259275284m_i^{31}m_j^9 + 456344983724m_i^{30}m_j^{10} \\
& + 1209532876244m_i^{29}m_j^{11} + 2088641437624m_i^{28}m_j^{12} + 539735465084m_i^{27}m_j^{13} \\
& - 97037916956m_i^{26}m_j^{14} - 2143728539388m_i^{25}m_j^{15} - 48353278656m_i^{24}m_j^{16} \\
& - 291225831620m_i^{23}m_j^{17} + 5471553183060m_i^{22}m_j^{18} - 5608736338588m_i^{21}m_j^{19} \\
& + 12590096338936m_i^{20}m_j^{20} - 22273439295756m_i^{19}m_j^{21} + 31540152535540m_i^{18}m_j^{22} \\
& - 40631876851812m_i^{17}m_j^{23} + 45523072784496m_i^{16}m_j^{24} - 40074853873084m_i^{15}m_j^{25} \\
& + 23322377466244m_i^{14}m_j^{26} - 97823446020m_i^{13}m_j^{27} - 23099595267128m_i^{12}m_j^{28} \\
& + 39181947562772m_i^{11}m_j^{29} - 4279146277652m_i^{10}m_j^{30} + 34752440575276m_i^9m_j^{31} \\
& - 22599523446240m_i^8m_j^{32} + 11152436588868m_i^7m_j^{33} - 3580809655592m_i^6m_j^{34} \\
& + 640392257228m_i^5m_j^{35} - 57560941984m_i^4m_j^{36} + 4037073704m_i^3m_j^{37} \\
& - 115075740m_i^2m_j^{38} + 3453184m_im_j^{39} + 1728m_i^{39} \\
& - 200160m_i^{38}m_j + 6624288m_i^{37}m_j^2 - 15043688m_i^{36}m_j^3 \\
& - 42769728m_i^{35}m_j^4 - 728270264m_i^{34}m_j^5 + 1987584960m_i^{33}m_j^6 \\
& + 4712712112m_i^{32}m_j^7 + 15534462232m_i^{31}m_j^8 - 56129001288m_i^{30}m_j^9 \\
& - 109728454800m_i^{29}m_j^{10} - 238620327744m_i^{28}m_j^{11} + 33026077064m_i^{27}m_j^{12} \\
& + 1938206024m_i^{26}m_j^{13} + 399827577056m_i^{25}m_j^{14} + 34583580752m_i^{24}m_j^{15} \\
& - 76995593640m_i^{23}m_j^{16} - 153497571016m_i^{22}m_j^{17} - 905678477616m_i^{21}m_j^{18} \\
& + 1872389687184m_i^{20}m_j^{19} - 3476830606840m_i^{19}m_j^{20} + 6236281996312m_i^{18}m_j^{21} \\
& - 8679833548736m_i^{17}m_j^{22} + 10075349258832m_i^{16}m_j^{23} - 9929278652792m_i^{15}m_j^{24}
\end{aligned}$$

$$\begin{aligned}
& + 7285876799080m_i^{14}m_j^{25} - 2408876386160m_i^{13}m_j^{26} - 2907921027520m_i^{12}m_j^{27} \\
& + 7098953309912m_i^{11}m_j^{28} - 9114253677224m_i^{10}m_j^{29} + 8496404652512m_i^9m_j^{30} \\
& - 6136564867408m_i^8m_j^{31} + 3567090510664m_i^7m_j^{32} - 1562485851768m_i^6m_j^{33} \\
& + 441884275472m_i^5m_j^{34} - 67183520808m_i^4m_j^{35} + 4540965592m_i^3m_j^{36} \\
& - 272094464m_i^2m_j^{37} + 3857216m_im_j^{38} - 116480m_j^{39} \\
& + 4608m_i^{38} - 297792m_i^{37}m_j + 965136m_i^{36}m_j^2 \\
& + 1443440m_i^{35}m_j^3 + 52720276m_i^{34}m_j^4 - 174157224m_i^{33}m_j^5 \\
& - 228186420m_i^{32}m_j^6 - 1571598816m_i^{31}m_j^7 + 5807751908m_i^{30}m_j^8 \\
& + 6238031944m_i^{29}m_j^9 + 25126807548m_i^{28}m_j^{10} - 17159581360m_i^{27}m_j^{11} \\
& + 14191092900m_i^{26}m_j^{12} - 71141275496m_i^{25}m_j^{13} + 37223117660m_i^{24}m_j^{14} \\
& - 72734523776m_i^{23}m_j^{15} + 176274451092m_i^{22}m_j^{16} - 213239067064m_i^{21}m_j^{17} \\
& + 417444706604m_i^{20}m_j^{18} - 731608009008m_i^{19}m_j^{19} + 1073203425244m_i^{18}m_j^{20} \\
& - 1547119079608m_i^{17}m_j^{21} + 1931623217316m_i^{16}m_j^{22} - 1978078887264m_i^{15}m_j^{23} \\
& + 1646714412140m_i^{14}m_j^{24} - 898118845416m_i^{13}m_j^{25} - 126729151180m_i^{12}m_j^{26} \\
& + 1039025380784m_i^{11}m_j^{27} - 1577367916724m_i^{10}m_j^{28} + 1667047742920m_i^9m_j^{29} \\
& - 1358423202252m_i^8m_j^{30} + 877972908736m_i^7m_j^{31} - 456903061220m_i^6m_j^{32} \\
& + 177996806296m_i^5m_j^{33} - 44320076140m_i^4m_j^{34} + 5655894464m_i^3m_j^{35} \\
& - 254681952m_i^2m_j^{36} + 13168640m_im_j^{37} + 6912m_i^{37} \\
& - 31104m_i^{36}m_j - 9792m_i^{35}m_j^2 - 2754816m_i^{34}m_j^3 \\
& + 10799392m_i^{33}m_j^4 + 2849536m_i^{32}m_j^5 + 131936672m_i^{31}m_j^6 \\
& - 490096832m_i^{30}m_j^7 - 2812704m_i^{29}m_j^8 - 2469088448m_i^{28}m_j^9 \\
& + 3296422368m_i^{27}m_j^{10} - 4638624768m_i^{26}m_j^{11} + 11919305504m_i^{25}m_j^{12} \\
& - 12795324800m_i^{24}m_j^{13} + 19023103392m_i^{23}m_j^{14} - 25854961472m_i^{22}m_j^{15} \\
& + 13276195552m_i^{21}m_j^{16} + 8873341632m_i^{20}m_j^{17} - 59958387616m_i^{19}m_j^{18} \\
& + 147057180800m_i^{18}m_j^{19} - 239706800288m_i^{17}m_j^{20} + 324292639232m_i^{16}m_j^{21} \\
& - 367308085792m_i^{15}m_j^{22} + 329589694656m_i^{14}m_j^{23} - 218446823008m_i^{13}m_j^{24} \\
& + 60006076096m_i^{12}m_j^{25} + 104829867168m_i^{11}m_j^{26} - 219700262144m_i^{10}m_j^{27} \\
& + 259518843232m_i^9m_j^{28} - 234173638528m_i^8m_j^{29} + 168542499552m_i^7m_j^{30} \\
& - 97604659392m_i^6m_j^{31} + 45433766816m_i^5m_j^{32} - 15747220288m_i^4m_j^{33} \\
& + 3453467488m_i^3m_j^{34} - 366329472m_i^2m_j^{35} + 8332032m_im_j^{36} \\
& - 416768m_j^{37} + 78336m_i^{34}m_j^2 - 373248m_i^{33}m_j^3 \\
& + 344704m_i^{32}m_j^4 - 8004096m_i^{31}m_j^5 + 31220416m_i^{30}m_j^6 \\
& - 37057408m_i^{29}m_j^7 + 202729600m_i^{28}m_j^8 - 379309952m_i^{27}m_j^9 \\
& + 663950272m_i^{26}m_j^{10} - 1377344256m_i^{25}m_j^{11} + 1931592320m_i^{24}m_j^{12} \\
& - 3004618496m_i^{23}m_j^{13} + 4517573056m_i^{22}m_j^{14} - 5862917760m_i^{21}m_j^{15} \\
& + 8617643136m_i^{20}m_j^{16} - 12705085056m_i^{19}m_j^{17} + 18773336768m_i^{18}m_j^{18} \\
& - 27892452352m_i^{17}m_j^{19} + 37115294592m_i^{16}m_j^{20} - 43212252160m_i^{15}m_j^{21} \\
& + 42719807552m_i^{14}m_j^{22} - 32569292928m_i^{13}m_j^{23} + 14911647104m_i^{12}m_j^{24} \\
& + 4957703040m_i^{11}m_j^{25} - 21389612224m_i^{10}m_j^{26} + 29645633792m_i^9m_j^{27} \\
& - 29153556608m_i^8m_j^{28} + 22921063680m_i^7m_j^{29} - 14676012224m_i^6m_j^{30} \\
& + 7626531456m_i^5m_j^{31} - 3175047808m_i^4m_j^{32} + 979856000m_i^3m_j^{33} \\
& - 189694912m_i^2m_j^{34} + 16625664m_im_j^{35} + 221184m_i^{31}m_j^4 \\
& - 1119744m_i^{30}m_j^5 + 3077120m_i^{29}m_j^6 - 12658176m_i^{28}m_j^7 \\
& + 33667072m_i^{27}m_j^8 - 80658432m_i^{26}m_j^9 + 181603328m_i^{25}m_j^{10}
\end{aligned}$$

$$\begin{aligned}
& -345007104m_i^{24}m_j^{11} + 595019776m_i^{23}m_j^{12} - 886184448m_i^{22}m_j^{13} \\
& + 1073519616m_i^{21}m_j^{14} - 979673600m_i^{20}m_j^{15} + 348270592m_i^{19}m_j^{16} \\
& + 924099584m_i^{18}m_j^{17} - 2657950720m_i^{17}m_j^{18} + 4448770048m_i^{16}m_j^{19} \\
& - 5662347264m_i^{15}m_j^{20} + 5789808128m_i^{14}m_j^{21} - 4720129024m_i^{13}m_j^{22} \\
& + 2745300480m_i^{12}m_j^{23} - 514516992m_i^{11}m_j^{24} - 1297426432m_i^{10}m_j^{25} \\
& + 2303276032m_i^9m_j^{26} - 2460001280m_i^8m_j^{27} + 2025351168m_i^7m_j^{28} \\
& - 1368000000m_i^6m_j^{29} + 765355008m_i^5m_j^{30} - 351170048m_i^4m_j^{31} \\
& + 129269760m_i^3m_j^{32} - 35453952m_i^2m_j^{33} + 6183936m_im_j^{34} \\
& - 495616m_j^{35}
\end{aligned}$$

$\text{Res}_{m_i}(\Omega, R)$

The univariate polynomial in  $m_j$  of degree 1179 whose real roots in  $[1, \infty)$  are 1, 1.178, 1.223 and 2.674; its coefficients have up to 390 digits. 760 terms. This polynomial is too long to typeset and is given only in the plain-text file.

## Appendix A.3.7

Polynomials of Appendix A.3.7. Throughout,  $c = m_j - 1$ ,  $d = m_i - m_j$ , and  $z = 1/\sqrt{m_i}$ ,  $\tau = m_j z$ ,  $\eta = \rho/z^2$  in the rescaled polynomials. The resultant  $\text{Res}_\rho(q(m_i, 1, \cdot), \Xi(m_i, 1, \cdot))$  has degree 19 and real roots  $-1.271$ ,  $0.373$  and  $2.786$ , so  $m_i^{\min} \approx 2.786$  with  $\rho \approx 0.331$ ; on that boundary  $\Xi(2, 1, \rho) \approx -48.904$  at  $\rho \approx 0.484$  and  $\Xi(3, 1, \rho) \approx 31.963$  at  $\rho \approx 0.304$ . The cutoff is  $\tilde{w}_i \approx 0.120$ , where  $w_i - x_{i1}^* = 1/(4(m_i^{\min})^2) \approx 0.032$ . The Krawczyk certification used 3000 initial slices of  $[0, z_{\text{start}}]$ , refined into 4959 boxes.

$P_M$

Numerator of  $\mathcal{M}$ ; its coefficients in  $\rho$  are  $G_4, \dots, G_0$ . 25 terms.

$$\begin{aligned}
P_M(m_i, m_j, \rho) = & 8m_i^2m_j^3\rho^3 - 8m_i^3m_j^3\rho - 2m_i^3m_j\rho^3 \\
& - 4m_i^2m_j^2\rho^3 - 2m_i^2m_j\rho^4 - 2m_im_j^3\rho^3 \\
& + 2m_i^4m_j\rho + 2m_i^3m_j\rho^2 + 2m_i^3\rho^3 \\
& + 2m_i^2m_j^3\rho - 4m_i^2m_j^2\rho^2 + 1m_i^2m_j\rho^3 \\
& + 2m_i^2\rho^4 + 1m_j^3\rho^3 - 2m_i^4\rho \\
& - 3m_i^3m_j\rho - 2m_i^3\rho^2 - 2m_i^2m_j\rho^2 \\
& - 1m_im_j^3\rho - 8m_i^3m_j + 2m_i^3\rho \\
& - 8m_i^2m_j\rho + 2m_i^2\rho^2 + 4m_i^3 \\
& + 4m_i^2\rho
\end{aligned}$$

$U_M$

Coefficient of  $\rho$  in  $(-A_2)^3T_M$ . Mixed signs. 163 terms.

$$\begin{aligned}
U_M(c, d) = & -4096c^{17}d^2 - 32768c^{16}d^3 - 114688c^{15}d^4 \\
& - 229376c^{14}d^5 - 286720c^{13}d^6 - 229376c^{12}d^7 \\
& - 114688c^{11}d^8 - 32768c^{10}d^9 - 4096c^9d^{10} \\
& + 4096c^{18} + 36864c^{17}d + 89088c^{16}d^2 \\
& - 95232c^{15}d^3 - 924672c^{14}d^4 - 2171904c^{13}d^5 \\
& - 2774016c^{12}d^6 - 2153472c^{11}d^7 - 1016832c^{10}d^8 \\
& - 269312c^9d^9 - 30720c^8d^{10} + 58368c^{17} \\
& + 497664c^{16}d + 1486848c^{15}d^2 + 1354752c^{14}d^3 \\
& - 2685184c^{13}d^4 - 9294336c^{12}d^5 - 12408576c^{11}d^6
\end{aligned}$$

$$\begin{aligned}
& -9347072c^{10}d^7 - 4147968c^9d^8 - 1012224c^8d^9 \\
& -104704c^7d^{10} + 396544c^{16} + 3192064c^{15}d \\
& + 9680640c^{14}d^2 + 12387072c^{13}d^3 - 538304c^{12}d^4 \\
& - 23367872c^{11}d^5 - 33936256c^{10}d^6 - 24812160c^9d^7 \\
& - 10282688c^8d^8 - 2291904c^7d^9 - 212736c^6d^{10} \\
& + 1712576c^{15} + 12965120c^{14}d + 38202240c^{13}d^2 \\
& + 52477440c^{12}d^3 + 21047680c^{11}d^4 - 36740512c^{10}d^5 \\
& - 63061952c^9d^6 - 44835008c^8d^7 - 17211712c^7d^8 \\
& - 3462048c^6d^9 - 283840c^5d^{10} + 5284960c^{14} \\
& + 37464512c^{13}d + 105190912c^{12}d^2 + 144308416c^{11}d^3 \\
& + 81109504c^{10}d^4 - 32624240c^9d^5 - 83404600c^8d^6 \\
& - 57996008c^7d^7 - 20414952c^6d^8 - 3644680c^5d^9 \\
& - 257824c^4d^{10} + 12403984c^{13} + 81924352c^{12}d \\
& + 216454472c^{11}d^2 + 286786320c^{10}d^3 + 177687432c^9d^4 \\
& - 1764560c^8d^5 - 79554880c^7d^6 - 54897784c^6d^7 \\
& - 17490156c^5d^8 - 2706880c^4d^9 - 159404c^3d^{10} \\
& + 22974192c^{12} + 140549424c^{11}d + 345791936c^{10}d^2 \\
& + 433055952c^9d^3 + 271403676c^8d^4 + 41458172c^7d^5 \\
& - 53768792c^6d^6 - 38133700c^5d^7 - 10805978c^4d^8 \\
& - 1399494c^3d^9 - 64680c^2d^{10} + 34334368c^{11} \\
& + 193225952c^{10}d + 438082272c^9d^2 + 509506824c^8d^3 \\
& + 307954512c^7d^4 + 68580180c^6d^5 - 23974622c^5d^6 \\
& - 19156792c^4d^7 - 4708064c^3d^8 - 481614c^2d^9 \\
& - 15631cd^{10} + 41924032c^{10} + 215267168c^9d \\
& + 444621552c^8d^2 + 471969856c^7d^3 + 265521076c^6d^4 \\
& + 64944196c^5d^5 - 5336472c^4d^6 - 6714952c^3d^7 \\
& - 1373813c^2d^8 - 99505cd^9 - 1715d^{10} \\
& + 42055904c^9 + 195070624c^8d + 362144944c^7d^2 \\
& + 344175224c^6d^3 + 174352912c^5d^4 + 41422544c^4d^5 \\
& + 775676c^3d^6 - 1531626c^2d^7 - 240795cd^8 \\
& - 9359d^9 + 34658464c^8 + 143440384c^7d \\
& + 235425808c^6d^2 + 195715888c^5d^3 + 86055012c^4d^4 \\
& + 18191500c^3d^5 + 958180c^2d^6 - 197408cd^7 \\
& - 19124d^8 + 23350848c^7 + 84893152c^6d \\
& + 120592064c^5d^2 + 85081400c^4d^3 + 30977552c^3d^4 \\
& + 5316108c^2d^5 + 279674cd^6 - 10042d^7 \\
& + 12737184c^6 + 39852448c^5d + 47611472c^4d^2 \\
& + 27322592c^3d^3 + 7686284c^2d^4 + 935396cd^5 \\
& + 30556d^6 + 5537808c^5 + 14499808c^4d \\
& + 13980552c^3d^2 + 6109496c^2d^3 + 1176008cd^4 \\
& + 75200d^5 + 1873904c^4 + 3942352c^3d \\
& + 2875056c^2d^2 + 849440cd^3 + 83648d^4 \\
& + 475616c^3 + 753856c^2d + 369504cd^2 \\
& + 55296d^3 + 85184c^2 + 90432cd
\end{aligned}$$

$$\begin{aligned}
& + 22336d^2 + 9600c + 5120d \\
& + 512
\end{aligned}$$

$V_M$

Constant term of  $(-A_2)^3 T_M$ . All coefficients positive. 143 terms.

$$\begin{aligned}
V_M(c, d) = & 4096c^{17}d + 32768c^{16}d^2 + 114688c^{15}d^3 \\
& + 229376c^{14}d^4 + 286720c^{13}d^5 + 229376c^{12}d^6 \\
& + 114688c^{11}d^7 + 32768c^{10}d^8 + 4096c^9d^9 \\
& + 57344c^{16}d + 431104c^{15}d^2 + 1412096c^{14}d^3 \\
& + 2630656c^{13}d^4 + 3046400c^{12}d^5 + 2243584c^{11}d^6 \\
& + 1025024c^{10}d^7 + 265216c^9d^8 + 29696c^8d^9 \\
& + 5120c^{16} + 423168c^{15}d + 2836480c^{14}d^2 \\
& + 8519936c^{13}d^3 + 14605824c^{12}d^4 + 15530240c^{11}d^5 \\
& + 10451968c^{10}d^6 + 4335360c^9d^7 + 1010176c^8d^8 \\
& + 100864c^7d^9 + 71424c^{15} + 2149248c^{14}d \\
& + 12373952c^{13}d^2 + 33465152c^{12}d^3 + 52150784c^{11}d^4 \\
& + 50417024c^{10}d^5 + 30733248c^9d^6 + 11473472c^8d^7 \\
& + 2385792c^7d^8 + 210304c^6d^9 + 465088c^{14} \\
& + 8092928c^{13}d + 39528512c^{12}d^2 + 95068192c^{11}d^3 \\
& + 133242880c^{10}d^4 + 115947520c^9d^5 + 63363776c^8d^6 \\
& + 21052064c^7d^7 + 3855680c^6d^8 + 295232c^5d^9 \\
& + 1878096c^{13} + 23138688c^{12}d + 96286080c^{11}d^2 \\
& + 204552704c^{10}d^3 + 255550808c^9d^4 + 198121640c^8d^5 \\
& + 95909016c^7d^6 + 27950600c^6d^7 + 4426976c^5d^8 \\
& + 287344c^4d^9 + 5268304c^{12} + 50881584c^{11}d \\
& + 182033672c^{10}d^2 + 340258256c^9d^3 + 375772576c^8d^4 \\
& + 256621932c^7d^5 + 108447024c^6d^6 + 27197004c^5d^7 \\
& + 3629552c^4d^8 + 192352c^3d^9 + 10893664c^{11} \\
& + 86965280c^{10}d + 269375576c^9d^2 + 441186632c^8d^3 \\
& + 426641764c^7d^4 + 253119052c^6d^5 + 91616400c^5d^6 \\
& + 19254496c^4d^7 + 2082798c^3d^8 + 84742c^2d^9 \\
& + 17188608c^{10} + 116506080c^9d + 313191736c^8d^2 \\
& + 446473920c^7d^3 + 373366076c^6d^4 + 188920660c^5d^5 \\
& + 57039154c^4d^6 + 9659996c^3d^7 + 795508c^2d^8 \\
& + 22141cd^9 + 21124960c^9 + 122978720c^8d \\
& + 286042296c^7d^2 + 351126680c^6d^3 + 249499736c^5d^4 \\
& + 104902008c^4d^5 + 25404926c^3d^6 + 3255178c^2d^7 \\
& + 181652cd^8 + 2597d^9 + 20450656c^8 \\
& + 102387456c^7d + 204249800c^6d^2 + 212345968c^5d^3 \\
& + 124975000c^4d^4 + 41994076c^3d^5 + 7658650c^2d^6 \\
& + 660660cd^7 + 18746d^8 + 15655936c^7 \\
& + 66963936c^6d + 112805160c^5d^2 + 96873400c^4d^3 \\
& + 45431572c^3d^4 + 11455252c^2d^5 + 1399914cd^6 \\
& + 61002d^7 + 9451648c^6 + 34044128c^5d
\end{aligned}$$

$$\begin{aligned}
& + 47260456c^4d^2 + 32262656c^3d^3 + 11322028c^2d^4 \\
& + 1906276cd^5 + 117140d^6 + 4454736c^5 \\
& + 13194592c^4d + 14530984c^3d^2 + 7406248c^2d^3 \\
& + 1730904cd^4 + 146048d^5 + 1607632c^4 \\
& + 3770832c^3d + 3094864c^2d^2 + 1048288cd^3 \\
& + 122464d^4 + 429600c^3 + 749824c^2d \\
& + 408288cd^2 + 68992d^3 + 80192c^2 \\
& + 92736cd + 25152d^2 + 9344c \\
& + 5376d + 512
\end{aligned}$$

$W_M$

$W_M = A_0U_M - A_1V_M$ . All coefficients positive. 220 terms.

$$\begin{aligned}
W_M(c, d) = & 32768c^{22} + 360448c^{21}d + 1802240c^{20}d^2 \\
& + 5406720c^{19}d^3 + 10813440c^{18}d^4 + 15138816c^{17}d^5 \\
& + 15138816c^{16}d^6 + 10813440c^{15}d^7 + 5406720c^{14}d^8 \\
& + 1802240c^{13}d^9 + 360448c^{12}d^{10} + 32768c^{11}d^{11} \\
& + 573440c^{21} + 6070272c^{20}d + 29188096c^{19}d^2 \\
& + 84172800c^{18}d^3 + 161832960c^{17}d^4 + 217989120c^{16}d^5 \\
& + 210223104c^{15}d^6 + 145539072c^{14}d^7 + 71270400c^{13}d^8 \\
& + 23797760c^{12}d^9 + 5038080c^{11}d^{10} + 581632c^{10}d^{11} \\
& + 24576c^9d^{12} + 4820992c^{20} + 49055744c^{19}d \\
& + 226541568c^{18}d^2 + 627019776c^{17}d^3 + 1156614144c^{16}d^4 \\
& + 1495019520c^{15}d^5 + 1385029632c^{14}d^6 + 923615232c^{13}d^7 \\
& + 438091776c^{12}d^8 + 143263744c^{11}d^9 + 30384128c^{10}d^{10} \\
& + 3698688c^9d^{11} + 190464c^8d^{12} + 25952256c^{19} \\
& + 253472256c^{18}d + 1122236416c^{17}d^2 + 2974630912c^{16}d^3 \\
& + 5249627648c^{15}d^4 + 6487020544c^{14}d^5 + 5743216640c^{13}d^6 \\
& + 3661103104c^{12}d^7 + 1662374912c^{11}d^8 + 522178048c^{10}d^9 \\
& + 107122688c^9d^{10} + 12790784c^8d^{11} + 665088c^7d^{12} \\
& + 100563072c^{18} + 940945408c^{17}d + 3984590720c^{16}d^2 \\
& + 10084449920c^{15}d^3 + 16962178688c^{14}d^4 + 19939248896c^{13}d^5 \\
& + 16759981056c^{12}d^6 + 10123012096c^{11}d^7 + 4346450560c^{10}d^8 \\
& + 1288574208c^9d^9 + 249097344c^8d^{10} + 28002688c^7d^{11} \\
& + 1371776c^6d^{12} + 298814528c^{17} + 2671796992c^{16}d \\
& + 10787707040c^{15}d^2 + 25967111520c^{14}d^3 + 41425286848c^{13}d^4 \\
& + 46039399936c^{12}d^5 + 36454903872c^{11}d^6 + 20655106496c^{10}d^7 \\
& + 8278281920c^9d^8 + 2277375488c^8d^9 + 405582368c^7d^{10} \\
& + 41626080c^6d^{11} + 1839936c^5d^{12} + 707828384c^{16} \\
& + 6028241344c^{15}d + 23114581040c^{14}d^2 + 52659654816c^{13}d^3 \\
& + 79199729248c^{12}d^4 + 82608247024c^{11}d^5 + 61062038416c^{10}d^6 \\
& + 32092731664c^9d^7 + 11839762416c^8d^8 + 2969836368c^7d^9 \\
& + 476470720c^6d^{10} + 43361872c^5d^{11} + 1662864c^4d^{12} \\
& + 1370911008c^{15} + 11075253824c^{14}d + 40130109232c^{13}d^2 \\
& + 86014794992c^{12}d^3 + 121090810288c^{11}d^4 + 117516485768c^{10}d^5
\end{aligned}$$

$$\begin{aligned}
& + 80248535248c^9d^6 + 38629601216c^8d^7 + 12914906040c^7d^8 \\
& + 2896560520c^6d^9 + 408279888c^5d^{10} + 31867136c^4d^{11} \\
& + 1011752c^3d^{12} + 2207671520c^{14} + 16833985984c^{13}d \\
& + 57298182176c^{12}d^2 + 114732748912c^{11}d^3 + 149924347856c^{10}d^4 \\
& + 134027635648c^9d^5 + 83537342640c^8d^6 + 36292309748c^7d^7 \\
& + 10795977416c^6d^8 + 2114822788c^5d^9 + 253842100c^4d^{10} \\
& + 16259880c^3d^{11} + 399252c^2d^{12} + 2988323232c^{13} \\
& + 21379490112c^{12}d + 67880230080c^{11}d^2 + 125935110720c^{10}d^3 \\
& + 151261534384c^9d^4 + 123111349072c^8d^5 + 69047613576c^7d^6 \\
& + 26598123464c^6d^7 + 6882330632c^5d^8 + 1142408258c^4d^9 \\
& + 111857558c^3d^{10} + 5500446c^2d^{11} + 92610cd^{12} \\
& + 3422205632c^{12} + 22811732096c^{11}d + 67010948064c^{10}d^2 \\
& + 114083286784c^9d^3 + 124511853872c^8d^4 + 90985866080c^7d^5 \\
& + 45130334104c^6d^6 + 15076228852c^5d^7 + 3293725474c^4d^8 \\
& + 444048852c^3d^9 + 33197969c^2d^{10} + 1112055cd^{11} \\
& + 9604d^{12} + 3325685824c^{11} + 20487507584c^{10}d \\
& + 55154224160c^9d^2 + 85193009824c^8d^3 + 83335965328c^7d^4 \\
& + 53749507840c^6d^5 + 23066846688c^5d^6 + 6488462208c^4d^7 \\
& + 1147557898c^3d^8 + 117655348c^2d^9 + 5962327cd^{10} \\
& + 101871d^{11} + 2742792768c^{10} + 15468577152c^9d \\
& + 37736933504c^8d^2 + 52171778208c^7d^3 + 44974169936c^6d^4 \\
& + 25052015584c^5d^5 + 9033867664c^4d^6 + 2052013340c^3d^7 \\
& + 275161470c^2d^8 + 19051064cd^9 + 490196d^{10} \\
& + 1914772352c^9 + 9776566144c^8d + 21324149728c^7d^2 \\
& + 25947727392c^6d^3 + 19290614736c^5d^4 + 9014192624c^4d^5 \\
& + 2619792712c^3d^6 + 450125736c^2d^7 + 40644006cd^8 \\
& + 1423842d^9 + 1125513248c^8 + 5133497536c^7d \\
& + 9845255088c^6d^2 + 10320040160c^5d^3 + 6427785456c^4d^4 \\
& + 2417055856c^3d^5 + 530391528c^2d^6 + 61229868cd^7 \\
& + 2789976d^8 + 552408992c^7 + 2213676352c^6d \\
& + 3653962288c^5d^2 + 3205531504c^4d^3 + 1604713984c^3d^4 \\
& + 454956600c^2d^5 + 66962880cd^6 + 3892944d^7 \\
& + 223641696c^6 + 770797760c^5d + 1063938656c^4d^2 \\
& + 749467888c^3d^3 + 282556320c^2d^4 + 53659296cd^5 \\
& + 3970864d^6 + 73392800c^5 + 211396672c^4d \\
& + 234126400c^3d^2 + 124094912c^2d^3 + 31304832cd^4 \\
& + 2985216d^5 + 19036288c^4 + 43970816c^3d \\
& + 36617408c^2d^2 + 12975872cd^3 + 1642048d^4 \\
& + 3756288c^3 + 6520832c^2d + 3628288cd^2 \\
& + 644480d^3 + 529920c^2 + 614400cd \\
& + 171264d^2 + 47616c + 27648d \\
& + 2048
\end{aligned}$$

$\Xi_{m_i}$ Partial derivative of  $\Xi$ , with respect to  $m_i$ . 45 terms.

$$\begin{aligned}
\Xi_{m_i}(m_i, m_j, \rho) = & -320m_i^4m_j^5\rho^2 + 96m_i^5m_j^3\rho^2 + 320m_i^4m_j^4\rho^2 \\
& + 80m_i^4m_j^3\rho^3 + 128m_i^3m_j^5\rho^2 - 96m_i^2m_j^5\rho^3 \\
& - 64m_i^4m_j^5\rho^4 - 144m_i^5m_j^2\rho^2 - 100m_i^4m_j^3\rho^2 \\
& - 120m_i^4m_j^2\rho^3 - 96m_i^3m_j^4\rho^2 + 144m_i^3m_j^3\rho^3 \\
& - 156m_i^2m_j^5\rho^2 + 96m_i^2m_j^3\rho^4 - 48m_i^2m_j^5\rho^3 \\
& + 16m_i^4m_j^4\rho^4 + 16m_i^3m_j^3\rho^5 + 8m_j^5\rho^4 \\
& + 60m_i^5m_j\rho^2 + 20m_i^4m_j^2\rho^2 + 50m_i^4m_j\rho^3 \\
& + 40m_i^3m_j^3\rho^2 - 80m_i^3m_j^2\rho^3 + 72m_i^2m_j^4\rho^2 \\
& + 18m_i^2m_j^3\rho^3 - 72m_i^2m_j^2\rho^4 + 24m_i^2m_j^5\rho^2 \\
& + 32m_i^4m_j\rho^3 - 16m_i^3m_j^2\rho^5 + 4m_j^5\rho^3 \\
& - 12m_i^5\rho^2 - 5m_i^4m_j\rho^2 - 10m_i^4\rho^3 \\
& - 24m_i^3m_j^2\rho^2 - 18m_i^2m_j^3\rho^2 - 30m_i^2m_j^2\rho^3 \\
& - 8m_i^3\rho^3 - 8m_i^2\rho^4 - 5m_j^5\rho^2 \\
& + 32m_i^3m_j^3 + 48m_i^2m_j^3\rho + 16m_i^2m_j^3\rho^2 \\
& - 16m_i^3m_j^2 - 24m_i^2m_j^2\rho - 8m_i^2m_j^2\rho^2
\end{aligned}$$

 $\Xi_{m_j}$ Partial derivative of  $\Xi$  with respect to  $m_j$ . 45 terms.

$$\begin{aligned}
\Xi_{m_j}(m_i, m_j, \rho) = & -320m_i^5m_j^4\rho^2 + 48m_i^6m_j^2\rho^2 + 256m_i^5m_j^3\rho^2 \\
& + 48m_i^5m_j^2\rho^3 + 160m_i^4m_j^4\rho^2 - 160m_i^3m_j^4\rho^3 \\
& - 160m_i^2m_j^4\rho^4 - 48m_i^6m_j\rho^2 - 60m_i^5m_j^2\rho^2 \\
& - 48m_i^5m_j\rho^3 - 96m_i^4m_j^3\rho^2 + 108m_i^4m_j^2\rho^3 \\
& - 260m_i^3m_j^4\rho^2 + 96m_i^3m_j^2\rho^4 - 120m_i^2m_j^4\rho^3 \\
& + 32m_i^2m_j^3\rho^4 + 24m_i^2m_j^2\rho^5 + 40m_i^2m_j^4\rho^4 \\
& + 10m_i^6\rho^2 + 8m_i^5m_j\rho^2 + 10m_i^5\rho^3 \\
& + 30m_i^4m_j^2\rho^2 - 40m_i^4m_j\rho^3 + 96m_i^3m_j^3\rho^2 \\
& + 18m_i^3m_j^2\rho^3 - 48m_i^3m_j\rho^4 + 60m_i^2m_j^4\rho^2 \\
& + 64m_i^2m_j^3\rho^3 - 16m_i^2m_j\rho^5 + 20m_i^2m_j^4\rho^3 \\
& - 20m_j^4\rho^4 - 1m_i^5\rho^2 - 12m_i^4m_j\rho^2 \\
& - 18m_i^3m_j^2\rho^2 - 20m_i^3m_j\rho^3 - 12m_i^2m_j^2\rho^3 \\
& - 8m_i^2m_j\rho^4 - 25m_i^2m_j^4\rho^2 - 20m_j^4\rho^3 \\
& + 24m_i^4m_j^2 + 48m_i^3m_j^2\rho + 24m_i^2m_j^2\rho^2 \\
& - 8m_i^4m_j - 16m_i^3m_j\rho - 8m_i^2m_j\rho^2
\end{aligned}$$

 $\Xi_\rho$ Partial derivative of  $\Xi$  with respect to  $\rho$ . 45 terms.

$$\begin{aligned}
\Xi_\rho(m_i, m_j, \rho) = & -128m_i^5m_j^5\rho + 32m_i^6m_j^3\rho + 128m_i^5m_j^4\rho \\
& + 48m_i^5m_j^3\rho^2 + 64m_i^4m_j^5\rho - 96m_i^3m_j^5\rho^2 \\
& - 128m_i^2m_j^5\rho^3 - 48m_i^6m_j^2\rho - 40m_i^5m_j^3\rho \\
& - 72m_i^5m_j^2\rho^2 - 48m_i^4m_j^4\rho + 108m_i^4m_j^3\rho^2 \\
& - 104m_i^3m_j^5\rho + 128m_i^3m_j^3\rho^3 - 72m_i^2m_j^5\rho^2
\end{aligned}$$

$$\begin{aligned}
& + 32m_i^2m_j^4\rho^3 + 40m_i^2m_j^3\rho^4 + 32m_im_j^5\rho^3 \\
& + 20m_i^6m_j\rho + 8m_i^5m_j^2\rho + 30m_i^5m_j\rho^2 \\
& + 20m_i^4m_j^3\rho - 60m_i^4m_j^2\rho^2 + 48m_i^3m_j^4\rho \\
& + 18m_i^3m_j^3\rho^2 - 96m_i^3m_j^2\rho^3 + 24m_i^2m_j^5\rho \\
& + 48m_i^2m_j^4\rho^2 - 40m_i^2m_j^2\rho^4 + 12m_im_j^5\rho^2 \\
& - 16m_j^5\rho^3 - 4m_i^6\rho - 2m_i^5m_j\rho \\
& - 6m_i^5\rho^2 - 12m_i^4m_j^2\rho - 12m_i^3m_j^3\rho \\
& - 30m_i^3m_j^2\rho^2 - 12m_i^2m_j^3\rho^2 - 16m_i^2m_j^2\rho^3 \\
& - 10m_im_j^5\rho - 12m_j^5\rho^2 + 16m_i^3m_j^3 \\
& + 16m_i^2m_j^3\rho - 8m_i^3m_j^2 - 8m_i^2m_j^2\rho
\end{aligned}$$

DetTotal

DetTotal =  $q_{m_i}W_\rho - q_\rho W_{m_i}$ . 106 terms.

$$\begin{aligned}
\text{DetTotal}(m_i, m_j, \rho) = & - 64m_i^6m_j^5\rho^2 + 64m_i^5m_j^6\rho^2 + 64m_i^5m_j^5\rho^3 \\
& - 128m_i^4m_j^6\rho^3 + 128m_i^4m_j^5\rho^4 - 256m_i^6m_j^5\rho \\
& + 48m_i^6m_j^4\rho^2 + 192m_i^5m_j^6\rho - 272m_i^5m_j^5\rho^2 \\
& - 48m_i^5m_j^4\rho^3 - 32m_i^4m_j^6\rho^2 + 96m_i^4m_j^5\rho^3 \\
& - 96m_i^4m_j^4\rho^4 + 64m_i^3m_j^6\rho^3 - 112m_i^3m_j^5\rho^4 \\
& + 16m_i^2m_j^6\rho^4 - 32m_i^2m_j^5\rho^5 + 192m_i^6m_j^4\rho \\
& - 24m_i^6m_j^3\rho^2 - 48m_i^5m_j^5\rho + 204m_i^5m_j^4\rho^2 \\
& + 24m_i^5m_j^3\rho^3 - 64m_i^4m_j^6\rho - 72m_i^4m_j^5\rho^2 \\
& - 56m_i^4m_j^4\rho^3 + 48m_i^4m_j^3\rho^4 + 84m_i^3m_j^6\rho^2 \\
& - 200m_i^3m_j^5\rho^3 + 36m_i^3m_j^4\rho^4 + 24m_i^2m_j^6\rho^3 \\
& - 56m_i^2m_j^5\rho^4 + 8m_i^2m_j^4\rho^5 - 4m_im_j^6\rho^4 \\
& + 16m_im_j^5\rho^5 - 96m_i^6m_j^3\rho + 6m_i^6m_j^2\rho^2 \\
& + 112m_i^5m_j^5 + 36m_i^5m_j^4\rho - 98m_i^5m_j^3\rho^2 \\
& - 6m_i^5m_j^2\rho^3 + 16m_i^4m_j^6 - 112m_i^4m_j^5\rho \\
& + 90m_i^4m_j^4\rho^2 + 12m_i^4m_j^3\rho^3 - 12m_i^4m_j^2\rho^4 \\
& + 84m_i^3m_j^6\rho - 138m_i^3m_j^5\rho^2 + 54m_i^3m_j^4\rho^3 \\
& - 14m_i^3m_j^3\rho^4 + 12m_i^2m_j^6\rho^2 + 8m_i^2m_j^5\rho^3 \\
& + 22m_i^2m_j^4\rho^4 - 4m_i^2m_j^3\rho^5 - 8m_im_j^6\rho^3 \\
& + 38m_im_j^5\rho^4 + 2m_j^6\rho^4 - 4m_j^5\rho^5 \\
& + 24m_i^6m_j^2\rho - 1m_i^6m_j\rho^2 - 124m_i^5m_j^4 \\
& - 6m_i^5m_j^3\rho + 25m_i^5m_j^2\rho^2 + 1m_i^5m_j\rho^3 \\
& - 48m_i^4m_j^5 + 28m_i^4m_j^4\rho - 58m_i^4m_j^3\rho^2 \\
& - 2m_i^4m_j^2\rho^3 + 2m_i^4m_j\rho^4 - 4m_i^3m_j^6 \\
& + 14m_i^3m_j^5\rho + 34m_i^3m_j^4\rho^2 - 22m_i^3m_j^3\rho^3 \\
& - 16m_i^2m_j^6\rho + 31m_i^2m_j^5\rho^2 + 20m_i^2m_j^4\rho^3 \\
& - 10m_i^2m_j^3\rho^4 + 1m_im_j^6\rho^2 + 9m_im_j^5\rho^3 \\
& + 6m_j^6\rho^3 - 12m_j^5\rho^4 - 4m_i^6m_j\rho \\
& + 62m_i^5m_j^3 + 3m_i^5m_j^2\rho - 4m_i^5m_j\rho^2 \\
& + 26m_i^4m_j^4 - 16m_i^4m_j^3\rho + 16m_i^4m_j^2\rho^2 \\
& + 18m_i^3m_j^5 + 10m_i^3m_j^4\rho - 16m_i^3m_j^3\rho^2 \\
& + 2m_i^2m_j^6 - 12m_i^2m_j^5\rho + 12m_i^2m_j^4\rho^2
\end{aligned}$$

$$\begin{aligned}
& -8m_i^2m_j^3\rho^3 + 7m_im_j^6\rho - 16m_im_j^5\rho^2 \\
& + 4m_j^6\rho^2 - 8m_j^5\rho^3 - 16m_i^5m_j^2 \\
& - 6m_i^4m_j^3 - 2m_i^4m_j\rho^2 - 8m_i^3m_j^4 \\
& - 2m_i^2m_j^5 - 2m_i^2m_j^3\rho^2 + 2m_i^5m_j \\
& + 2m_i^3m_j^3
\end{aligned}$$

NumTotal

Numerator of  $d\Xi/dm_j$  after clearing DetTotal. 596 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$\Xi_n$

$\Xi_n = z^{11}\Xi$  after the substitution  $m_i = 1/z^2$ ,  $m_j = \tau/z$ ,  $\rho = \eta z^2$ . 47 terms.

$$\begin{aligned}
\Xi_n(z, \tau, \eta) = & -4z^{14}\tau^5\eta^4 - 8z^{15}\tau^2\eta^5 + 8z^{14}\tau^3\eta^5 \\
& + 8z^{12}\tau^5\eta^4 - 4z^{12}\tau^5\eta^3 - 4z^{13}\tau^2\eta^4 \\
& + 8z^{11}\tau^4\eta^4 - 32z^{10}\tau^5\eta^4 + 4z^{10}\tau^5\eta^3 \\
& - 24z^{11}\tau^2\eta^4 + 32z^{10}\tau^3\eta^4 - 4z^{10}\tau^3\eta^3 \\
& + 16z^9\tau^4\eta^3 - 24z^8\tau^5\eta^3 - 5z^8\tau^5\eta^2 \\
& - 10z^9\tau^2\eta^3 + 6z^8\tau^3\eta^3 - 32z^6\tau^5\eta^3 \\
& - 4z^9\tau^2\eta^2 + 8z^8\tau^3\eta^2 + 12z^6\tau^5\eta^2 \\
& - 20z^7\tau^2\eta^3 + 36z^6\tau^3\eta^3 - 6z^6\tau^3\eta^2 \\
& + 24z^5\tau^4\eta^2 - 52z^4\tau^5\eta^2 - 2z^7\eta^3 \\
& + 10z^6\tau\eta^3 - 24z^5\tau^2\eta^3 + 16z^4\tau^3\eta^3 \\
& - 6z^5\tau^2\eta^2 + 10z^4\tau^3\eta^2 - 24z^3\tau^4\eta^2 \\
& + 32z^2\tau^5\eta^2 - 8z^5\tau^2\eta + 16z^4\tau^3\eta \\
& - 1z^4\tau\eta^2 + 4z^3\tau^2\eta^2 - 20z^2\tau^3\eta^2 \\
& + 64z\tau^4\eta^2 - 64\tau^5\eta^2 - 2z^3\eta^2 \\
& + 10z^2\tau\eta^2 - 24z\tau^2\eta^2 + 16\tau^3\eta^2 \\
& - 4z\tau^2 + 8\tau^3
\end{aligned}$$

DetTotal<sub>n</sub>

DetTotal<sub>n</sub> =  $z^{15}$ DetTotal. 106 terms.

$$\begin{aligned}
\text{DetTotal}_n(z, \tau, \eta) = & -4z^{20}\tau^5\eta^5 + 16z^{18}\tau^5\eta^5 - 12z^{18}\tau^5\eta^4 \\
& + 2z^{17}\tau^6\eta^4 - 4z^{18}\tau^3\eta^5 + 8z^{17}\tau^4\eta^5 \\
& - 32z^{16}\tau^5\eta^5 + 38z^{16}\tau^5\eta^4 - 4z^{15}\tau^6\eta^4 \\
& - 8z^{16}\tau^5\eta^3 + 6z^{15}\tau^6\eta^3 - 10z^{16}\tau^3\eta^4 \\
& + 22z^{15}\tau^4\eta^4 - 56z^{14}\tau^5\eta^4 + 16z^{13}\tau^6\eta^4 \\
& + 9z^{14}\tau^5\eta^3 - 8z^{13}\tau^6\eta^3 - 14z^{14}\tau^3\eta^4 \\
& + 4z^{13}\tau^6\eta^2 + 36z^{13}\tau^4\eta^4 - 112z^{12}\tau^5\eta^4 \\
& - 8z^{14}\tau^3\eta^3 + 20z^{13}\tau^4\eta^3 + 8z^{12}\tau^5\eta^3 \\
& + 24z^{11}\tau^6\eta^3 + 2z^{14}\tau\eta^4 - 12z^{13}\tau^2\eta^4 \\
& - 16z^{12}\tau^5\eta^2 + 48z^{12}\tau^3\eta^4 + 1z^{11}\tau^6\eta^2 \\
& - 96z^{11}\tau^4\eta^4 + 128z^{10}\tau^5\eta^4 - 22z^{12}\tau^3\eta^3 \\
& + 54z^{11}\tau^4\eta^3 - 200z^{10}\tau^5\eta^3 + 64z^9\tau^6\eta^3 \\
& - 2z^{12}\tau^3\eta^2 + 12z^{11}\tau^4\eta^2 + 31z^{10}\tau^5\eta^2
\end{aligned}$$

$$\begin{aligned}
& + 12z^9\tau^6\eta^2 - 2z^{11}\tau^2\eta^3 + 12z^{10}\tau^3\eta^3 \\
& + 7z^9\tau^6\eta - 56z^9\tau^4\eta^3 + 96z^8\tau^5\eta^3 \\
& - 128z^7\tau^6\eta^3 - 16z^{10}\tau^3\eta^2 + 34z^9\tau^4\eta^2 \\
& - 138z^8\tau^5\eta^2 + 84z^7\tau^6\eta^2 + 1z^{10}\tau\eta^3 \\
& - 6z^9\tau^2\eta^3 - 12z^8\tau^5\eta + 24z^8\tau^3\eta^3 \\
& - 16z^7\tau^6\eta - 48z^7\tau^4\eta^3 + 64z^6\tau^5\eta^3 \\
& - 2z^{10}\tau\eta^2 + 16z^9\tau^2\eta^2 - 58z^8\tau^3\eta^2 \\
& + 90z^7\tau^4\eta^2 - 72z^6\tau^5\eta^2 - 32z^5\tau^6\eta^2 \\
& + 10z^7\tau^4\eta + 14z^6\tau^5\eta + 84z^5\tau^6\eta \\
& - 4z^8\tau\eta^2 + 25z^7\tau^2\eta^2 - 2z^6\tau^5 \\
& - 98z^6\tau^3\eta^2 + 2z^5\tau^6 + 204z^5\tau^4\eta^2 \\
& - 272z^4\tau^5\eta^2 + 64z^3\tau^6\eta^2 - 16z^6\tau^3\eta \\
& + 28z^5\tau^4\eta - 112z^4\tau^5\eta - 64z^3\tau^6\eta \\
& + 2z^6\tau^3 - 1z^6\tau\eta^2 - 8z^5\tau^4 \\
& + 6z^5\tau^2\eta^2 + 18z^4\tau^5 - 24z^4\tau^3\eta^2 \\
& - 4z^3\tau^6 + 48z^3\tau^4\eta^2 - 64z^2\tau^5\eta^2 \\
& + 3z^5\tau^2\eta - 6z^4\tau^3\eta + 36z^3\tau^4\eta \\
& - 48z^2\tau^5\eta + 192z\tau^6\eta - 6z^4\tau^3 \\
& + 26z^3\tau^4 - 48z^2\tau^5 + 16z\tau^6 \\
& - 4z^4\tau\eta + 24z^3\tau^2\eta - 96z^2\tau^3\eta \\
& + 192z\tau^4\eta - 256\tau^5\eta + 2z^4\tau \\
& - 16z^3\tau^2 + 62z^2\tau^3 - 124z\tau^4 \\
& + 112\tau^5
\end{aligned}$$

NumTotal<sub>n</sub>

NumTotal<sub>n</sub> =  $z^{25}$ NumTotal. 596 terms.

$$\begin{aligned}
\text{NumTotal}_n(z, \tau, \eta) = & 16z^{34}\tau^9\eta^9 - 96z^{35}\tau^6\eta^{10} + 64z^{34}\tau^7\eta^{10} \\
& - 96z^{32}\tau^9\eta^9 + 384z^{33}\tau^6\eta^{10} + 80z^{32}\tau^9\eta^8 \\
& - 256z^{32}\tau^7\eta^{10} + 56z^{31}\tau^{10}\eta^8 - 288z^{33}\tau^6\eta^9 \\
& + 448z^{32}\tau^7\eta^9 - 288z^{31}\tau^8\eta^9 + 384z^{30}\tau^9\eta^9 \\
& + 64z^{33}\tau^4\eta^{10} - 64z^{32}\tau^5\eta^{10} - 736z^{31}\tau^6\eta^{10} \\
& - 392z^{30}\tau^9\eta^8 + 512z^{30}\tau^7\eta^{10} - 224z^{29}\tau^{10}\eta^8 \\
& + 736z^{31}\tau^6\eta^9 + 192z^{30}\tau^9\eta^7 - 944z^{30}\tau^7\eta^9 \\
& + 200z^{29}\tau^{10}\eta^7 + 576z^{29}\tau^8\eta^9 - 768z^{28}\tau^9\eta^9 \\
& - 112z^{31}\tau^6\eta^8 + 904z^{30}\tau^7\eta^8 - 952z^{29}\tau^8\eta^8 \\
& + 1136z^{28}\tau^9\eta^8 + 1120z^{27}\tau^{10}\eta^8 + 160z^{31}\tau^4\eta^9 \\
& - 32z^{30}\tau^5\eta^9 - 1152z^{29}\tau^6\eta^9 - 792z^{28}\tau^9\eta^7 \\
& + 3008z^{28}\tau^7\eta^9 - 728z^{27}\tau^{10}\eta^7 - 2304z^{27}\tau^8\eta^9 \\
& + 1024z^{26}\tau^9\eta^9 - 584z^{29}\tau^6\eta^8 + 304z^{28}\tau^9\eta^6 \\
& - 256z^{28}\tau^7\eta^8 + 208z^{27}\tau^{10}\eta^6 + 848z^{27}\tau^8\eta^8 \\
& - 1088z^{26}\tau^9\eta^8 - 1792z^{25}\tau^{10}\eta^8 + 448z^{29}\tau^6\eta^7 \\
& + 224z^{29}\tau^4\eta^9 + 528z^{28}\tau^7\eta^7 - 80z^{28}\tau^5\eta^9 \\
& - 1296z^{27}\tau^8\eta^7 - 3744z^{27}\tau^6\eta^9 + 2064z^{26}\tau^9\eta^7 \\
& + 2432z^{26}\tau^7\eta^9 + 3840z^{25}\tau^{10}\eta^7 + 128z^{29}\tau^4\eta^8
\end{aligned}$$

$$\begin{aligned}
& -8z^{28}\tau^5\eta^8 + 1856z^{27}\tau^6\eta^8 - 894z^{26}\tau^9\eta^6 \\
& + 3824z^{26}\tau^7\eta^8 - 470z^{25}\tau^{10}\eta^6 - 4096z^{25}\tau^8\eta^8 \\
& - 2944z^{24}\tau^9\eta^8 + 3584z^{23}\tau^{10}\eta^8 + 64z^{29}\tau^2\eta^9 \\
& - 272z^{28}\tau^3\eta^9 - 1540z^{27}\tau^6\eta^7 + 896z^{27}\tau^4\eta^9 \\
& + 272z^{26}\tau^9\eta^5 + 2252z^{26}\tau^7\eta^7 - 1664z^{26}\tau^5\eta^9 \\
& + 64z^{25}\tau^{10}\eta^5 - 488z^{25}\tau^8\eta^7 + 512z^{25}\tau^6\eta^9 \\
& - 1216z^{24}\tau^9\eta^7 + 1024z^{24}\tau^7\eta^9 - 5472z^{23}\tau^{10}\eta^7 \\
& + 512z^{27}\tau^6\eta^6 + 664z^{27}\tau^4\eta^8 - 8z^{26}\tau^7\eta^6 \\
& + 16z^{26}\tau^5\eta^8 - 1488z^{25}\tau^8\eta^6 - 10616z^{25}\tau^6\eta^8 \\
& + 2966z^{24}\tau^9\eta^6 + 15936z^{24}\tau^7\eta^8 + 2820z^{23}\tau^{10}\eta^6 \\
& - 11520z^{23}\tau^8\eta^8 + 9728z^{22}\tau^9\eta^8 + 64z^{27}\tau^4\eta^7 \\
& - 752z^{26}\tau^5\eta^7 + 3656z^{25}\tau^6\eta^7 - 384z^{24}\tau^9\eta^5 \\
& + 684z^{24}\tau^7\eta^7 + 163z^{23}\tau^{10}\eta^5 - 2464z^{23}\tau^8\eta^7 \\
& - 6528z^{22}\tau^9\eta^7 + 13056z^{21}\tau^{10}\eta^7 + 112z^{27}\tau^2\eta^8 \\
& - 640z^{26}\tau^3\eta^8 + 584z^{25}\tau^6\eta^6 + 2944z^{25}\tau^4\eta^8 \\
& + 96z^{24}\tau^9\eta^4 + 1586z^{24}\tau^7\eta^6 - 6304z^{24}\tau^5\eta^8 \\
& - 1034z^{23}\tau^8\eta^6 + 560z^{23}\tau^6\eta^8 - 224z^{22}\tau^9\eta^6 \\
& - 1536z^{22}\tau^7\eta^8 + 424z^{21}\tau^{10}\eta^6 + 8192z^{21}\tau^8\eta^8 \\
& - 12288z^{20}\tau^9\eta^8 + 160z^{25}\tau^6\eta^5 + 680z^{25}\tau^4\eta^7 \\
& + 112z^{24}\tau^7\eta^5 + 80z^{24}\tau^5\eta^7 - 1504z^{23}\tau^8\eta^5 \\
& - 5220z^{23}\tau^6\eta^7 + 2137z^{22}\tau^9\eta^5 + 27320z^{22}\tau^7\eta^7 \\
& - 248z^{21}\tau^{10}\eta^5 - 28128z^{21}\tau^8\eta^7 + 11776z^{20}\tau^9\eta^7 \\
& + 512z^{19}\tau^{10}\eta^7 + 64z^{25}\tau^4\eta^6 + 288z^{25}\tau^2\eta^8 \\
& - 1144z^{24}\tau^5\eta^6 - 1256z^{24}\tau^3\eta^8 - 788z^{23}\tau^6\eta^6 \\
& + 3800z^{23}\tau^4\eta^8 + 262z^{22}\tau^9\eta^4 + 3560z^{22}\tau^7\eta^6 \\
& - 6240z^{22}\tau^5\eta^8 + 132z^{21}\tau^{10}\eta^4 - 5168z^{21}\tau^8\eta^6 \\
& + 704z^{21}\tau^6\eta^8 - 3816z^{20}\tau^9\eta^6 + 4608z^{20}\tau^7\eta^8 \\
& + 2800z^{19}\tau^{10}\eta^6 - 96z^{25}\tau^2\eta^7 + 576z^{24}\tau^3\eta^7 \\
& + 1732z^{23}\tau^6\eta^5 + 384z^{23}\tau^4\eta^7 - 1032z^{22}\tau^7\eta^5 \\
& - 2800z^{22}\tau^5\eta^7 - 524z^{21}\tau^8\eta^5 - 12368z^{21}\tau^6\eta^7 \\
& + 3972z^{20}\tau^9\eta^5 + 15232z^{20}\tau^7\eta^7 + 5928z^{19}\tau^{10}\eta^5 \\
& - 2560z^{19}\tau^8\eta^7 - 5120z^{18}\tau^9\eta^7 + 8192z^{17}\tau^{10}\eta^7 \\
& + 80z^{23}\tau^6\eta^4 + 544z^{23}\tau^4\eta^6 - 3310z^{22}\tau^5\eta^6 \\
& - 672z^{21}\tau^8\eta^4 + 13786z^{21}\tau^6\eta^6 - 238z^{20}\tau^9\eta^4 \\
& + 13440z^{20}\tau^7\eta^6 - 295z^{19}\tau^{10}\eta^4 - 31192z^{19}\tau^8\eta^6 \\
& + 15712z^{18}\tau^9\eta^6 + 25152z^{17}\tau^{10}\eta^6 + 32z^{23}\tau^4\eta^5 \\
& + 508z^{23}\tau^2\eta^7 - 512z^{22}\tau^5\eta^5 - 2820z^{22}\tau^3\eta^7 \\
& - 1000z^{21}\tau^6\eta^5 + 10232z^{21}\tau^4\eta^7 + 248z^{20}\tau^9\eta^3 \\
& + 5814z^{20}\tau^7\eta^5 - 19880z^{20}\tau^5\eta^7 - 8280z^{19}\tau^8\eta^5 \\
& + 10320z^{19}\tau^6\eta^7 - 2036z^{18}\tau^9\eta^5 + 3008z^{18}\tau^7\eta^7 \\
& - 6816z^{17}\tau^{10}\eta^5 + 2560z^{17}\tau^8\eta^7 - 20480z^{16}\tau^9\eta^7 \\
& - 224z^{23}\tau^2\eta^6 + 1504z^{22}\tau^3\eta^6 + 416z^{21}\tau^6\eta^4 \\
& - 2944z^{21}\tau^4\eta^6 - 44z^{20}\tau^7\eta^4 + 4932z^{20}\tau^5\eta^6 \\
& - 1982z^{19}\tau^8\eta^4 - 19792z^{19}\tau^6\eta^6 + 7133z^{18}\tau^9\eta^4
\end{aligned}$$

$$\begin{aligned}
& + 37808z^{18}\tau^7\eta^6 + 1704z^{17}\tau^{10}\eta^4 - 28096z^{17}\tau^8\eta^6 \\
& + 320z^{16}\tau^9\eta^6 - 1792z^{15}\tau^{10}\eta^6 + 64z^{21}\tau^6\eta^3 \\
& + 528z^{21}\tau^4\eta^5 + 576z^{21}\tau^2\eta^7 - 128z^{20}\tau^7\eta^3 \\
& - 5300z^{20}\tau^5\eta^5 - 2604z^{20}\tau^3\eta^7 + 12518z^{19}\tau^6\eta^5 \\
& + 6272z^{19}\tau^4\eta^7 - 580z^{18}\tau^9\eta^3 + 1796z^{18}\tau^7\eta^5 \\
& - 5344z^{18}\tau^5\eta^7 + 82z^{17}\tau^{10}\eta^3 - 23648z^{17}\tau^8\eta^5 \\
& - 11136z^{17}\tau^6\eta^7 + 24992z^{16}\tau^9\eta^5 + 14080z^{16}\tau^7\eta^7 \\
& + 24368z^{15}\tau^{10}\eta^5 - 344z^{21}\tau^2\eta^6 - 208z^{20}\tau^5\eta^4 \\
& + 2278z^{20}\tau^3\eta^6 + 2756z^{19}\tau^6\eta^4 - 3646z^{19}\tau^4\eta^6 \\
& - 1968z^{18}\tau^7\eta^4 - 1472z^{18}\tau^5\eta^6 - 146z^{17}\tau^8\eta^4 \\
& - 6952z^{17}\tau^6\eta^6 - 3918z^{16}\tau^9\eta^4 + 29824z^{16}\tau^7\eta^6 \\
& + 664z^{15}\tau^{10}\eta^4 - 35328z^{15}\tau^8\eta^6 + 9728z^{14}\tau^9\eta^6 \\
& + 17408z^{13}\tau^{10}\eta^6 - 64z^{21}\tau^2\eta^5 - 16z^{21}\eta^7 \\
& + 352z^{20}\tau^3\eta^5 + 144z^{20}\tau\eta^7 + 64z^{19}\tau^6\eta^3 \\
& + 480z^{19}\tau^4\eta^5 - 580z^{19}\tau^2\eta^7 + 304z^{18}\tau^7\eta^3 \\
& - 5748z^{18}\tau^5\eta^5 + 1024z^{18}\tau^3\eta^7 - 1768z^{17}\tau^8\eta^3 \\
& + 20240z^{17}\tau^6\eta^5 + 384z^{17}\tau^4\eta^7 + 3438z^{16}\tau^9\eta^3 \\
& + 12596z^{16}\tau^7\eta^5 - 5120z^{16}\tau^5\eta^7 - 332z^{15}\tau^{10}\eta^3 \\
& - 40928z^{15}\tau^8\eta^5 + 8960z^{15}\tau^6\eta^7 + 15520z^{14}\tau^9\eta^5 \\
& - 4096z^{14}\tau^7\eta^7 + 10496z^{13}\tau^{10}\eta^5 + 240z^{19}\tau^4\eta^4 \\
& + 1076z^{19}\tau^2\eta^6 - 2136z^{18}\tau^5\eta^4 - 5968z^{18}\tau^3\eta^6 \\
& - 322z^{17}\tau^6\eta^4 + 16912z^{17}\tau^4\eta^6 + 198z^{16}\tau^9\eta^2 \\
& + 10518z^{16}\tau^7\eta^4 - 19512z^{16}\tau^5\eta^6 - 24356z^{15}\tau^8\eta^4 \\
& - 14784z^{15}\tau^6\eta^6 + 19852z^{14}\tau^9\eta^4 + 43328z^{14}\tau^7\eta^6 \\
& + 32z^{13}\tau^{10}\eta^4 - 22528z^{13}\tau^8\eta^6 - 15360z^{12}\tau^9\eta^6 \\
& - 812z^{19}\tau^2\eta^5 - 128z^{18}\tau^5\eta^3 + 5472z^{18}\tau^3\eta^5 \\
& + 1152z^{17}\tau^6\eta^3 - 14268z^{17}\tau^4\eta^5 - 1932z^{16}\tau^7\eta^3 \\
& + 18256z^{16}\tau^5\eta^5 + 2116z^{15}\tau^8\eta^3 - 27916z^{15}\tau^6\eta^5 \\
& - 578z^{14}\tau^9\eta^3 + 35240z^{14}\tau^7\eta^5 + 1800z^{13}\tau^{10}\eta^3 \\
& - 37248z^{13}\tau^8\eta^5 + 21376z^{12}\tau^9\eta^5 - 3584z^{11}\tau^{10}\eta^5 \\
& - 16z^{19}\tau^2\eta^4 - 16z^{19}\eta^6 + 48z^{18}\tau^3\eta^4 \\
& + 116z^{18}\tau\eta^6 + 192z^{17}\tau^6\eta^2 + 1424z^{17}\tau^4\eta^4 \\
& + 292z^{17}\tau^2\eta^6 - 384z^{16}\tau^7\eta^2 - 11284z^{16}\tau^5\eta^4 \\
& - 3488z^{16}\tau^3\eta^6 + 34960z^{15}\tau^6\eta^4 + 12712z^{15}\tau^4\eta^6 \\
& - 798z^{14}\tau^9\eta^2 - 20072z^{14}\tau^7\eta^4 - 24032z^{14}\tau^5\eta^6 \\
& + 10z^{13}\tau^{10}\eta^2 - 18176z^{13}\tau^8\eta^4 + 22592z^{13}\tau^6\eta^6 \\
& + 35464z^{12}\tau^9\eta^4 - 25600z^{12}\tau^7\eta^6 + 19472z^{11}\tau^{10}\eta^4 \\
& + 18432z^{11}\tau^8\eta^6 + 32z^{17}\tau^4\eta^3 - 344z^{17}\tau^2\eta^5 \\
& - 576z^{16}\tau^5\eta^3 + 2666z^{16}\tau^3\eta^5 + 2232z^{15}\tau^6\eta^3 \\
& - 7544z^{15}\tau^4\eta^5 + 1672z^{14}\tau^7\eta^3 + 14140z^{14}\tau^5\eta^5 \\
& - 10964z^{13}\tau^8\eta^3 - 34848z^{13}\tau^6\eta^5 + 6432z^{12}\tau^9\eta^3 \\
& + 93184z^{12}\tau^7\eta^5 - 3440z^{11}\tau^{10}\eta^3 - 100608z^{11}\tau^8\eta^5 \\
& + 30464z^{10}\tau^9\eta^5 + 28672z^9\tau^{10}\eta^5 - 200z^{17}\tau^2\eta^4 \\
& - 32z^{17}\eta^6 + 988z^{16}\tau^3\eta^4 + 286z^{16}\tau\eta^6
\end{aligned}$$

$$\begin{aligned}
& -288z^{15}\tau^6\eta^2 - 594z^{15}\tau^4\eta^4 - 1188z^{15}\tau^2\eta^6 \\
& + 920z^{14}\tau^7\eta^2 - 6572z^{14}\tau^5\eta^4 + 2368z^{14}\tau^3\eta^6 \\
& - 1420z^{13}\tau^8\eta^2 + 19908z^{13}\tau^6\eta^4 - 640z^{13}\tau^4\eta^6 \\
& + 4270z^{12}\tau^9\eta^2 - 15712z^{12}\tau^7\eta^4 - 7168z^{12}\tau^5\eta^6 \\
& - 44z^{11}\tau^{10}\eta^2 - 7792z^{11}\tau^8\eta^4 + 14592z^{11}\tau^6\eta^6 \\
& - 11712z^{10}\tau^9\eta^4 - 7168z^{10}\tau^7\eta^6 - 7936z^9\tau^{10}\eta^4 \\
& - 32z^{17}\tau^2\eta^3 - 24z^{17}\eta^5 + 256z^{16}\tau^3\eta^3 \\
& + 268z^{16}\tau\eta^5 - 392z^{15}\tau^4\eta^3 - 193z^{15}\tau^2\eta^5 \\
& - 1960z^{14}\tau^5\eta^3 - 3220z^{14}\tau^3\eta^5 + 4056z^{13}\tau^6\eta^3 \\
& + 13600z^{13}\tau^4\eta^5 + 40z^{12}\tau^9\eta + 1930z^{12}\tau^7\eta^3 \\
& - 17864z^{12}\tau^5\eta^5 - 12912z^{11}\tau^8\eta^3 - 11472z^{11}\tau^6\eta^5 \\
& + 18216z^{10}\tau^9\eta^3 + 32704z^{10}\tau^7\eta^5 + 9280z^9\tau^{10}\eta^3 \\
& - 32768z^9\tau^8\eta^5 + 24576z^8\tau^9\eta^5 - 20480z^7\tau^{10}\eta^5 \\
& + 16z^{15}\tau^4\eta^2 - 972z^{15}\tau^2\eta^4 - 464z^{14}\tau^5\eta^2 \\
& + 6552z^{14}\tau^3\eta^4 + 2752z^{13}\tau^6\eta^2 - 17454z^{13}\tau^4\eta^4 \\
& - 4602z^{12}\tau^7\eta^2 + 22208z^{12}\tau^5\eta^4 + 3950z^{11}\tau^8\eta^2 \\
& - 26036z^{11}\tau^6\eta^4 - 7760z^{10}\tau^9\eta^2 + 60416z^{10}\tau^7\eta^4 \\
& + 232z^9\tau^{10}\eta^2 - 82784z^9\tau^8\eta^4 + 64064z^8\tau^9\eta^4 \\
& + 15872z^7\tau^{10}\eta^4 - 32z^{15}\tau^2\eta^3 - 32z^{15}\eta^5 \\
& + 96z^{14}\tau^3\eta^3 + 267z^{14}\tau\eta^5 + 192z^{13}\tau^6\eta \\
& + 1384z^{13}\tau^4\eta^3 - 984z^{13}\tau^2\eta^5 - 384z^{12}\tau^7\eta \\
& - 10492z^{12}\tau^5\eta^3 + 2100z^{12}\tau^3\eta^5 + 33552z^{11}\tau^6\eta^3 \\
& - 2704z^{11}\tau^4\eta^5 - 176z^{10}\tau^9\eta - 33652z^{10}\tau^7\eta^3 \\
& + 736z^{10}\tau^5\eta^5 - 400z^9\tau^8\eta^3 + 4096z^9\tau^6\eta^5 \\
& + 19008z^8\tau^9\eta^3 - 14592z^8\tau^7\eta^5 - 9600z^7\tau^{10}\eta^3 \\
& + 8192z^7\tau^8\eta^5 + 4096z^6\tau^9\eta^5 - 24z^{15}\eta^4 \\
& + 258z^{14}\tau\eta^4 + 48z^{13}\tau^4\eta^2 - 1410z^{13}\tau^2\eta^4 \\
& - 348z^{12}\tau^5\eta^2 + 4682z^{12}\tau^3\eta^4 - 220z^{11}\tau^6\eta^2 \\
& - 9868z^{11}\tau^4\eta^4 + 6282z^{10}\tau^7\eta^2 + 14228z^{10}\tau^5\eta^4 \\
& - 16168z^9\tau^8\eta^2 - 17664z^9\tau^6\eta^4 + 19432z^8\tau^9\eta^2 \\
& + 38848z^8\tau^7\eta^4 - 464z^7\tau^{10}\eta^2 - 50048z^7\tau^8\eta^4 \\
& + 15104z^6\tau^9\eta^4 - 4096z^5\tau^{10}\eta^4 - 176z^{13}\tau^2\eta^3 \\
& - 16z^{13}\eta^5 + 548z^{12}\tau^3\eta^3 + 140z^{12}\tau\eta^5 \\
& - 448z^{11}\tau^6\eta + 1212z^{11}\tau^4\eta^3 - 636z^{11}\tau^2\eta^5 \\
& + 920z^{10}\tau^7\eta - 9646z^{10}\tau^5\eta^3 + 1664z^{10}\tau^3\eta^5 \\
& - 200z^9\tau^8\eta + 17512z^9\tau^6\eta^3 - 2432z^9\tau^4\eta^5 \\
& + 928z^8\tau^9\eta - 14000z^8\tau^7\eta^3 + 1024z^8\tau^5\eta^5 \\
& + 1920z^7\tau^8\eta^3 + 2304z^7\tau^6\eta^5 - 21312z^6\tau^9\eta^3 \\
& - 2048z^6\tau^7\eta^5 + 22016z^5\tau^{10}\eta^3 - 80z^{13}\tau^2\eta^2 \\
& - 40z^{13}\eta^4 + 624z^{12}\tau^3\eta^2 + 458z^{12}\tau\eta^4 \\
& - 1656z^{11}\tau^4\eta^2 - 1817z^{11}\tau^2\eta^4 + 458z^{10}\tau^5\eta^2 \\
& + 3512z^{10}\tau^3\eta^4 + 6038z^9\tau^6\eta^2 - 2920z^9\tau^4\eta^4 \\
& - 13148z^8\tau^7\eta^2 + 4728z^8\tau^5\eta^4 + 15432z^7\tau^8\eta^2 \\
& - 22960z^7\tau^6\eta^4 - 13312z^6\tau^9\eta^2 + 34752z^6\tau^7\eta^4
\end{aligned}$$

$$\begin{aligned}
& + 1088z^5\tau^{10}\eta^2 - 22016z^5\tau^8\eta^4 + 9216z^4\tau^9\eta^4 \\
& + 4096z^3\tau^{10}\eta^4 - 8z^{12}\tau\eta^3 + 32z^{11}\tau^4\eta \\
& - 392z^{11}\tau^2\eta^3 - 544z^{10}\tau^5\eta + 2824z^{10}\tau^3\eta^3 \\
& + 2928z^9\tau^6\eta - 7308z^9\tau^4\eta^3 - 4048z^8\tau^7\eta \\
& + 7376z^8\tau^5\eta^3 + 768z^7\tau^8\eta - 2320z^7\tau^6\eta^3 \\
& - 1856z^6\tau^9\eta + 27008z^6\tau^7\eta^3 - 61568z^5\tau^8\eta^3 \\
& + 70656z^4\tau^9\eta^3 - 22528z^3\tau^{10}\eta^3 - 16z^{11}\tau^2\eta^2 \\
& - 16z^{11}\eta^4 + 72z^{10}\tau^3\eta^2 + 143z^{10}\tau\eta^4 \\
& + 64z^9\tau^6 + 236z^9\tau^4\eta^2 - 1036z^9\tau^2\eta^4 \\
& - 128z^8\tau^7 - 2038z^8\tau^5\eta^2 + 5248z^8\tau^3\eta^4 \\
& + 4724z^7\tau^6\eta^2 - 17680z^7\tau^4\eta^4 + 2768z^6\tau^7\eta^2 \\
& + 36288z^6\tau^5\eta^4 - 25952z^5\tau^8\eta^2 - 43136z^5\tau^6\eta^4 \\
& + 27584z^4\tau^9\eta^2 + 25600z^4\tau^7\eta^4 - 768z^3\tau^{10}\eta^2 \\
& - 10240z^3\tau^8\eta^4 + 4096z^2\tau^9\eta^4 - 40z^{11}\eta^3 \\
& + 388z^{10}\tau\eta^3 - 1642z^9\tau^2\eta^3 + 168z^8\tau^5\eta \\
& + 3038z^8\tau^3\eta^3 - 1568z^7\tau^6\eta + 672z^7\tau^4\eta^3 \\
& + 3504z^6\tau^7\eta - 16600z^6\tau^5\eta^3 - 3168z^5\tau^8\eta \\
& + 40000z^5\tau^6\eta^3 + 4352z^4\tau^9\eta - 53696z^4\tau^7\eta^3 \\
& + 51200z^3\tau^8\eta^3 - 56320z^2\tau^9\eta^3 + 24576z\tau^{10}\eta^3 \\
& - 40z^9\tau^2\eta^2 - 182z^8\tau^3\eta^2 - 2z^8\tau\eta^4 \\
& - 176z^7\tau^6 + 2418z^7\tau^4\eta^2 - 28z^7\tau^2\eta^4 \\
& + 320z^6\tau^7 - 8784z^6\tau^5\eta^2 + 320z^6\tau^3\eta^4 \\
& + 48z^5\tau^8 + 16152z^5\tau^6\eta^2 - 1408z^5\tau^4\eta^4 \\
& - 16064z^4\tau^7\eta^2 + 3072z^4\tau^5\eta^4 + 16000z^3\tau^8\eta^2 \\
& - 3328z^3\tau^6\eta^4 - 15872z^2\tau^9\eta^2 + 1024z^2\tau^7\eta^4 \\
& + 1024z\tau^{10}\eta^2 - 64z^9\tau^2\eta - 16z^9\eta^3 \\
& + 480z^8\tau^3\eta + 190z^8\tau\eta^3 - 1296z^7\tau^4\eta \\
& - 948z^7\tau^2\eta^3 + 768z^6\tau^5\eta + 3012z^6\tau^3\eta^3 \\
& + 3456z^5\tau^6\eta - 7632z^5\tau^4\eta^3 - 7488z^4\tau^7\eta \\
& + 19712z^4\tau^5\eta^3 + 4608z^3\tau^8\eta - 43904z^3\tau^6\eta^3 \\
& - 3072z^2\tau^9\eta + 65792z^2\tau^7\eta^3 - 51200z\tau^8\eta^3 \\
& + 24576\tau^9\eta^3 - 10z^8\tau\eta^2 + 16z^7\tau^4 \\
& + 12z^7\tau^2\eta^2 - 208z^6\tau^5 + 174z^6\tau^3\eta^2 \\
& + 1072z^5\tau^6 - 784z^5\tau^4\eta^2 - 1328z^4\tau^7 \\
& + 840z^4\tau^5\eta^2 - 96z^3\tau^8 + 432z^3\tau^6\eta^2 \\
& + 3648z^2\tau^7\eta^2 - 19968z\tau^8\eta^2 + 17408\tau^9\eta^2 \\
& + 24z^6\tau^3\eta - 8z^6\tau\eta^3 - 152z^5\tau^4\eta \\
& - 112z^5\tau^2\eta^3 + 816z^4\tau^5\eta + 1280z^4\tau^3\eta^3 \\
& - 2656z^3\tau^6\eta - 5632z^3\tau^4\eta^3 + 6016z^2\tau^7\eta \\
& + 12288z^2\tau^5\eta^3 - 6656z\tau^8\eta - 13312z\tau^6\eta^3 \\
& + 4096\tau^9\eta + 4096\tau^7\eta^3 - 16z^7\eta^2 \\
& + 132z^6\tau\eta^2 - 16z^5\tau^4 - 352z^5\tau^2\eta^2 \\
& + 160z^4\tau^5 - 548z^4\tau^3\eta^2 - 784z^3\tau^6 \\
& + 6072z^3\tau^4\eta^2 + 768z^2\tau^7 - 16768z^2\tau^5\eta^2
\end{aligned}$$

$$\begin{aligned}
& + 384z\tau^8 + 21824z\tau^6\eta^2 - 9984\tau^7\eta^2 \\
& - 96z^4\tau^3\eta + 544z^3\tau^4\eta - 1536z^2\tau^5\eta \\
& + 2816z\tau^6\eta - 3072\tau^7\eta - 16z^5\tau^2 \\
& + 112z^4\tau^3 - 240z^3\tau^4 - 80z^2\tau^5 \\
& + 1248z\tau^6 - 1408\tau^7
\end{aligned}$$

$$\text{Res}_\rho(q, \Xi)|_{m_j=1}$$

Primitive part of the resultant on the boundary  $m_j = 1$ , a polynomial in  $m_i$  of degree 19. 20 terms.

$$\begin{aligned}
\text{Res}_\rho(q, \Xi)|_{m_j=1}(m_i) = & - 240100m_i^{19} + 1604505m_i^{18} - 4608296m_i^{17} \\
& + 7726144m_i^{16} - 6794896m_i^{15} + 529340m_i^{14} \\
& + 8748776m_i^{13} - 15862432m_i^{12} + 17804328m_i^{11} \\
& - 14975722m_i^{10} + 10070520m_i^9 - 5557776m_i^8 \\
& + 2548400m_i^7 - 972804m_i^6 + 307512m_i^5 \\
& - 79360m_i^4 + 16284m_i^3 - 2535m_i^2 \\
& + 272m_i - 16
\end{aligned}$$

## Appendix A.3.9

Polynomials of Appendix A.3.9. Here  $m_i = 1/z$  and  $m_j = \tau/z$  in the rescaled polynomials. The Krawczyk certification cut  $z \in [0, 1]$  into 2000 pieces, refined into 8219 boxes, the first and last containing the exact ends ( $z$ ,  $\tau$ ,  $\rho$ ) = (1, 1, 1) and (0, 1,  $\frac{1}{13}$ ).

$$\Pi_{\text{poly}}$$

$\Pi = \Pi_{\text{poly}}/(\rho(1 + \rho)^3 A_0^2)$ ;  $du_i^\# / dw_i = \Pi / \Delta$ . 108 terms.

$$\begin{aligned}
\Pi_{\text{poly}}(m_i, m_j, \rho) = & 64m_i^9m_j^5\rho^3 + 256m_i^8m_j^7\rho^2 - 128m_i^8m_j^6\rho^3 \\
& + 128m_i^8m_j^5\rho^4 - 128m_i^7m_j^6\rho^4 + 64m_i^7m_j^5\rho^5 \\
& - 256m_i^8m_j^6\rho^2 + 32m_i^8m_j^5\rho^3 + 64m_i^8m_j^4\rho^4 \\
& - 384m_i^7m_j^7\rho^2 + 64m_i^7m_j^6\rho^3 - 192m_i^7m_j^5\rho^4 \\
& + 192m_i^7m_j^4\rho^5 + 128m_i^6m_j^7\rho^3 + 96m_i^6m_j^6\rho^4 \\
& - 352m_i^6m_j^5\rho^5 + 192m_i^6m_j^4\rho^6 + 128m_i^5m_j^7\rho^4 \\
& + 32m_i^5m_j^6\rho^5 - 128m_i^5m_j^5\rho^6 + 64m_i^5m_j^4\rho^7 \\
& + 160m_i^9m_j^4\rho^2 + 8m_i^9m_j^3\rho^3 + 48m_i^8m_j^5\rho^2 \\
& + 192m_i^8m_j^4\rho^3 + 16m_i^8m_j^3\rho^4 + 352m_i^7m_j^6\rho^2 \\
& - 200m_i^7m_j^5\rho^3 + 176m_i^7m_j^4\rho^4 + 8m_i^7m_j^3\rho^5 \\
& + 336m_i^6m_j^7\rho^2 - 48m_i^6m_j^6\rho^3 - 80m_i^6m_j^5\rho^4 \\
& + 208m_i^6m_j^4\rho^5 - 32m_i^5m_j^7\rho^3 - 112m_i^5m_j^6\rho^4 \\
& + 104m_i^5m_j^5\rho^5 + 32m_i^5m_j^4\rho^6 - 160m_i^4m_j^7\rho^4 \\
& - 32m_i^4m_j^6\rho^5 + 64m_i^4m_j^5\rho^6 - 32m_i^4m_j^4\rho^7 \\
& + 64m_i^9m_j^4\rho - 64m_i^9m_j^3\rho^2 + 128m_i^8m_j^5\rho \\
& - 208m_i^8m_j^4\rho^2 - 56m_i^8m_j^3\rho^3 + 80m_i^7m_j^5\rho^2 \\
& - 192m_i^7m_j^4\rho^3 + 8m_i^7m_j^3\rho^4 - 192m_i^6m_j^6\rho^2 \\
& + 264m_i^6m_j^5\rho^3 + 48m_i^6m_j^4\rho^4 - 256m_i^5m_j^7\rho^2 \\
& - 64m_i^5m_j^6\rho^3 + 216m_i^5m_j^5\rho^4 - 64m_i^5m_j^4\rho^5
\end{aligned}$$

$$\begin{aligned}
& -112m_i^4m_j^7\rho^3 + 32m_i^4m_j^6\rho^4 + 48m_i^4m_j^5\rho^5 \\
& -64m_i^4m_j^4\rho^6 + 48m_i^3m_j^7\rho^4 - 32m_i^9m_j^3\rho \\
& + 16m_i^9m_j^2\rho^2 - 256m_i^8m_j^4\rho + 68m_i^8m_j^3\rho^2 \\
& + 16m_i^8m_j^2\rho^3 - 64m_i^7m_j^5\rho - 48m_i^7m_j^4\rho^2 \\
& + 96m_i^7m_j^3\rho^3 - 72m_i^6m_j^5\rho^2 + 192m_i^6m_j^4\rho^3 \\
& + 96m_i^5m_j^6\rho^2 - 64m_i^5m_j^5\rho^3 - 64m_i^5m_j^4\rho^4 \\
& + 68m_i^4m_j^7\rho^2 + 64m_i^4m_j^6\rho^3 - 32m_i^4m_j^5\rho^4 \\
& - 80m_i^4m_j^4\rho^5 + 32m_i^3m_j^7\rho^3 - 16m_i^2m_j^7\rho^4 \\
& + 8m_i^9m_j^2\rho + 16m_i^8m_j^4 + 112m_i^8m_j^3\rho \\
& - 16m_i^8m_j^2\rho^2 - 32m_i^7m_j^5 + 152m_i^7m_j^4\rho \\
& + 108m_i^7m_j^3\rho^2 - 24m_i^7m_j^2\rho^3 - 16m_i^6m_j^5\rho \\
& + 64m_i^6m_j^4\rho^2 + 24m_i^5m_j^5\rho^2 - 184m_i^5m_j^4\rho^3 \\
& + 16m_i^4m_j^5\rho^3 - 112m_i^4m_j^4\rho^4 - 20m_i^3m_j^7\rho^2 \\
& - 16m_i^2m_j^7\rho^3 - 24m_i^8m_j^2\rho - 24m_i^7m_j^2\rho^2 \\
& + 32m_i^6m_j^5 - 104m_i^6m_j^4\rho + 64m_i^5m_j^5\rho \\
& - 184m_i^5m_j^4\rho^2 + 32m_i^4m_j^5\rho^2 - 80m_i^4m_j^4\rho^3 \\
& - 16m_i^6m_j^4 - 32m_i^5m_j^4\rho - 16m_i^4m_j^4\rho^2
\end{aligned}$$

$z^5q$

$q$  after  $m_i = 1/z$ ,  $m_j = \tau/z$ , cleared by  $z^5$ . 18 terms.

$$\begin{aligned}
z^5q(z, \tau, \rho) = & -z^3\tau^2\rho^2 + z^2\tau^3\rho + 4z^2\tau^2\rho^2 \\
& + z^3\tau^2 - z^3\rho^2 - z^2\tau^2\rho \\
& + 2z^2\tau\rho^2 - 2z\tau^3\rho - 8z\tau^2\rho^2 \\
& - 2z^2\tau^2 + z^2\tau\rho + 8\tau^3\rho \\
& + z^3 - 4z^2\tau - z^2\rho \\
& + 8z\tau^2 + 2z\tau\rho - 8\tau^2\rho
\end{aligned}$$

$z^5\Upsilon$

$\Upsilon$  after the same substitution, cleared by  $z^5$ . 14 terms.

$$\begin{aligned}
z^5\Upsilon(z, \tau, \rho) = & -2z^3\tau^2\rho + z^3\rho^2 + 4z^2\tau^2\rho \\
& - 4z^2\tau\rho^2 + 6z\tau^2\rho^2 - z^3\rho \\
& - z^2\tau^2 + 4z^2\tau\rho - 10z\tau^2\rho \\
& + z^2\rho + 2z\tau^2 - 4z\tau\rho \\
& + 6\tau^2\rho - 2\tau^2
\end{aligned}$$

$z^{15}\Pi_{\text{poly}}$

$\Pi_{\text{poly}}$  after the same substitution, cleared by  $z^{15}$ ; its  $z = 0$  part is  $256\rho^2\tau^7$ . 108 terms.

$$\begin{aligned}
z^{15}\Pi_{\text{poly}}(z, \tau, \rho) = & -32z^7\tau^4\rho^7 - 64z^7\tau^4\rho^6 - 16z^6\tau^7\rho^4 \\
& + 64z^6\tau^5\rho^6 + 64z^6\tau^4\rho^7 - 80z^7\tau^4\rho^5 \\
& - 16z^6\tau^7\rho^3 + 48z^6\tau^5\rho^5 + 32z^6\tau^4\rho^6 \\
& + 48z^5\tau^7\rho^4 - 32z^5\tau^6\rho^5 - 128z^5\tau^5\rho^6 \\
& - 112z^7\tau^4\rho^4 - 32z^6\tau^5\rho^4 - 64z^6\tau^4\rho^5 \\
& + 32z^5\tau^7\rho^3 + 32z^5\tau^6\rho^4 + 104z^5\tau^5\rho^5
\end{aligned}$$

$$\begin{aligned}
& + 192z^5\tau^4\rho^6 - 160z^4\tau^7\rho^4 + 32z^4\tau^6\rho^5 \\
& - 80z^7\tau^4\rho^3 + 16z^6\tau^5\rho^3 - 64z^6\tau^4\rho^4 \\
& - 20z^5\tau^7\rho^2 + 64z^5\tau^6\rho^3 + 216z^5\tau^5\rho^4 \\
& + 208z^5\tau^4\rho^5 - 112z^4\tau^7\rho^3 - 112z^4\tau^6\rho^4 \\
& - 352z^4\tau^5\rho^5 + 128z^3\tau^7\rho^4 - 16z^7\tau^4\rho^2 \\
& + 32z^6\tau^5\rho^2 - 184z^6\tau^4\rho^3 - 64z^5\tau^5\rho^3 \\
& + 48z^5\tau^4\rho^4 + 8z^5\tau^3\rho^5 + 68z^4\tau^7\rho^2 \\
& - 64z^4\tau^6\rho^3 - 80z^4\tau^5\rho^4 + 192z^4\tau^4\rho^5 \\
& - 32z^3\tau^7\rho^3 + 96z^3\tau^6\rho^4 + 64z^3\tau^5\rho^5 \\
& - 184z^6\tau^4\rho^2 + 24z^5\tau^5\rho^2 + 192z^5\tau^4\rho^3 \\
& + 8z^5\tau^3\rho^4 + 96z^4\tau^6\rho^2 + 264z^4\tau^5\rho^3 \\
& + 176z^4\tau^4\rho^4 - 256z^3\tau^7\rho^2 - 48z^3\tau^6\rho^3 \\
& - 192z^3\tau^5\rho^4 + 128z^2\tau^7\rho^3 - 128z^2\tau^6\rho^4 \\
& - 32z^6\tau^4\rho - 24z^6\tau^2\rho^3 + 64z^5\tau^5\rho \\
& + 64z^5\tau^4\rho^2 + 96z^5\tau^3\rho^3 - 72z^4\tau^5\rho^2 \\
& - 192z^4\tau^4\rho^3 + 16z^4\tau^3\rho^4 - 192z^3\tau^6\rho^2 \\
& - 200z^3\tau^5\rho^3 + 64z^3\tau^4\rho^4 + 336z^2\tau^7\rho^2 \\
& + 64z^2\tau^6\rho^3 + 128z^2\tau^5\rho^4 - 24z^6\tau^2\rho^2 \\
& - 104z^5\tau^4\rho + 108z^5\tau^3\rho^2 + 16z^5\tau^2\rho^3 \\
& - 16z^4\tau^5\rho - 48z^4\tau^4\rho^2 - 56z^4\tau^3\rho^3 \\
& + 80z^3\tau^5\rho^2 + 192z^3\tau^4\rho^3 + 352z^2\tau^6\rho^2 \\
& + 32z^2\tau^5\rho^3 - 384z\tau^7\rho^2 - 128z\tau^6\rho^3 \\
& - 16z^5\tau^4 - 16z^5\tau^2\rho^2 + 32z^4\tau^5 \\
& + 152z^4\tau^4\rho + 68z^4\tau^3\rho^2 - 64z^3\tau^5\rho \\
& - 208z^3\tau^4\rho^2 + 8z^3\tau^3\rho^3 + 48z^2\tau^5\rho^2 \\
& - 256z\tau^6\rho^2 + 64z\tau^5\rho^3 + 256\tau^7\rho^2 \\
& - 24z^5\tau^2\rho + 112z^4\tau^3\rho + 16z^4\tau^2\rho^2 \\
& - 32z^3\tau^5 - 256z^3\tau^4\rho - 64z^3\tau^3\rho^2 \\
& + 128z^2\tau^5\rho + 160z^2\tau^4\rho^2 + 8z^4\tau^2\rho \\
& + 16z^3\tau^4 - 32z^3\tau^3\rho + 64z^2\tau^4\rho
\end{aligned}$$

## Appendix A.4.2, the one-sided crossing $w^{os}$

Polynomials of Appendix A.4.2. Here  $u = 2\sqrt{x_{i1}}$ ,  $s = 2\sqrt{w_i - x_{i1}}$  and  $r = \sqrt{\Psi(s)}$  with  $\Psi(s) = s^2 - 4s + 7$ . The resultant of  $K$  and  $8h$  with respect to  $u$  is a polynomial in  $(s, r)$  which, reduced by  $r^2 = \Psi(s)$ , equals  $\Psi(s)^2 Q(s) r$  with  $Q$  as in the appendix. Numerically  $s^* \approx 0.64209$ ,  $u^* \approx 0.85528$ ,  $w^{os} \approx 0.2859$  and  $w_j^{un}(w^{os}) \approx 0.5377$ ; at  $(u^*, s^*)$ ,  $dx_{j1}^*/dw_i \approx -0.094$ .

$K$

$K = N - 2\Psi G$  with  $r = \sqrt{\Psi(s)}$ ;  $dTI/dw_i - 1 = K/(2\Psi G)$ . 24 terms.

$$\begin{aligned}
K(u, s, r) = & - 4u^3s^4 - 6u^2s^4r + 14us^6 \\
& - 2s^6r + 30u^3s^3 + 23u^2s^3r \\
& - 137us^5 + 32s^5r - 102u^3s^2 \\
& - 37u^2s^2r + 623us^4 - 173s^4r \\
& + 170u^3s + 1u^2sr - 1584us^3
\end{aligned}$$

$$\begin{aligned}
& + 512s^3r - 126u^3 + 19u^2r \\
& + 2398us^2 - 876s^2r - 2007us \\
& + 872sr + 917u - 413r
\end{aligned}$$

$\text{Res}_u(K, 8h)$

Resultant of  $K$  and  $8h$  with respect to  $u$ , before reduction by  $r^2 = \Psi(s)$ . 54 terms.

$$\begin{aligned}
\text{Res}_u(K, 8h)(s, r) = & -48s^{19}r + 192s^{17}r^3 - 64s^{15}r^5 \\
& + 980s^{18}r - 3912s^{16}r^3 + 1168s^{14}r^5 \\
& - 10106s^{17}r + 40658s^{15}r^3 - 10953s^{13}r^5 \\
& + 67914s^{16}r - 279114s^{14}r^3 + 67733s^{12}r^5 \\
& - 327288s^{15}r + 1402354s^{13}r^3 - 304654s^{11}r^5 \\
& + 1175480s^{14}r - 5430962s^{12}r^3 + 1045438s^{10}r^5 \\
& - 3156744s^{13}r + 16702914s^{11}r^3 - 2809619s^9r^5 \\
& + 6055624s^{12}r - 41511642s^{10}r^3 + 5995519s^8r^5 \\
& - 6735656s^{11}r + 84215282s^9r^3 - 10220788s^7r^5 \\
& - 2425744s^{10}r - 140186402s^8r^3 + 13918468s^6r^5 \\
& + 29218468s^9r + 191945606s^7r^3 - 15049503s^5r^5 \\
& - 68252836s^8r - 216383406s^6r^3 + 12734211s^4r^5 \\
& + 89164408s^7r + 200949958s^5r^3 - 8199646s^3r^5 \\
& - 50505048s^6r - 153504582s^4r^3 + 3815630s^2r^5 \\
& - 55956040s^5r + 95702502s^3r^3 - 1153509sr^5 \\
& + 174150088s^4r - 47297838s^2r^3 + 170569r^5 \\
& - 222444712s^3r + 16739478sr^3 + 173509980s^2r \\
& - 3035550r^3 - 82426330sr + 17849034r
\end{aligned}$$

### Sturm sequence of $Q$

The sequence  $Q_0 = Q$ ,  $Q_1 = Q'$ ,  $Q_{k+1} = -\text{rem}(Q_{k-1}, Q_k)$  has 16 terms. Their coefficients grow to more than sixty digits, so the polynomials are given only in the plain-text file, each scaled by a positive constant to a primitive integer polynomial. The table lists the degree of each term and its sign at  $s = 0$  and  $s = 1$ ; the numbers of sign changes are 8 and 7.

|           |    |    |    |    |    |    |   |   |   |   |    |    |    |    |    |    |
|-----------|----|----|----|----|----|----|---|---|---|---|----|----|----|----|----|----|
| $k$       | 0  | 1  | 2  | 3  | 4  | 5  | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| degree    | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5  | 4  | 3  | 2  | 1  | 0  |
| sign at 0 | +  | -  | -  | -  | +  | +  | - | - | + | + | -  | -  | +  | -  | -  | +  |
| sign at 1 | -  | -  | +  | +  | -  | -  | + | + | - | - | +  | +  | -  | +  | +  | +  |

### Appendix A.4.3, single crossing of $dTI/dw_i$

Polynomials of Appendix A.4.3 and of the endpoint analysis in Appendix D. Here  $a = 1/m_i$  and  $b = 1/m_j$  in the rescaled polynomials. Numerically  $m_i^* \approx 1.519306$ ,  $\rho^* \approx 0.652536$ ,  $\hat{w}_j \approx 0.536763$ ,  $T(2, 2, 1) \approx -2.929$  and  $T(m_i^*, 1, \rho^*) \approx -3.625$ . The Krawczyk certification cut  $[m_i^* + 10^{-7}, 2 - 10^{-7}]$  into 24000 pieces, refined into 26219 boxes, plus the two endpoint boxes of half-width  $2 \cdot 10^{-4}$ , 26221 boxes in all. The exhaustive search tested 68004 boxes in the chart  $(t, b)$ , of which 25361 were discarded because  $H_0 \neq 0$  and 8642 lay inside the certified boxes, and 11 and 9 boxes in the two corner charts, all discarded.

$H_{\text{poly}}$

$dTI/dw_i - 1 = H_{\text{poly}}/(\rho(1 + \rho)^3 A_0^3 \Delta)$ . 206 terms.

$$H_{\text{poly}}(m_i, m_j, \rho) = 512m_i^{10}m_j^8\rho^3 + 512m_i^9m_j^8\rho^4 + 128m_i^{11}m_j^6\rho^3$$

$$\begin{aligned}
& + 512m_i^{10}m_j^8\rho^2 - 896m_i^{10}m_j^7\rho^3 + 256m_i^{10}m_j^6\rho^4 \\
& + 128m_i^9m_j^8\rho^3 - 128m_i^9m_j^7\rho^4 + 128m_i^9m_j^6\rho^5 \\
& + 512m_i^8m_j^9\rho^3 - 512m_i^8m_j^8\rho^4 + 1280m_i^8m_j^7\rho^5 \\
& - 512m_i^7m_j^9\rho^4 - 128m_i^7m_j^8\rho^5 + 384m_i^7m_j^7\rho^6 \\
& - 1024m_i^6m_j^9\rho^5 - 128m_i^6m_j^7\rho^7 - 1152m_i^{10}m_j^7\rho^2 \\
& + 320m_i^{10}m_j^6\rho^3 + 128m_i^{10}m_j^5\rho^4 - 128m_i^9m_j^8\rho^2 \\
& - 448m_i^9m_j^7\rho^3 - 1120m_i^9m_j^6\rho^4 + 320m_i^9m_j^5\rho^5 \\
& + 128m_i^8m_j^9\rho^2 - 448m_i^8m_j^8\rho^3 + 640m_i^8m_j^7\rho^4 \\
& - 2432m_i^8m_j^6\rho^5 + 256m_i^8m_j^5\rho^6 - 1536m_i^7m_j^9\rho^3 \\
& + 608m_i^7m_j^8\rho^4 - 544m_i^7m_j^7\rho^5 - 992m_i^7m_j^6\rho^6 \\
& + 64m_i^7m_j^5\rho^7 - 1152m_i^6m_j^9\rho^4 + 928m_i^6m_j^8\rho^5 \\
& - 448m_i^6m_j^7\rho^6 + 512m_i^5m_j^9\rho^5 + 32m_i^5m_j^7\rho^7 \\
& + 320m_i^{11}m_j^5\rho^2 + 16m_i^{11}m_j^4\rho^3 - 512m_i^{10}m_j^7\rho \\
& + 384m_i^{10}m_j^6\rho^2 + 560m_i^{10}m_j^5\rho^3 + 32m_i^{10}m_j^4\rho^4 \\
& + 96m_i^9m_j^7\rho^2 - 864m_i^9m_j^6\rho^3 + 816m_i^9m_j^5\rho^4 \\
& + 16m_i^9m_j^4\rho^5 - 64m_i^8m_j^8\rho^2 + 816m_i^8m_j^7\rho^3 \\
& - 1856m_i^8m_j^6\rho^4 + 864m_i^8m_j^5\rho^5 - 672m_i^7m_j^9\rho^2 \\
& + 1360m_i^7m_j^8\rho^3 - 208m_i^7m_j^7\rho^4 - 480m_i^7m_j^6\rho^5 \\
& + 240m_i^7m_j^5\rho^6 + 96m_i^6m_j^9\rho^3 + 1184m_i^6m_j^8\rho^4 \\
& - 288m_i^6m_j^7\rho^5 + 160m_i^6m_j^6\rho^6 - 48m_i^6m_j^5\rho^7 \\
& + 448m_i^5m_j^9\rho^4 - 240m_i^5m_j^8\rho^5 + 112m_i^5m_j^7\rho^6 \\
& + 32m_i^5m_j^6\rho^7 - 320m_i^4m_j^9\rho^5 - 16m_i^4m_j^7\rho^7 \\
& + 128m_i^{11}m_j^5\rho - 128m_i^{11}m_j^4\rho^2 + 1152m_i^{10}m_j^6\rho \\
& - 48m_i^{10}m_j^5\rho^2 - 216m_i^{10}m_j^4\rho^3 + 384m_i^9m_j^7\rho \\
& + 1232m_i^9m_j^6\rho^2 + 48m_i^9m_j^5\rho^3 - 184m_i^9m_j^4\rho^4 \\
& + 256m_i^8m_j^7\rho^2 - 320m_i^8m_j^6\rho^3 + 448m_i^8m_j^5\rho^4 \\
& - 160m_i^8m_j^4\rho^5 + 592m_i^7m_j^8\rho^2 - 912m_i^7m_j^7\rho^3 \\
& - 768m_i^7m_j^6\rho^4 + 272m_i^7m_j^5\rho^5 - 64m_i^7m_j^4\rho^6 \\
& + 304m_i^6m_j^9\rho^2 + 248m_i^6m_j^8\rho^3 - 752m_i^6m_j^7\rho^4 \\
& - 336m_i^6m_j^6\rho^5 + 48m_i^6m_j^5\rho^6 - 160m_i^5m_j^9\rho^3 \\
& - 232m_i^5m_j^8\rho^4 - 16m_i^5m_j^7\rho^5 + 32m_i^5m_j^6\rho^6 \\
& - 400m_i^4m_j^9\rho^4 + 112m_i^4m_j^8\rho^5 - 48m_i^4m_j^7\rho^6 \\
& + 64m_i^3m_j^9\rho^5 - 64m_i^{11}m_j^4\rho + 32m_i^{11}m_j^3\rho^2 \\
& + 640m_i^{10}m_j^6 - 896m_i^{10}m_j^5\rho - 88m_i^{10}m_j^4\rho^2 \\
& + 56m_i^{10}m_j^3\rho^3 - 256m_i^9m_j^7 + 64m_i^9m_j^6\rho \\
& - 944m_i^9m_j^5\rho^2 - 104m_i^9m_j^4\rho^3 + 24m_i^9m_j^3\rho^4 \\
& - 576m_i^8m_j^7\rho - 1632m_i^8m_j^6\rho^2 + 392m_i^8m_j^5\rho^3 \\
& - 160m_i^8m_j^4\rho^4 - 448m_i^7m_j^7\rho^2 - 1568m_i^7m_j^6\rho^3 \\
& + 824m_i^7m_j^5\rho^4 - 112m_i^7m_j^4\rho^5 - 136m_i^6m_j^8\rho^2 \\
& - 40m_i^6m_j^7\rho^3 - 432m_i^6m_j^6\rho^4 + 384m_i^6m_j^5\rho^5 \\
& - 32m_i^6m_j^4\rho^6 - 176m_i^5m_j^9\rho^2 + 40m_i^5m_j^8\rho^3 \\
& + 40m_i^5m_j^7\rho^4 + 48m_i^5m_j^6\rho^5 - 88m_i^4m_j^9\rho^3 \\
& + 176m_i^4m_j^8\rho^4 - 48m_i^4m_j^7\rho^5 - 32m_i^4m_j^6\rho^6 \\
& + 72m_i^3m_j^9\rho^4 - 16m_i^2m_j^9\rho^5 + 16m_i^{11}m_j^3\rho
\end{aligned}$$

$$\begin{aligned}
& -576m_i^{10}m_j^5 + 336m_i^{10}m_j^4\rho + 24m_i^{10}m_j^3\rho^2 \\
& -288m_i^9m_j^6 - 288m_i^9m_j^5\rho + 352m_i^9m_j^4\rho^2 \\
& + 8m_i^9m_j^3\rho^3 + 320m_i^8m_j^7 - 368m_i^8m_j^6\rho \\
& + 880m_i^8m_j^5\rho^2 - 80m_i^8m_j^4\rho^3 + 656m_i^7m_j^7\rho \\
& - 128m_i^7m_j^6\rho^2 + 984m_i^7m_j^5\rho^3 - 192m_i^7m_j^4\rho^4 \\
& + 344m_i^6m_j^7\rho^2 + 16m_i^6m_j^6\rho^3 + 392m_i^6m_j^5\rho^4 \\
& - 96m_i^6m_j^4\rho^5 + 64m_i^5m_j^8\rho^2 - 40m_i^5m_j^7\rho^3 \\
& - 32m_i^5m_j^6\rho^4 + 32m_i^4m_j^9\rho^2 + 64m_i^4m_j^8\rho^3 \\
& - 48m_i^4m_j^7\rho^4 - 96m_i^4m_j^6\rho^5 + 8m_i^3m_j^9\rho^3 \\
& - 24m_i^2m_j^9\rho^4 + 272m_i^{10}m_j^4 - 72m_i^{10}m_j^3\rho \\
& + 160m_i^9m_j^5 + 224m_i^9m_j^4\rho - 72m_i^9m_j^3\rho^2 \\
& + 80m_i^8m_j^6 + 256m_i^8m_j^5\rho - 176m_i^8m_j^4\rho^2 \\
& - 96m_i^7m_j^7 + 232m_i^7m_j^5\rho^2 - 224m_i^7m_j^4\rho^3 \\
& - 184m_i^6m_j^7\rho - 144m_i^6m_j^6\rho^2 + 136m_i^6m_j^5\rho^3 \\
& - 96m_i^6m_j^4\rho^4 - 88m_i^5m_j^7\rho^2 - 160m_i^5m_j^6\rho^3 \\
& - 96m_i^4m_j^6\rho^4 - 8m_i^3m_j^9\rho^2 - 8m_i^2m_j^9\rho^3 \\
& - 64m_i^{10}m_j^3 + 8m_i^{10}m_j^2\rho - 48m_i^9m_j^4 \\
& - 64m_i^9m_j^3\rho + 8m_i^9m_j^2\rho^2 - 32m_i^8m_j^5 \\
& - 80m_i^8m_j^4\rho - 48m_i^7m_j^6 - 64m_i^7m_j^4\rho^2 \\
& + 32m_i^6m_j^7 - 88m_i^6m_j^6\rho + 32m_i^6m_j^5\rho^2 \\
& - 32m_i^6m_j^4\rho^3 + 64m_i^5m_j^7\rho - 72m_i^5m_j^6\rho^2 \\
& + 32m_i^4m_j^7\rho^2 - 32m_i^4m_j^6\rho^3 + 8m_i^{10}m_j^2 \\
& + 8m_i^9m_j^2\rho + 16m_i^8m_j^4 + 16m_i^7m_j^4\rho \\
& + 8m_i^6m_j^6 + 8m_i^5m_j^6\rho
\end{aligned}$$

$D$

$\Delta = 4m_i^2m_j^2D/(\rho(1+\rho)^2A_0^2)$ . 62 terms.

$$\begin{aligned}
D(m_i, m_j, \rho) = & 64m_i^5m_j^5\rho^2 - 64m_i^5m_j^4\rho^2 - 8m_i^5m_j^2\rho^4 \\
& - 32m_i^4m_j^5\rho^2 + 4m_i^4m_j^3\rho^4 - 16m_i^4m_j^2\rho^5 \\
& + 32m_i^3m_j^5\rho^3 + 8m_i^3m_j^3\rho^5 - 8m_i^3m_j^2\rho^6 \\
& + 32m_i^2m_j^5\rho^4 + 4m_i^2m_j^3\rho^6 + 40m_i^6m_j^2\rho^2 \\
& + 12m_i^5m_j^3\rho^2 + 48m_i^5m_j^2\rho^3 + 24m_i^4m_j^4\rho^2 \\
& - 40m_i^4m_j^3\rho^3 + 12m_i^4m_j^2\rho^4 + 52m_i^3m_j^5\rho^2 \\
& - 20m_i^3m_j^3\rho^4 + 8m_i^3m_j^2\rho^5 + 24m_i^2m_j^5\rho^3 \\
& - 8m_i^2m_j^4\rho^4 + 4m_i^2m_j^2\rho^6 - 8m_i^6m_j^5\rho^4 \\
& + 32m_i^6m_j^2\rho - 16m_i^6m_j\rho^2 + 20m_i^5m_j^2\rho^2 \\
& - 16m_i^5m_j\rho^3 + 16m_i^4m_j^2\rho^3 - 24m_i^3m_j^4\rho^2 \\
& + 12m_i^3m_j^3\rho^3 + 36m_i^3m_j^2\rho^4 - 12m_i^2m_j^5\rho^2 \\
& - 16m_i^2m_j^4\rho^3 + 8m_i^2m_j^3\rho^4 + 16m_i^2m_j^2\rho^5 \\
& - 4m_i^5m_j^3\rho^3 + 4m_i^5\rho^4 - 16m_i^6m_j\rho \\
& + 4m_i^6\rho^2 - 8m_i^5m_j^2\rho - 15m_i^5m_j\rho^2 \\
& + 4m_i^5\rho^3 + 20m_i^4m_j^2\rho^2 + 6m_i^3m_j^3\rho^2 \\
& + 52m_i^3m_j^2\rho^3 + 4m_i^2m_j^3\rho^3 + 24m_i^2m_j^2\rho^4 \\
& + 5m_i^5m_j^2\rho^2 + 4m_i^5\rho^3 + 4m_i^6\rho
\end{aligned}$$

$$\begin{aligned}
& + 4m_i^5 m_j^2 + 4m_i^5 \rho^2 - 8m_i^4 m_j^3 \\
& + 28m_i^4 m_j^2 \rho - 16m_i^3 m_j^3 \rho + 40m_i^3 m_j^2 \rho^2 \\
& - 8m_i^2 m_j^3 \rho^2 + 16m_i^2 m_j^2 \rho^3 + 4m_i^4 m_j^2 \\
& + 8m_i^3 m_j^2 \rho + 4m_i^2 m_j^2 \rho^2
\end{aligned}$$

$T_{\text{ess}}$

$T = 8m_i^2 m_j^2 (m_i + \rho)^2 T_{\text{ess}} / ((1 + \rho) A_0^2 D^3)$ . 2475 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$P_1$

$\text{Res}_\rho(q_1, H_{\text{poly}}(\cdot, 1, \cdot)) = 64(m_i - 1)^3 m_i^4 (5m_i^2 - 2m_i + 1)^4 P_1$ . 22 terms.

$$\begin{aligned}
P_1(m_i) = & 283318m_i^{21} - 2651390m_i^{20} + 14815983m_i^{19} \\
& - 52141117m_i^{18} + 128684696m_i^{17} - 235884014m_i^{16} \\
& + 330453124m_i^{15} - 365487967m_i^{14} + 325747092m_i^{13} \\
& - 237768827m_i^{12} + 143608690m_i^{11} - 72226573m_i^{10} \\
& + 30302288m_i^9 - 10578007m_i^8 + 3047604m_i^7 \\
& - 713317m_i^6 + 131798m_i^5 - 18179m_i^4 \\
& + 1639m_i^3 - 50m_i^2 - 8m_i \\
& + 1
\end{aligned}$$

$Q_0$

$a^3 b^3 q = t b^2 Q_0(t, b, \sigma)$  with  $a = bt = 1/m_i$ ,  $b = 1/m_j$ ,  $\rho = b\sigma$ ; the search uses the positive root  $\sigma$  of  $Q_0$ . 18 terms.

$$\begin{aligned}
Q_0(t, b, \sigma) = & -1t^4 b^4 \sigma^2 + 4t^3 b^3 \sigma^2 - 1t^2 b^4 \sigma^2 \\
& + 2t^2 b^3 \sigma^2 + 1t^3 b^2 \sigma - 8t^2 b^2 \sigma^2 \\
& - 1t^2 b^2 \sigma + 1t^2 b^2 - 2t^2 b \sigma \\
& + 1tb^2 \sigma - 1b^2 \sigma - 2tb \\
& + 8t\sigma + 1b^2 + 2b\sigma \\
& - 4b - 8\sigma + 8
\end{aligned}$$

$H_0$

$a^{11} b^9 H_{\text{poly}} = t b^4 H_0(t, b, \sigma)$ , the polynomial tested in the chart  $(t, b)$ . 206 terms.

$$\begin{aligned}
H_0(t, b, \sigma) = & -16t^{13} b^{12} \sigma^7 + 32t^{12} b^{12} \sigma^7 - 32t^{12} b^{12} \sigma^6 \\
& + 32t^{12} b^{11} \sigma^7 - 48t^{11} b^{12} \sigma^7 - 48t^{12} b^{11} \sigma^6 \\
& - 16t^{13} b^{10} \sigma^5 + 32t^{11} b^{11} \sigma^6 - 128t^{11} b^{10} \sigma^7 \\
& - 32t^{10} b^{12} \sigma^6 + 64t^{10} b^{11} \sigma^7 - 96t^{11} b^{11} \sigma^5 \\
& + 112t^{11} b^{10} \sigma^6 + 48t^{10} b^{11} \sigma^6 + 64t^{12} b^9 \sigma^5 \\
& - 48t^{11} b^{10} \sigma^5 + 160t^{10} b^{10} \sigma^6 - 64t^9 b^{11} \sigma^6 \\
& - 24t^{12} b^9 \sigma^4 + 112t^{11} b^9 \sigma^5 + 48t^{10} b^{10} \sigma^5 \\
& - 448t^{10} b^9 \sigma^6 - 96t^9 b^{11} \sigma^5 + 240t^9 b^{10} \sigma^6 \\
& - 320t^{11} b^8 \sigma^5 - 96t^{10} b^{10} \sigma^4 - 16t^{10} b^9 \sigma^5 \\
& + 384t^9 b^{10} \sigma^5 - 992t^9 b^9 \sigma^6 + 72t^{11} b^8 \sigma^4 \\
& - 48t^{10} b^9 \sigma^4 - 240t^{10} b^8 \sigma^5 - 336t^9 b^9 \sigma^5 \\
& + 384t^9 b^8 \sigma^6 - 112t^8 b^{10} \sigma^5 + 256t^8 b^9 \sigma^6 \\
& - 8t^{11} b^8 \sigma^3 + 176t^{10} b^8 \sigma^4 + 512t^{10} b^7 \sigma^5
\end{aligned}$$

$$\begin{aligned}
& -32t^9b^9\sigma^4 - 288t^9b^8\sigma^5 - 96t^8b^{10}\sigma^4 \\
& + 272t^8b^9\sigma^5 - 400t^{10}b^7\sigma^4 - 32t^9b^9\sigma^3 \\
& + 40t^9b^8\sigma^4 + 928t^9b^7\sigma^5 + 392t^8b^9\sigma^4 \\
& - 480t^8b^8\sigma^5 - 160t^7b^9\sigma^5 + 8t^{10}b^7\sigma^3 \\
& - 232t^9b^7\sigma^4 - 1024t^9b^6\sigma^5 - 432t^8b^8\sigma^4 \\
& - 544t^8b^7\sigma^5 - 192t^7b^9\sigma^4 + 864t^7b^8\sigma^5 \\
& + 64t^9b^7\sigma^3 + 448t^9b^6\sigma^4 - 160t^8b^8\sigma^3 \\
& - 752t^8b^7\sigma^4 - 128t^8b^6\sigma^5 - 32t^7b^9\sigma^3 \\
& + 824t^7b^8\sigma^4 - 2432t^7b^7\sigma^5 + 16t^6b^8\sigma^5 \\
& - 88t^9b^6\sigma^3 - 40t^8b^7\sigma^3 + 1184t^8b^6\sigma^4 \\
& + 136t^7b^8\sigma^3 - 768t^7b^7\sigma^4 + 1280t^7b^6\sigma^5 \\
& - 160t^6b^8\sigma^4 + 320t^6b^7\sigma^5 - 8t^9b^6\sigma^2 \\
& + 32t^8b^7\sigma^2 + 40t^8b^6\sigma^3 - 1152t^8b^5\sigma^4 \\
& + 16t^7b^7\sigma^3 - 208t^7b^6\sigma^4 - 224t^6b^8\sigma^3 \\
& + 448t^6b^7\sigma^4 + 128t^6b^6\sigma^5 + 24t^5b^8\sigma^4 \\
& - 160t^8b^5\sigma^3 - 72t^7b^7\sigma^2 - 40t^7b^6\sigma^3 \\
& + 608t^7b^5\sigma^4 + 984t^6b^7\sigma^3 - 1856t^6b^6\sigma^4 \\
& - 184t^5b^7\sigma^4 + 32t^8b^5\sigma^2 - 88t^7b^6\sigma^2 \\
& + 248t^7b^5\sigma^3 - 512t^7b^4\sigma^4 + 32t^6b^7\sigma^2 \\
& - 1568t^6b^6\sigma^3 + 640t^6b^5\sigma^4 - 80t^5b^7\sigma^3 \\
& + 816t^5b^6\sigma^4 + 64t^7b^5\sigma^2 + 96t^7b^4\sigma^3 \\
& - 144t^6b^6\sigma^2 - 912t^6b^5\sigma^3 - 512t^6b^4\sigma^4 \\
& - 64t^5b^7\sigma^2 + 392t^5b^6\sigma^3 - 1120t^5b^5\sigma^4 \\
& + 8t^4b^7\sigma^3 + 32t^4b^6\sigma^4 - 176t^7b^4\sigma^2 \\
& + 8t^6b^6\sigma + 344t^6b^5\sigma^2 + 1360t^6b^4\sigma^3 \\
& + 232t^5b^6\sigma^2 - 320t^5b^5\sigma^3 - 128t^5b^4\sigma^4 \\
& - 104t^4b^6\sigma^3 + 128t^4b^5\sigma^4 + 64t^6b^5\sigma \\
& - 136t^6b^4\sigma^2 - 1536t^6b^3\sigma^3 - 128t^5b^5\sigma^2 \\
& + 816t^5b^4\sigma^3 + 512t^5b^3\sigma^4 - 176t^4b^6\sigma^2 \\
& + 48t^4b^5\sigma^3 + 256t^4b^4\sigma^4 + 8t^3b^7\sigma^2 \\
& + 56t^3b^6\sigma^3 + 304t^6b^3\sigma^2 - 88t^5b^5\sigma \\
& - 448t^5b^4\sigma^2 - 448t^5b^3\sigma^3 + 16t^4b^6\sigma \\
& + 880t^4b^5\sigma^2 - 864t^4b^4\sigma^3 - 72t^3b^6\sigma^2 \\
& - 216t^3b^5\sigma^3 - 184t^5b^4\sigma + 592t^5b^3\sigma^2 \\
& + 512t^5b^2\sigma^3 - 1632t^4b^4\sigma^2 - 448t^4b^3\sigma^3 \\
& + 352t^3b^5\sigma^2 + 560t^3b^4\sigma^3 - 672t^5b^2\sigma^2 \\
& + 256t^4b^3\sigma^2 + 128t^4b^2\sigma^3 - 80t^3b^5\sigma \\
& - 944t^3b^4\sigma^2 + 320t^3b^3\sigma^3 + 8t^2b^6\sigma \\
& + 24t^2b^5\sigma^2 + 16t^2b^4\sigma^3 + 8t^4b^4 \\
& + 656t^4b^3\sigma - 64t^4b^2\sigma^2 + 256t^3b^4\sigma \\
& + 1232t^3b^3\sigma^2 - 896t^3b^2\sigma^3 - 64t^2b^5\sigma \\
& - 88t^2b^4\sigma^2 + 32t^4b^3 + 128t^4b\sigma^2 \\
& - 368t^3b^3\sigma + 96t^3b^2\sigma^2 + 512t^3b\sigma^3 \\
& + 224t^2b^4\sigma - 48t^2b^3\sigma^2 + 128t^2b^2\sigma^3
\end{aligned}$$

$$\begin{aligned}
& + 8tb^5\sigma + 32tb^4\sigma^2 - 48t^3b^3 \\
& - 576t^3b^2\sigma - 128t^3b\sigma^2 + 16t^2b^4 \\
& - 288t^2b^3\sigma + 384t^2b^2\sigma^2 - 72tb^4\sigma \\
& - 128tb^3\sigma^2 - 96t^3b^2 - 32t^2b^3 \\
& + 64t^2b^2\sigma - 1152t^2b\sigma^2 + 336tb^3\sigma \\
& + 320tb^2\sigma^2 + 80t^2b^2 + 384t^2b\sigma \\
& + 512t^2\sigma^2 - 48tb^3 - 896tb^2\sigma \\
& + 8b^4 + 16b^3\sigma + 320t^2b \\
& + 160tb^2 + 1152tb\sigma - 64b^3 \\
& - 64b^2\sigma - 288tb - 512t\sigma \\
& + 272b^2 + 128b\sigma - 256t \\
& - 576b + 640
\end{aligned}$$

$Q_1$

Corner chart 1:  $u = 1 - t$ ,  $b = u\lambda$ ,  $\sigma = \hat{\sigma}/u$ ;  $\hat{\sigma}$  is the positive root of  $Q_1$ . 33 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$H_1$

The polynomial tested in corner chart 1. 488 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$Q_2$

Corner chart 2:  $1 - t = b\mu$ ,  $\sigma = \hat{\sigma}/b$ ;  $\hat{\sigma}$  is the positive root of  $Q_2$ . 33 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$H_2$

The polynomial tested in corner chart 2. 488 terms. This polynomial is too long to typeset and is given only in the plain-text file.

## Appendix A.4.4, the response of total investment to the richer contestant's budget

Polynomials of Appendix A.4.4. Here  $a = 1/m_i$ ,  $b = 1/m_j$ , and on the arc  $b = \beta^2$ ,  $a = bs$ ,  $s = \beta\sigma$ ,  $\rho = at$ . The remaining charts use  $\rho = at$  (chart 1),  $a = \sigma\rho$  (chart 2),  $b = \nu^2s^2$ ,  $a = bs$ ,  $\rho = at$  (chart 3b),  $\rho = b\tau$ ,  $a = \theta\rho$  (chart 4) and  $b = \mu\rho$ ,  $a = bs$  (chart 5). Numerically the boundary root is  $m_i \approx 2.8639$  and  $T^{(j)}(2, 2, 1) \approx -2.929$ . The Krawczyk certification used 20000 slices of  $[0, \sqrt{1/2} - 10^{-7}]$ , refined into 20000 boxes, plus the endpoint box. Leading forms on the boundary of each chart. arc:  $q$ -form  $-8(t-1)$ ,  $H$ -form  $128(4\sigma^2t^2 - 8t - 1)$ . chart 1:  $q$ -form  $-b(b^2t - b^2 - 2bt + 4b + 8t - 8)$ ,  $H$ -form  $-16b^3(b^2 - 4b + 8)(b^2t - 4bt + 8t + 1)$ . chart 2:  $q$ -form  $b(b^2\sigma - b^2 - 4b\sigma + 2b + 8\sigma - 8)$ ,  $H$ -form  $-16b^3\sigma(b^2 - 4b + 8)(b^2 - 4b + \sigma + 8)$ . chart 3b, s=0:  $q$ -form  $-8(t-1)$ ,  $H$ -form  $128(-8\nu^2t - \nu^2 + 4t^2)$ . chart 3b, nu=0:  $q$ -form  $8(st - t + 1)$ ,  $H$ -form  $512t^2$ . chart 4:  $q$ -form  $8(\tau\theta + \theta - 1)$ ,  $H$ -form  $512\tau^2\theta^2$ . chart 5:  $q$ -form  $8(\mu s + s - 1)$ ,  $H$ -form  $512s^2$ . Box counts of the exhaustive search. chart 0 ( $a, b, \rho$ ),  $a \geq 10^{-2}$ : 1874138 bisections, 1708316 boxes excluded, 165823 inside the certified boxes. chart 1 ( $\beta, \sigma, t$ ),  $b \geq 10^{-3}$ ,  $a \leq 10^{-2}$ ,  $\rho \leq 3a$ : 15810 bisections, 14353 boxes excluded, 1458 inside the certified boxes. chart 2 ( $\sigma, b, \rho$ ),  $a = \sigma\rho \leq 10^{-2}$ ,  $b \geq 10^{-3}$ ,  $\rho \geq 3a$ : 3231 bisections, 3232 boxes excluded, 0 inside the certified boxes. chart 3a ( $\beta, \sigma, t$ ),  $b = \beta^2 \leq 10^{-3}$ ,  $s \leq 3\sqrt{b}$ ,  $\rho \leq 3a$ : 1277 bisections, 1020 boxes excluded, 258 inside the certified boxes. chart 3b ( $\nu, s, t$ ),  $b = \nu^2s^2 \leq 10^{-3}$ ,  $s \geq 3\sqrt{b}$ ,  $\rho \leq 3a$ : 14 bisections, 15 boxes excluded, 0 inside the certified boxes. chart 4 ( $b, \tau, \theta$ ),  $b \leq 10^{-3}$ ,  $3a \leq \rho \leq 10b$ : 106 bisections, 107 boxes excluded, 0 inside the certified boxes. chart 5 ( $\mu, s, \rho$ ),  $b = \mu\rho$ ,  $\rho \geq 10b$ : 4 bisections, 5 boxes excluded, 0 inside the certified boxes.

$$H_{\text{poly}}^{(j)}$$

$$dTI/dw_j - 1 = H_{\text{poly}}^{(j)}/(\rho(1+\rho)^3 A_0^3 \Delta). \text{ 199 terms.}$$

$$\begin{aligned} H_{\text{poly}}^{(j)}(m_i, m_j, \rho) = & 512m_i^9 m_j^9 \rho^3 - 256m_i^9 m_j^7 \rho^5 + 128m_i^8 m_j^8 \rho^5 \\ & - 512m_i^8 m_j^7 \rho^6 + 256m_i^7 m_j^8 \rho^6 - 256m_i^7 m_j^7 \rho^7 \\ & + 512m_i^6 m_j^9 \rho^6 + 128m_i^6 m_j^8 \rho^7 + 128m_i^{11} m_j^6 \rho^3 \\ & + 128m_i^{10} m_j^7 \rho^3 + 256m_i^{10} m_j^6 \rho^4 + 512m_i^9 m_j^9 \rho^2 \\ & - 640m_i^9 m_j^8 \rho^3 + 384m_i^9 m_j^7 \rho^4 + 512m_i^9 m_j^6 \rho^5 \\ & - 256m_i^8 m_j^9 \rho^3 - 128m_i^8 m_j^8 \rho^4 + 256m_i^8 m_j^7 \rho^5 \\ & + 768m_i^8 m_j^6 \rho^6 + 128m_i^7 m_j^9 \rho^4 + 96m_i^7 m_j^8 \rho^5 \\ & - 128m_i^7 m_j^7 \rho^6 + 384m_i^7 m_j^6 \rho^7 + 768m_i^6 m_j^9 \rho^5 \\ & - 448m_i^6 m_j^8 \rho^6 - 128m_i^6 m_j^7 \rho^7 - 128m_i^5 m_j^9 \rho^6 \\ & - 32m_i^5 m_j^8 \rho^7 - 192m_i^{11} m_j^5 \rho^3 + 384m_i^{10} m_j^7 \rho^2 \\ & - 608m_i^{10} m_j^6 \rho^3 - 384m_i^{10} m_j^5 \rho^4 - 384m_i^9 m_j^8 \rho^2 \\ & + 480m_i^9 m_j^7 \rho^3 - 480m_i^9 m_j^6 \rho^4 - 352m_i^9 m_j^5 \rho^5 \\ & - 640m_i^8 m_j^9 \rho^2 + 224m_i^8 m_j^8 \rho^3 - 768m_i^8 m_j^7 \rho^4 \\ & + 400m_i^8 m_j^6 \rho^5 - 320m_i^8 m_j^5 \rho^6 + 160m_i^7 m_j^9 \rho^3 \\ & - 224m_i^7 m_j^8 \rho^4 - 1216m_i^7 m_j^7 \rho^5 + 320m_i^7 m_j^6 \rho^6 \\ & - 160m_i^7 m_j^5 \rho^7 + 608m_i^6 m_j^9 \rho^4 - 720m_i^6 m_j^8 \rho^5 \\ & - 384m_i^6 m_j^7 \rho^6 + 48m_i^6 m_j^6 \rho^7 - 64m_i^5 m_j^9 \rho^5 \\ & + 128m_i^5 m_j^8 \rho^6 - 32m_i^5 m_j^7 \rho^7 + 128m_i^4 m_j^9 \rho^6 \\ & + 16m_i^4 m_j^8 \rho^7 + 80m_i^{11} m_j^4 \rho^3 - 2080m_i^{10} m_j^6 \rho^2 \\ & + 560m_i^{10} m_j^5 \rho^3 + 160m_i^{10} m_j^4 \rho^4 + 256m_i^9 m_j^7 \rho^2 \\ & - 1824m_i^9 m_j^6 \rho^3 + 464m_i^9 m_j^5 \rho^4 + 112m_i^9 m_j^4 \rho^5 \\ & + 352m_i^8 m_j^8 \rho^2 + 208m_i^8 m_j^7 \rho^3 + 400m_i^8 m_j^6 \rho^4 \\ & - 288m_i^8 m_j^5 \rho^5 + 64m_i^8 m_j^4 \rho^6 + 384m_i^7 m_j^9 \rho^2 \\ & - 144m_i^7 m_j^8 \rho^3 + 288m_i^7 m_j^7 \rho^4 + 128m_i^7 m_j^6 \rho^5 \\ & - 240m_i^7 m_j^5 \rho^6 + 32m_i^7 m_j^4 \rho^7 + 96m_i^6 m_j^9 \rho^3 \\ & - 304m_i^6 m_j^8 \rho^4 + 352m_i^6 m_j^7 \rho^5 + 48m_i^6 m_j^6 \rho^6 \\ & - 48m_i^6 m_j^5 \rho^7 - 144m_i^5 m_j^9 \rho^4 + 144m_i^5 m_j^8 \rho^5 \\ & + 64m_i^5 m_j^6 \rho^7 + 128m_i^4 m_j^9 \rho^5 - 48m_i^4 m_j^8 \rho^6 \\ & - 16m_i^4 m_j^7 \rho^7 - 16m_i^3 m_j^9 \rho^6 - 16m_i^{11} m_j^3 \rho^3 \\ & - 1024m_i^{10} m_j^6 \rho + 1808m_i^{10} m_j^5 \rho^2 - 272m_i^{10} m_j^4 \rho^3 \\ & - 32m_i^{10} m_j^3 \rho^4 + 640m_i^9 m_j^7 \rho - 464m_i^9 m_j^6 \rho^2 \\ & + 1640m_i^9 m_j^5 \rho^3 - 272m_i^9 m_j^4 \rho^4 - 16m_i^9 m_j^3 \rho^5 \\ & + 832m_i^8 m_j^7 \rho^2 - 592m_i^8 m_j^6 \rho^3 - 176m_i^8 m_j^5 \rho^4 \\ & + 16m_i^8 m_j^4 \rho^5 - 112m_i^7 m_j^8 \rho^2 - 1888m_i^7 m_j^6 \rho^4 \\ & + 64m_i^7 m_j^5 \rho^5 + 16m_i^7 m_j^4 \rho^6 - 176m_i^6 m_j^9 \rho^2 \\ & - 192m_i^6 m_j^7 \rho^4 - 624m_i^6 m_j^6 \rho^5 + 72m_i^6 m_j^5 \rho^6 \\ & - 88m_i^5 m_j^9 \rho^3 + 16m_i^5 m_j^8 \rho^4 - 16m_i^5 m_j^7 \rho^5 \\ & + 112m_i^5 m_j^6 \rho^6 + 80m_i^4 m_j^9 \rho^4 - 96m_i^4 m_j^8 \rho^5 \\ & - 16m_i^4 m_j^7 \rho^6 + 8m_i^2 m_j^9 \rho^6 + 1024m_i^{10} m_j^5 \rho \\ & - 848m_i^{10} m_j^4 \rho^2 + 64m_i^{10} m_j^3 \rho^3 - 256m_i^9 m_j^6 \rho \\ & + 760m_i^9 m_j^5 \rho^2 - 832m_i^9 m_j^4 \rho^3 + 64m_i^9 m_j^3 \rho^4 \end{aligned}$$

$$\begin{aligned}
& -608m_i^8m_j^7\rho - 2192m_i^8m_j^6\rho^2 + 352m_i^8m_j^5\rho^3 \\
& -16m_i^8m_j^4\rho^4 - 992m_i^7m_j^7\rho^2 - 2432m_i^7m_j^6\rho^3 \\
& + 1040m_i^7m_j^5\rho^4 - 64m_i^7m_j^4\rho^5 + 32m_i^6m_j^8\rho^2 \\
& - 192m_i^6m_j^7\rho^3 - 368m_i^6m_j^6\rho^4 + 424m_i^6m_j^5\rho^5 \\
& - 32m_i^6m_j^4\rho^6 + 40m_i^5m_j^9\rho^2 + 208m_i^5m_j^7\rho^4 \\
& + 96m_i^5m_j^6\rho^5 + 32m_i^4m_j^9\rho^3 - 32m_i^4m_j^8\rho^4 \\
& + 16m_i^4m_j^7\rho^5 - 32m_i^4m_j^6\rho^6 + 8m_i^2m_j^9\rho^5 \\
& - 512m_i^{10}m_j^4\rho + 200m_i^{10}m_j^3\rho^2 - 8m_i^{10}m_j^2\rho^3 \\
& - 128m_i^9m_j^6 + 48m_i^9m_j^5\rho - 432m_i^9m_j^4\rho^2 \\
& + 200m_i^9m_j^3\rho^3 - 8m_i^9m_j^2\rho^4 - 512m_i^8m_j^6\rho \\
& + 920m_i^8m_j^5\rho^2 - 48m_i^8m_j^4\rho^3 + 208m_i^7m_j^7\rho \\
& - 80m_i^7m_j^6\rho^2 + 1240m_i^7m_j^5\rho^3 - 224m_i^7m_j^4\rho^4 \\
& + 360m_i^6m_j^7\rho^2 + 344m_i^6m_j^6\rho^3 + 368m_i^6m_j^5\rho^4 \\
& - 96m_i^6m_j^4\rho^5 + 104m_i^5m_j^7\rho^3 - 56m_i^5m_j^6\rho^4 \\
& - 8m_i^4m_j^9\rho^2 - 48m_i^4m_j^7\rho^4 - 96m_i^4m_j^6\rho^5 \\
& - 8m_i^3m_j^9\rho^3 + 128m_i^{10}m_j^3\rho - 24m_i^{10}m_j^2\rho^2 \\
& + 64m_i^9m_j^5 + 16m_i^9m_j^4\rho + 128m_i^9m_j^3\rho^2 \\
& - 24m_i^9m_j^2\rho^3 + 32m_i^8m_j^6 + 320m_i^8m_j^5\rho \\
& - 176m_i^8m_j^4\rho^2 + 144m_i^7m_j^6\rho + 320m_i^7m_j^5\rho^2 \\
& - 288m_i^7m_j^4\rho^3 - 64m_i^6m_j^7\rho + 8m_i^6m_j^6\rho^2 \\
& + 64m_i^6m_j^5\rho^3 - 96m_i^6m_j^4\rho^4 - 128m_i^5m_j^7\rho^2 \\
& - 200m_i^5m_j^6\rho^3 - 64m_i^4m_j^7\rho^3 - 96m_i^4m_j^6\rho^4 \\
& - 16m_i^{10}m_j^2\rho - 16m_i^9m_j^4 - 16m_i^9m_j^2\rho^2 \\
& - 96m_i^8m_j^4\rho - 16m_i^7m_j^6 - 112m_i^7m_j^4\rho^2 \\
& - 80m_i^6m_j^6\rho - 32m_i^6m_j^4\rho^3 - 96m_i^5m_j^6\rho^2 \\
& - 32m_i^4m_j^6\rho^3
\end{aligned}$$

$T_{\text{ess}}^{(j)}$

$T^{(j)} = 8m_i^2m_j^2(m_i + \rho)^2T_{\text{ess}}^{(j)}/((1 + \rho)A_0^2D^3)$ . 2416 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$P^{(j)}$

Degree-18 factor of  $\text{Res}_\rho(q_1, H_{\text{poly}}^{(j)}(\cdot, 1, \cdot))$ . 19 terms.

$$\begin{aligned}
P^{(j)}(m_i) = & 153664m_i^{18} - 1001560m_i^{17} + 2771636m_i^{16} \\
& - 4745657m_i^{15} + 5209386m_i^{14} - 3945997m_i^{13} \\
& + 1949571m_i^{12} - 517369m_i^{11} - 16800m_i^{10} \\
& + 46059m_i^9 + 19069m_i^8 - 35427m_i^7 \\
& + 20978m_i^6 - 7183m_i^5 + 1453m_i^4 \\
& - 107m_i^3 - 28m_i^2 + 9m_i \\
& - 1
\end{aligned}$$

$\hat{q}$

$a^3b^3q = \beta^5\sigma\hat{q}$  on the arc chart. 18 terms.

$$\hat{q}(\beta, \sigma, t) = -\beta^{12}\sigma^4t^2 - \beta^{10}\sigma^2t^2 + 4\beta^9\sigma^3t^2$$

$$\begin{aligned}
& + 2\beta^8\sigma^2t^2 + \beta^7\sigma^3t - 8\beta^6\sigma^2t^2 \\
& - \beta^6\sigma^2t + \beta^6\sigma^2 + \beta^5\sigma t \\
& - 2\beta^4\sigma^2t - \beta^4t + \beta^4 \\
& - 2\beta^3\sigma + 2\beta^2t + 8\beta\sigma t \\
& - 4\beta^2 - 8t + 8
\end{aligned}$$

$\hat{H}$

$$a^{11}b^9H_{\text{poly}}^{(j)} = \beta^{12}\sigma^2\hat{H}. \text{ 199 terms.}$$

$$\begin{aligned}
\hat{H}(\beta, \sigma, t) = & -16\beta^{34}\sigma^{12}t^7 + 8\beta^{33}\sigma^{13}t^6 + 64\beta^{33}\sigma^{11}t^7 \\
& + 16\beta^{32}\sigma^{12}t^7 - 32\beta^{33}\sigma^{11}t^6 - 48\beta^{32}\sigma^{10}t^7 \\
& - 32\beta^{31}\sigma^{11}t^7 - 16\beta^{31}\sigma^{11}t^6 - 16\beta^{30}\sigma^{12}t^6 \\
& + 32\beta^{31}\sigma^9t^7 + 8\beta^{30}\sigma^{12}t^5 + 48\beta^{30}\sigma^{10}t^7 \\
& - 32\beta^{29}\sigma^{11}t^7 - 32\beta^{31}\sigma^9t^6 + 112\beta^{30}\sigma^{10}t^6 \\
& - 48\beta^{29}\sigma^{11}t^6 - 96\beta^{30}\sigma^{10}t^5 - 160\beta^{29}\sigma^9t^7 \\
& - 128\beta^{28}\sigma^{10}t^7 + 72\beta^{29}\sigma^9t^6 + 128\beta^{27}\sigma^{11}t^6 \\
& + 16\beta^{28}\sigma^{10}t^5 + 384\beta^{27}\sigma^9t^7 + 128\beta^{26}\sigma^{10}t^7 \\
& + 16\beta^{28}\sigma^8t^6 + 48\beta^{27}\sigma^9t^6 + 128\beta^{26}\sigma^{10}t^6 \\
& - 96\beta^{28}\sigma^8t^5 + 96\beta^{27}\sigma^9t^5 - 96\beta^{26}\sigma^{10}t^5 \\
& - 256\beta^{25}\sigma^9t^7 - 96\beta^{27}\sigma^9t^4 - 240\beta^{26}\sigma^8t^6 \\
& - 384\beta^{25}\sigma^9t^6 - 128\beta^{24}\sigma^{10}t^6 + 424\beta^{26}\sigma^8t^5 \\
& - 16\beta^{25}\sigma^9t^5 + 128\beta^{24}\sigma^{10}t^5 - 48\beta^{25}\sigma^9t^4 \\
& + 64\beta^{25}\sigma^7t^6 + 320\beta^{24}\sigma^8t^6 - 448\beta^{23}\sigma^9t^6 \\
& - 64\beta^{25}\sigma^7t^5 - 624\beta^{24}\sigma^8t^5 + 144\beta^{23}\sigma^9t^5 \\
& - 96\beta^{25}\sigma^7t^4 - 56\beta^{24}\sigma^8t^4 - 32\beta^{23}\sigma^9t^4 \\
& - 320\beta^{23}\sigma^7t^6 - 128\beta^{22}\sigma^8t^6 + 512\beta^{21}\sigma^9t^6 \\
& - 32\beta^{24}\sigma^8t^3 + 64\beta^{23}\sigma^7t^5 + 352\beta^{22}\sigma^8t^5 \\
& - 64\beta^{21}\sigma^9t^5 + 368\beta^{23}\sigma^7t^4 + 208\beta^{22}\sigma^8t^4 \\
& + 80\beta^{21}\sigma^9t^4 + 768\beta^{21}\sigma^7t^6 + 256\beta^{20}\sigma^8t^6 \\
& - 64\beta^{22}\sigma^8t^3 + 16\beta^{22}\sigma^6t^5 - 8\beta^{21}\sigma^9t^3 \\
& + 128\beta^{21}\sigma^7t^5 - 720\beta^{20}\sigma^8t^5 - 224\beta^{22}\sigma^6t^4 \\
& - 368\beta^{21}\sigma^7t^4 + 16\beta^{20}\sigma^8t^4 - 512\beta^{19}\sigma^7t^6 \\
& - 32\beta^{22}\sigma^6t^3 - 200\beta^{21}\sigma^7t^3 - 16\beta^{21}\sigma^5t^5 \\
& - 288\beta^{20}\sigma^6t^5 - 1216\beta^{19}\sigma^7t^5 + 768\beta^{18}\sigma^8t^5 \\
& + 1040\beta^{20}\sigma^6t^4 - 192\beta^{19}\sigma^7t^4 - 144\beta^{18}\sigma^8t^4 \\
& + 64\beta^{20}\sigma^6t^3 + 104\beta^{19}\sigma^7t^3 + 112\beta^{19}\sigma^5t^5 \\
& + 32\beta^{18}\sigma^8t^3 + 400\beta^{18}\sigma^6t^5 + 96\beta^{17}\sigma^7t^5 \\
& - 8\beta^{20}\sigma^4t^4 - 16\beta^{19}\sigma^5t^4 - 1888\beta^{18}\sigma^6t^4 \\
& - 304\beta^{17}\sigma^7t^4 - 288\beta^{19}\sigma^5t^3 + 344\beta^{18}\sigma^6t^3 \\
& - 352\beta^{17}\sigma^5t^5 + 256\beta^{16}\sigma^6t^5 - 96\beta^{18}\sigma^6t^2 \\
& + 64\beta^{18}\sigma^4t^4 - 176\beta^{17}\sigma^5t^4 + 288\beta^{16}\sigma^6t^4 \\
& + 608\beta^{15}\sigma^7t^4 + 1240\beta^{17}\sigma^5t^3 - 192\beta^{16}\sigma^6t^3 \\
& - 88\beta^{15}\sigma^7t^3 + 512\beta^{15}\sigma^5t^5 + 128\beta^{14}\sigma^6t^5 \\
& - 128\beta^{16}\sigma^6t^2 - 272\beta^{16}\sigma^4t^4 - 8\beta^{15}\sigma^7t^2 \\
& + 400\beta^{15}\sigma^5t^4 - 224\beta^{14}\sigma^6t^4 - 24\beta^{17}\sigma^3t^3
\end{aligned}$$

$$\begin{aligned}
& -48\beta^{16}\sigma^4t^3 - 2432\beta^{15}\sigma^5t^3 - 256\beta^{13}\sigma^5t^5 \\
& -112\beta^{16}\sigma^4t^2 + 8\beta^{15}\sigma^5t^2 - 32\beta^{15}\sigma^3t^4 \\
& + 464\beta^{14}\sigma^4t^4 - 768\beta^{13}\sigma^5t^4 + 128\beta^{12}\sigma^6t^4 \\
& + 200\beta^{15}\sigma^3t^3 + 352\beta^{14}\sigma^4t^3 + 96\beta^{12}\sigma^6t^3 \\
& + 320\beta^{14}\sigma^4t^2 + 360\beta^{13}\sigma^5t^2 + 160\beta^{13}\sigma^3t^4 \\
& + 40\beta^{12}\sigma^6t^2 - 480\beta^{12}\sigma^4t^4 - 128\beta^{11}\sigma^5t^4 \\
& - 8\beta^{14}\sigma^2t^3 - 832\beta^{13}\sigma^3t^3 - 592\beta^{12}\sigma^4t^3 \\
& - 144\beta^{11}\sigma^5t^3 - 16\beta^{14}\sigma^2t^2 - 176\beta^{13}\sigma^3t^2 \\
& - 80\beta^{12}\sigma^4t^2 + 32\beta^{11}\sigma^5t^2 - 384\beta^{11}\sigma^3t^4 \\
& + 384\beta^{10}\sigma^4t^4 - 80\beta^{12}\sigma^4t + 64\beta^{12}\sigma^2t^3 \\
& + 1640\beta^{11}\sigma^3t^3 + 208\beta^{10}\sigma^4t^3 + 160\beta^9\sigma^5t^3 \\
& + 128\beta^{12}\sigma^2t^2 + 920\beta^{11}\sigma^3t^2 - 992\beta^{10}\sigma^4t^2 \\
& - 176\beta^9\sigma^5t^2 + 256\beta^9\sigma^3t^4 - 64\beta^{10}\sigma^4t \\
& - 272\beta^{10}\sigma^2t^3 - 1824\beta^9\sigma^3t^3 + 224\beta^8\sigma^4t^3 \\
& - 24\beta^{11}\sigma t^2 - 432\beta^{10}\sigma^2t^2 - 2192\beta^9\sigma^3t^2 \\
& - 112\beta^8\sigma^4t^2 - 96\beta^{10}\sigma^2t + 144\beta^9\sigma^3t \\
& - 16\beta^9\sigma t^3 + 560\beta^8\sigma^2t^3 + 480\beta^7\sigma^3t^3 \\
& - 256\beta^6\sigma^4t^3 + 200\beta^9\sigma t^2 + 760\beta^8\sigma^2t^2 \\
& + 832\beta^7\sigma^3t^2 + 384\beta^6\sigma^4t^2 + 320\beta^8\sigma^2t \\
& + 208\beta^7\sigma^3t + 80\beta^7\sigma t^3 - 608\beta^6\sigma^2t^3 \\
& - 640\beta^5\sigma^3t^3 - 848\beta^7\sigma t^2 - 464\beta^6\sigma^2t^2 \\
& + 352\beta^5\sigma^3t^2 - 16\beta^8t + 16\beta^7\sigma t \\
& - 512\beta^6\sigma^2t - 192\beta^5\sigma t^3 + 128\beta^4\sigma^2t^3 \\
& + 512\beta^3\sigma^3t^3 - 16\beta^6\sigma^2 + 1808\beta^5\sigma t^2 \\
& + 256\beta^4\sigma^2t^2 - 640\beta^3\sigma^3t^2 + 128\beta^6t \\
& + 48\beta^5\sigma t - 608\beta^4\sigma^2t + 128\beta^3\sigma t^3 \\
& - 2080\beta^3\sigma t^2 - 384\beta^2\sigma^2t^2 - 512\beta^4t \\
& - 256\beta^3\sigma t - 16\beta^4 + 32\beta^3\sigma \\
& + 384\beta\sigma t^2 + 512\sigma^2t^2 + 1024\beta^2t \\
& + 640\beta\sigma t + 64\beta^2 - 1024t \\
& - 128
\end{aligned}$$

$\hat{T}$

$a^{20}b^{17}T_{\text{ess}}^{(j)} = \beta^{29}\sigma^6\hat{T}$ . 2416 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$\tilde{q}$

Chart 0:  $\tilde{q} = a^3b^3q$ . 18 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$\tilde{H}$

Chart 0:  $\tilde{H} = a^{11}b^9H_{\text{poly}}^{(j)}$ . 199 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$Q_2$

Chart 2 polynomial (reduced). 18 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$H_2$

Chart 2 polynomial (reduced). 199 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$Q_3$

Chart 3b polynomial (reduced). 18 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$H_3$

Chart 3b polynomial (reduced). 199 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$Q_4$

Chart 4 polynomial (reduced). 18 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$H_4$

Chart 4 polynomial (reduced). 199 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$Q_5$

Chart 5 polynomial (reduced). 18 terms. This polynomial is too long to typeset and is given only in the plain-text file.

$H_5$

Chart 5 polynomial (reduced). 199 terms. This polynomial is too long to typeset and is given only in the plain-text file.